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OPTICS

ADDISON-WESLEY PUBLISHING COMPANY

READING, MASSACHUSETTS
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Interference 9

The intricate color patterns shimmering across an oil slick on a wet asphalt pavement result from one of the more common manifestations of the phenomenon of interference.* On a macroscopic scale we might consider the related problem of the interaction of surface ripples in a pool of water. Our everyday experience with this kind of situation allows us to envision a complex distribution of disturbances (as shown e.g. in Fig. 9.1). There might be regions where two (or more) waves have overlapped, partially or even completely canceling each other. Still other regions might exist in the pattern, where the resultant troughs and crests were even more pronounced than those of any of the constituent waves. After being superimposed, the individual waves separate and continue on, completely unaffected by their previous encounter.

Phenomena arising from optical interference would, of course, be quite difficult to interpret in terms of a purely corpuscular model. The wave theory of the electromagnetic nature of light, however, provides a natural basis from which to proceed. Recall that the expression describing the optical disturbance is a second-order homogeneous linear partial differential equation (3.22). As we have seen, it therefore obeys the important *principle of superposition*. Accordingly, the resultant electric-field intensity **E**, at a point in space where two or more light waves overlap, is equal to the *vector sum* of the individual constituent disturbances. Briefly then, *optical interference may be termed an interaction of two or more light waves yielding a resultant irradiance which deviates from the sum of the component irradiances.*

Out of the multitude of optical systems which produce

* The layer of water on the asphalt allows the oil film to assume the shape of a smooth planar surface. The black asphalt absorbs the transmitted light preventing back reflection which would tend to obscure the fringes.

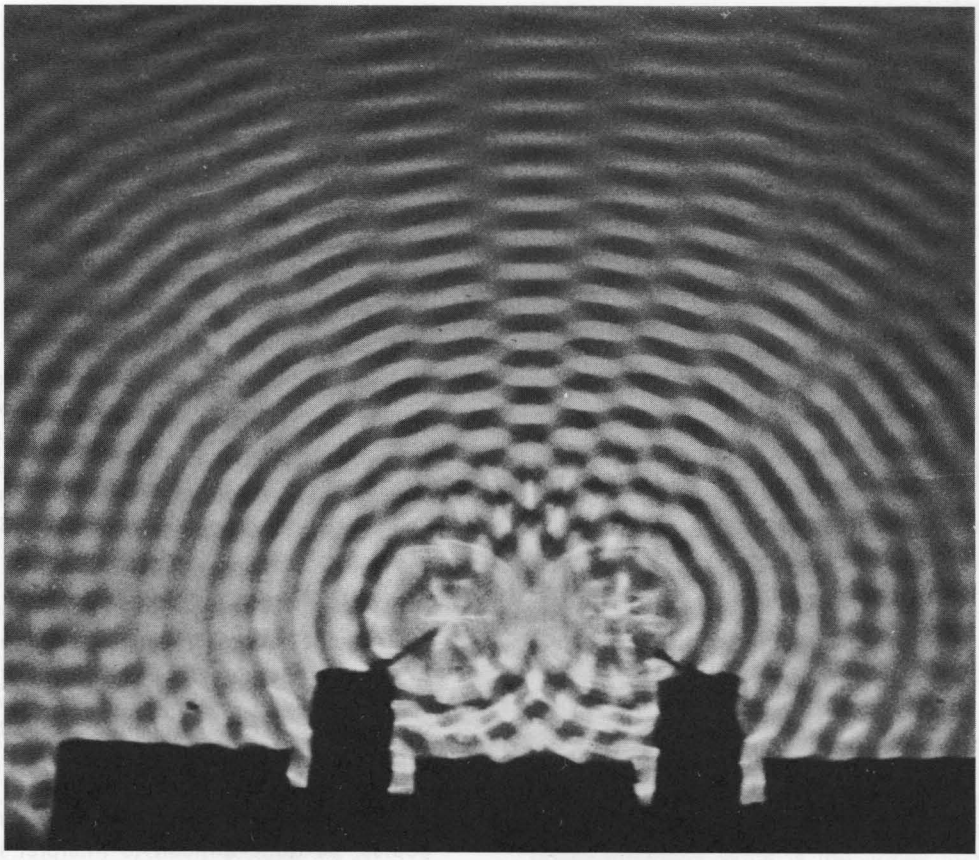


Fig. 9.1 Water waves from two point sources in a ripple tank.

interference, we will choose a few of the more important to examine. Interferometric devices will be divided, for the sake of discussion, into two groups, *wavefront splitting* and *amplitude splitting*. In the first instance, portions of the primary wavefront are used either directly as sources to emit secondary waves, or, in conjunction with optical devices, to produce virtual sources of secondary waves. These secondary waves are then brought together, thereupon to interfere. In the latter case of amplitude splitting, the primary wave itself is divided into two segments which travel different paths before recombining and interfering.

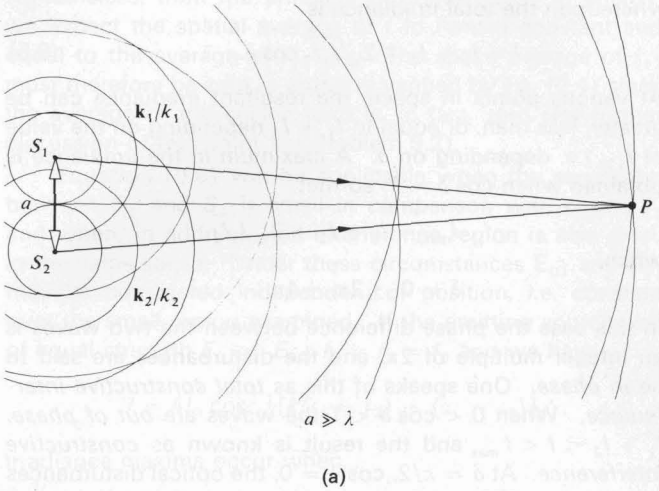
9.1 GENERAL CONSIDERATIONS

We have already examined the problem of the superposition of two scalar waves (Section 7.1) and in many respects those results will again be applicable. But light is, of course, a vector phenomenon; the electric and magnetic fields are vector fields. And an appreciation of this fact is fundamental to any kind of intuitive understanding of optics. Needless to say, there are many situations in which the particular optical system is so configured that the vector nature of light is of little practical significance. We will therefore derive the basic interference equations within the context of the vector model, thereafter delineating the conditions under which the scalar treatment is applicable.

In accordance with the principle of superposition, the electric-field intensity $\mathbf{E}$, at a point in space, arising from the separate fields $\mathbf{E}_1, \mathbf{E}_2, \ldots$ of various contributing sources is given by

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \cdots .$$

Once again we point out that the optical disturbance, or



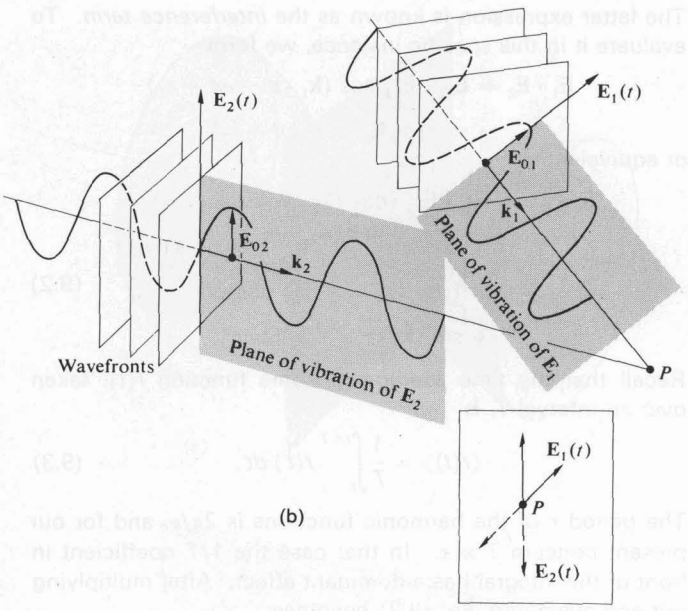


Fig. 9.2 Waves from two point sources overlapping in space.

light field **E**, varies in time at an exceedingly rapid rate, roughly

$$4.3 \times 10^{14}\ \text{Hz} \qquad \text{to} \qquad 7.5 \times 10^{14}\ \text{Hz},$$

making the actual field an impractical quantity to detect. On the other hand, the irradiance I can be measured directly using a wide variety of sensors (e.g. photocells, bolometers, photographic emulsions or eyes). Indeed then, if we are to study interference, we had best approach the problem by way of the irradiance.

Much of the analysis to follow can be performed without specifying the particular shape of the wavefronts and the results are therefore quite general in their applicability (Problem 9.1). For the sake of simplicity, however, consider two point sources S_1 and S_2 emitting monochromatic waves of the same frequency, in a homogeneous medium. Furthermore, let their separation a be much greater than λ. Locate the point of observation P, far enough away from the sources so that at P the wavefronts will be planes (Fig. 9.2). For the moment, we will only consider linearly polarized waves of the form

$$\mathbf{E}_1(\mathbf{r}, t) = \mathbf{E}_{01} \cos (\mathbf{k}_1 \cdot \mathbf{r} - \omega t + \varepsilon_1)$$

and

$$\mathbf{E}_2(\mathbf{r}, t) = \mathbf{E}_{02} \cos (\mathbf{k}_2 \cdot \mathbf{r} - \omega t + \varepsilon_2).$$

We saw in Chapter 3 that the irradiance at P is given by

$$I = \epsilon v \langle \mathbf{E}^2 \rangle.$$

Inasmuch as we will be concerned only with relative irradiances within the same medium we will, for the time being at least, simply neglect the constants and set

$$I = \langle \mathbf{E}^2 \rangle.$$

What is meant by $\langle \mathbf{E}^2 \rangle$ is of course the time average of the magnitude of the electric field intensity squared or $\langle \mathbf{E} \cdot \mathbf{E} \rangle$. Accordingly

$$\mathbf{E}^2 = \mathbf{E} \cdot \mathbf{E},$$

where now

$$\mathbf{E}^2 = (\mathbf{E}_1 + \mathbf{E}_2) \cdot (\mathbf{E}_1 + \mathbf{E}_2)$$

and thus

$$\mathbf{E}^2 = \mathbf{E}_1^2 + \mathbf{E}_2^2 + 2\mathbf{E}_1 \cdot \mathbf{E}_2.$$

Taking the time average of both sides, the irradiance becomes

$$I = I_1 + I_2 + I_{12} \tag{9.1}$$

provided that

$$I_1 = \langle \mathbf{E}_1^2 \rangle,$$

$$I_2 = \langle \mathbf{E}_2^2 \rangle$$

and

$$I_{12} = 2\langle \mathbf{E}_1 \cdot \mathbf{E}_2 \rangle.$$

The latter expression is known as the *interference term*. To evaluate it in this specific instance, we form

$$\mathbf{E}_1 \cdot \mathbf{E}_2 = \mathbf{E}_{01} \cdot \mathbf{E}_{02} \cos(\mathbf{k}_1 \cdot \mathbf{r} - \omega t + \varepsilon_1) \times \cos(\mathbf{k}_2 \cdot \mathbf{r} - \omega t + \varepsilon_2)$$

or equivalently

$$\begin{aligned}\mathbf{E}_1 \cdot \mathbf{E}_2 = \mathbf{E}_{01} \cdot \mathbf{E}_{02} [&\cos(\mathbf{k}_1 \cdot \mathbf{r} + \varepsilon_1) \\ &\times \cos \omega t + \sin(\mathbf{k}_1 \cdot \mathbf{r} + \varepsilon_1) \sin \omega t] \\ &\times [\cos(\mathbf{k}_2 \cdot \mathbf{r} + \varepsilon_2) \cos \omega t \\ &+ \sin(\mathbf{k}_2 \cdot \mathbf{r} + \varepsilon_2) \sin \omega t].\end{aligned} \tag{9.2}$$

Recall that the time average of some function $f(t)$, taken over an interval T, is

$$\langle f(t) \rangle = \frac{1}{T} \int_t^{t+T} f(t')\, dt'. \tag{9.3}$$

The period τ of the harmonic functions is $2\pi/\omega$ and for our present concern $T \gg \tau$. In that case the $1/T$ coefficient in front of the integral has a dominant effect. After multiplying out and averaging, Eq. (9.2) becomes

$$\langle \mathbf{E}_1 \cdot \mathbf{E}_2 \rangle = \tfrac{1}{2} \mathbf{E}_{01} \cdot \mathbf{E}_{02} \cos(\mathbf{k}_1 \cdot \mathbf{r} + \varepsilon_1 - \mathbf{k}_2 \cdot \mathbf{r} - \varepsilon_2),$$

where use was made of the fact that $\langle \cos^2 \omega t \rangle = \frac{1}{2}$, $\langle \sin^2 \omega t \rangle = \frac{1}{2}$ and $\langle \cos \omega t \sin \omega t \rangle = 0$. The interference term is then

$$I_{12} = \mathbf{E}_{01} \cdot \mathbf{E}_{02} \cos \delta \tag{9.4}$$

and δ, equal to $(\mathbf{k}_1 \cdot \mathbf{r} - \mathbf{k}_2 \cdot \mathbf{r} + \varepsilon_1 - \varepsilon_2)$, is the *phase difference* arising from a combined path-length and epoch-angle difference. Notice that if $\mathbf{E}_{01}$ and $\mathbf{E}_{02}$ (and therefore $\mathbf{E}_1$ and $\mathbf{E}_2$) are perpendicular, $I_{12} = 0$ and $I = I_1 + I_2$. Two such orthogonal $\mathscr{P}$-states will combine to yield an $\mathscr{R}$-, $\mathscr{L}$-, $\mathscr{P}$- or $\mathscr{E}$-state but the flux-density distribution will be unaltered.

By far the most commonly occurring situation in the work to follow corresponds to $\mathbf{E}_{01}$ parallel to $\mathbf{E}_{02}$. In that case, the irradiance reduces to the value found in the scalar treatment of Section 7.1. Under those conditions

$$I_{12} = E_{01} E_{02} \cos \delta.$$

This can be written in a more convenient way by noticing that

$$I_1 = \langle \mathbf{E}_1^2 \rangle = \frac{E_{01}^2}{2}$$

and

$$I_2 = \langle \mathbf{E}_2^2 \rangle = \frac{E_{02}^2}{2}.$$

The interference term becomes

$$I_{12} = 2\sqrt{I_1 I_2} \cos \delta,$$

whereupon the total irradiance is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta. \tag{9.5}$$

At various points in space, the resultant irradiance can be greater, less than, or equal to $I_1 + I_2$ depending on the value of I_{12}, i.e. depending on δ. A maximum in the irradiance is obtained when $\cos \delta = 1$, so that

$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

when

$$\delta = 0, \pm 2\pi, \pm 4\pi, \ldots .$$

In this case the phase difference between the two waves is an integer multiple of 2π, and the disturbances are said to be *in phase*. One speaks of this as *total constructive interference*. When $0 < \cos \delta < 1$ the waves are *out of phase*, $I_1 + I_2 < I < I_{\max}$ and the result is known as *constructive interference*. At $\delta = \pi/2$, $\cos \delta = 0$, the optical disturbances are said to be 90° out of phase and $I = I_1 + I_2$. For $0 > \cos \delta > -1$ we have the condition of *destructive* interference, $I_1 + I_2 > I > I_{\min}$. The minimum in the irradiance results when the waves are 180° out of phase, troughs overlap crests, $\cos \delta = -1$, and

$$I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}.$$

This, of course, occurs when $\delta = \pm\pi, \pm 3\pi, \pm 5\pi, \ldots$, and it is referred to as *total destructive interference*.

Another somewhat special, yet very important case arises when the amplitudes of both waves reaching P in Fig. 9.2 are equal, i.e. $\mathbf{E}_{01} = \mathbf{E}_{02}$. Since the irradiance contributions from both sources are then equal, let $I_1 = I_2 = I_0$. Equation (9.5) can now be written as

$$I = 2I_0(1 + \cos \delta) = 4I_0 \cos^2 \frac{\delta}{2} \tag{9.6}$$

from which it follows that $I_{\min} = 0$ and $I_{\max} = 4I_0$.

Equation (9.5) holds equally well for the spherical waves emitted by S_1 and S_2. Such waves can be expressed as

$$\mathbf{E}_1(r_1, t) = \mathbf{E}_{01}(r_1) \exp[i(kr_1 - \omega t + \varepsilon_1)]$$

and

$$\mathbf{E}_2(r_2, t) = \mathbf{E}_{02}(r_2) \exp[i(kr_2 - \omega t + \varepsilon_2)].$$

The terms r_1 and r_2 are the radii of the spherical wavefronts overlapping at P, i.e. they specify the distances from the sources to P. In this case

$$\delta = k(r_1 - r_2) + (\varepsilon_1 - \varepsilon_2).$$

The flux density in the region surrounding S_1 and S_2 will certainly vary from point to point as $(r_1 - r_2)$ varies. Nonetheless, from the principle of conservation of energy, we expect the spatial average of I to remain constant and equal to the average of $I_1 + I_2$. The space average of I_{12} must therefore be zero, a property verified by Eq. (9.4) since the average of the cosine term is, in fact, zero (for further discussion of this point see Problem 9.2).

Equation (9.6) will be applicable when the separation between S_1 and S_2 is small in comparison with r_1 and r_2 and when, in addition, the interference region is also small in the same sense. Under these circumstances $\mathbf{E}_{01}$ and $\mathbf{E}_{02}$ may be considered independent of position, i.e. constant over the small region examined. If the emitting sources are of equal strength $E_{01} = E_{02}$, $I_1 = I_2 = I_0$ and we have

$$I = 4I_0 \cos^2 \tfrac{1}{2}[k(r_1 - r_2) + (\varepsilon_1 - \varepsilon_2)].$$

Irradiance maxima occur when

$$\delta = 2\pi m$$

provided that $m = 0, \pm 1, \pm 2, \ldots$. Similarly minima, for which $I = 0$, arise when

$$\delta = \pi(2m + 1).$$

These expressions can be rewritten such that maximum irradiance occurs when

$$(r_1 - r_2) = [2\pi m + (\varepsilon_2 - \varepsilon_1)]/k \qquad (9.7a)$$

and minimum when

$$(r_1 - r_2) = [\pi(2m + 1) + (\varepsilon_2 - \varepsilon_1)]/k. \qquad (9.7b)$$

Either one of these equations defines a family of surfaces, each of which is a hyperboloid of revolution. The vertices of the hyperboloids are separated by distances equal to the right-hand sides of Eqs. (9.7). The foci are located at S_1 and S_2. If the waves are in phase at the emitter $\varepsilon_1 - \varepsilon_2 = 0$, and Eqs. (9.7) simplify to

$$(r_1 - r_2) = 2\pi m/k = m\lambda$$

$$(r_1 - r_2) = \pi(2m + 1)/k = (m + 1/2)\lambda$$

for maximum and minimum irradiance, respectively. Figure 9.3(a) shows a few of the surfaces over which there are irradiance maxima. The dark and light zones which would be seen on a screen placed in the region of interference are known as *interference fringes* [Fig. 9.3(b)].

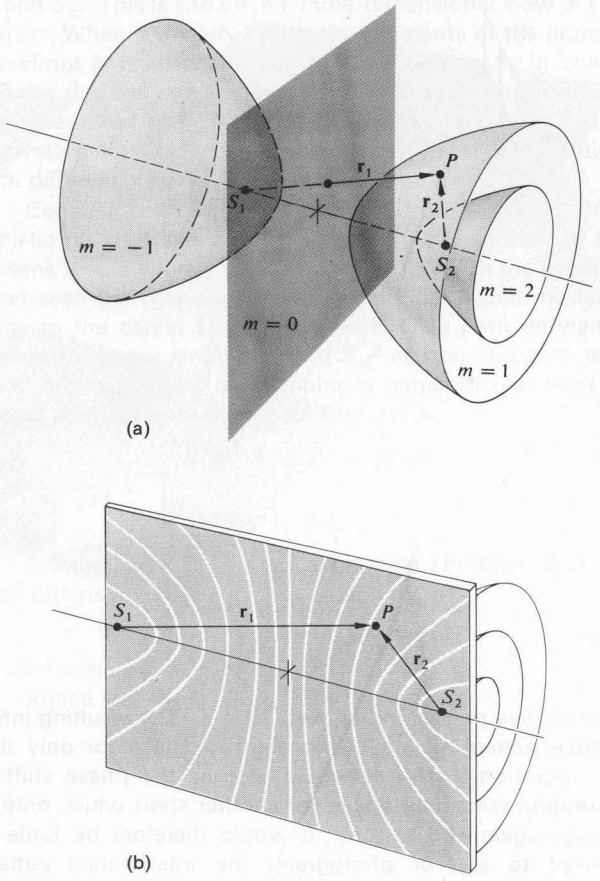


Fig. 9.3 Hyperboloidal surfaces of maximum irradiance for two point sources.

9.2 CONDITIONS FOR INTERFERENCE

If the interference pattern corresponding to Eqs. (9.7) is to be observable, the phase difference $(\varepsilon_1 - \varepsilon_2)$ between the two sources must remain fairly constant in time. Such sources are said to be coherent.* Two overlapping beams arising from separate emitters will interfere, but the resultant pattern will not be sustained long enough to be readily observable. A typical source contains a large number of excited atoms, each capable of radiating a wave train for about 10^{-8} s. Two distinct sources could therefore maintain

* Chapter 10 is devoted to the study of coherence and so here we'll merely touch on those aspects which are immediately pertinent.

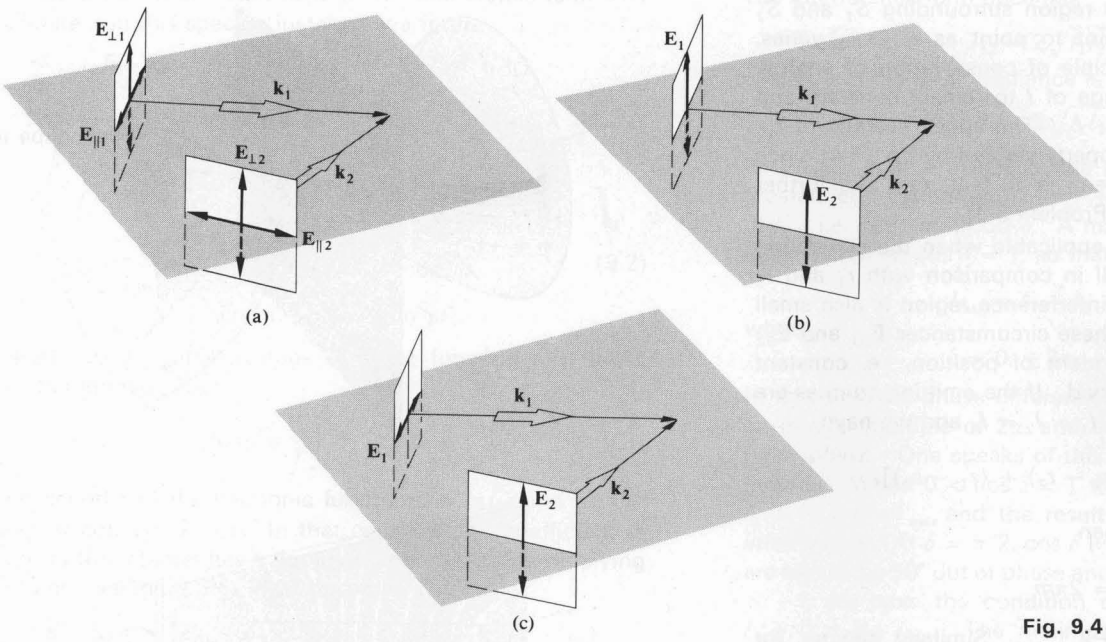


Fig. 9.4 Interference of polarized light.

their relative phases for, at best, 10^{-8} s. The resulting interference pattern would be constant in space for only that brief duration, before it would vary as the phase shifted, thereupon remaining stable for another short while, only to change again and so on. It would therefore be futile to attempt to see or photograph the interference pattern resulting from two light bulbs. Two separate lasers have been used to generate interference patterns, but we'll come back to that later. The most common means of overcoming this problem, as we shall see, is to make one source serve to produce two coherent secondary sources.

If two beams are to interfere to produce a stable pattern, they must have very nearly the same frequency. A significant frequency difference would result in a rapidly varying, time dependent, phase difference which in turn would cause I_{12} to average to zero during the detection interval (see Section 7.2.1).

The clearest patterns will exist when the interfering waves have equal or very nearly equal amplitudes. The central regions of the dark and light fringes will then correspond to complete destructive and constructive interference, respectively, yielding maximum contrast.

In the previous section, we assumed that the two overlapping optical disturbance vectors were linearly polarized and parallel. Nonetheless, the formulae of Section 9.1 apply as well to more complicated situations; indeed the treatment is applicable regardless of the polarization state of the waves. To appreciate this, recall that any polarization state can be synthesized out of two orthogonal $\mathscr{P}$-states. For natural (unpolarized) light these $\mathscr{P}$-states are mutually incoherent, but that represents no particular difficulty.

Suppose that every wave has its propagation vector in the same plane so that we can label the constituent orthogonal $\mathscr{P}$-states with respect to that plane, e.g. $\mathbf{E}_{\parallel}$ and $\mathbf{E}_{\perp}$, which are parallel and perpendicular to the plane, respectively [Fig. 9.4(a)]. Thus any plane wave, whether polarized or not, can be written in the form $(\mathbf{E}_{\parallel} + \mathbf{E}_{\perp})$. Imagine then, that the waves $(\mathbf{E}_{\parallel 1} + \mathbf{E}_{\perp 1})$ and $(\mathbf{E}_{\parallel 2} + \mathbf{E}_{\perp 2})$ emitted from two identical coherent sources superimpose in some region of space. The resulting flux-density distribution would consist of two independent, precisely overlapping, interference patterns, $\langle(\mathbf{E}_{\parallel 1} + \mathbf{E}_{\parallel 2})^2\rangle$ and $\langle(\mathbf{E}_{\perp 1} + \mathbf{E}_{\perp 2})^2\rangle$. Therefore, while we derived the equations of the previous section specifically for linear light, they are applicable, as well, for any polarization state including natural light.

Notice that even though $\mathbf{E}_{\perp 1}$ and $\mathbf{E}_{\perp 2}$ are always parallel to each other $\mathbf{E}_{\parallel 1}$ and $\mathbf{E}_{\parallel 2}$, which are in the reference plane,

need not be. They will only be parallel when the two beams are themselves parallel (i.e. $\mathbf{k}_1 = \mathbf{k}_2$). The inherent vector nature of the interference process as manifest in the dot-product representation (9.4) of I_{12} cannot therefore be ignored. As we shall see, there are a great many practical situations in which the beams approach being parallel and for these the scalar theory will do rather nicely. Even so, Figs. 9.4(b) and (c) are included as an urge to caution. They depict the imminent overlapping of two coherent linearly polarized waves. In Fig. 9.4(b) the optical vectors are parallel, even though the beams aren't, and interference would nonetheless result. In Fig. 9.4(c) the optical vectors are perpendicular and $I_{12} = 0$, which would be the case here even if the beams were parallel.

Fresnel and Arago made an extensive study of the conditions under which the interference of polarized light occurs and their conclusions summarize some of the above considerations. The *Fresnel-Arago laws* are as follows:

1. Two orthogonal, coherent $\mathscr{P}$-states cannot interfere in the sense that $I_{12} = 0$ and no fringes result.

2. Two parallel, coherent $\mathscr{P}$-states will interfere in the same way as will natural light.

3. The two constituent orthogonal $\mathscr{P}$-states of natural light cannot interfere to form a readily observable fringe pattern even if rotated into alignment. This last point is understandable since these $\mathscr{P}$-states are incoherent.

9.3 WAVEFRONT-SPLITTING INTERFEROMETERS

Return for a moment to Fig. (9.3) where the equation

$$(r_1 - r_2) = m\lambda$$

determined the surfaces of maximum irradiance. Since the wavelength λ for light is very small, a large number of surfaces corresponding to the lower values of m will exist close to, and on either side of, the plane $m = 0$. A number of fairly straight parallel fringes will therefore appear on a screen placed perpendicular to that ($m = 0$) plane and in the vicinity of it, and for this case the approximation $r_1 \approx r_2$ will hold. If S_1 and S_2 are then displaced normal to the $\overline{S_1S_2}$ line, the fringes will merely be displaced parallel to themselves. Practically, two narrow slits will therefore increase the irradiance leaving the central region of the two-point source pattern otherwise essentially unchanged.

Consider a hypothetical monochromatic plane wave illuminating a long narrow slit. From that primary slit a cylindrical wave will emerge and suppose that this wave, in turn, falls on two parallel, narrow, closely spaced slits S_1 and S_2. This is shown in a three-dimensional view in Fig. 9.5(a). When symmetry exists, the segments of the primary wavefront arriving at the two slits will be exactly in phase, and the slits will constitute two coherent secondary sources. We expect that wherever the two waves coming from S_1 and S_2 overlap, interference will occur (provided that the optical path difference is less than the coherence length, $c\,\Delta t$).

Consider the construction shown in Fig. 9.5(c). In a realistic physical situation the distance between each of the screens would be very large in comparison with the distance a between the two slits, and all the fringes would be fairly close to the center O of the screen. The path difference between the rays along $\overline{S_1P}$ and $\overline{S_2P}$ can be obtained, to a good approximation, by dropping a perpendicular from S_2 onto $\overline{S_1P}$. This path difference is given by

$$(\overline{S_1B}) = (\overline{S_1P}) - (\overline{S_2P}) \tag{9.8}$$

or

$$(\overline{S_1B}) = r_1 - r_2.$$

Continuing with this approximation (Problem 9.4) the path difference can be expressed as

$$r_1 - r_2 = a\theta \tag{9.9}$$

since $\theta \approx \sin\theta$.

Notice that

$$\theta = \frac{y}{s} \tag{9.10}$$

and so

$$r_1 - r_2 = \frac{a}{s}y.$$

In accordance with Section 9.1, *constructive* interference will occur when

$$r_1 - r_2 = m\lambda. \tag{9.11}$$

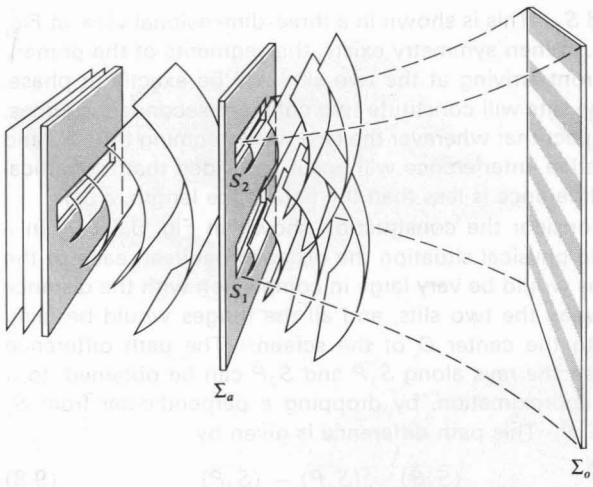
Thus, from the last two relations we obtain

$$y_m = \frac{s}{a}m\lambda. \tag{9.12}$$

This gives the position of the mth bright fringe on the screen if we count the maximum at 0 as the zeroth fringe. The angular position of the fringe is obtained by substituting the last expression into Eq. (9.10); thus

$$\theta_m = \frac{m\lambda}{a}. \tag{9.13}$$

This relationship can be obtained directly by inspecting Fig. 9.5(c). For the mth-order interference, m whole wavelengths



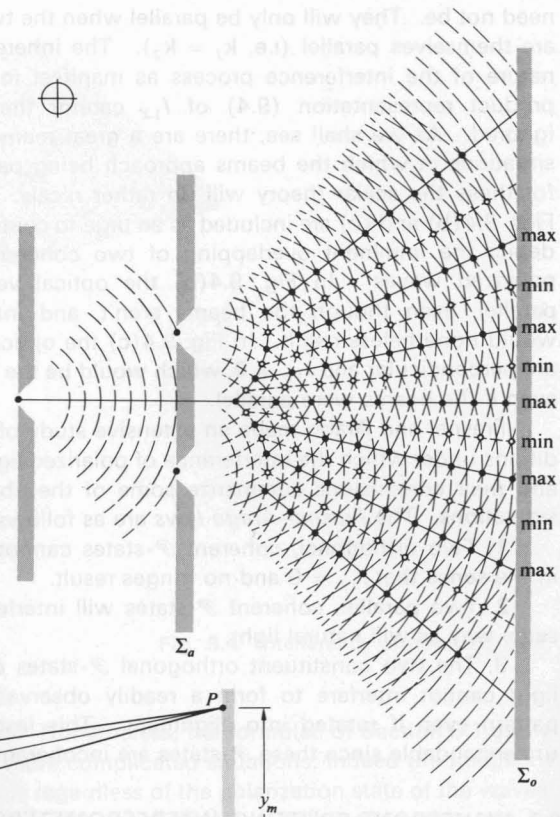
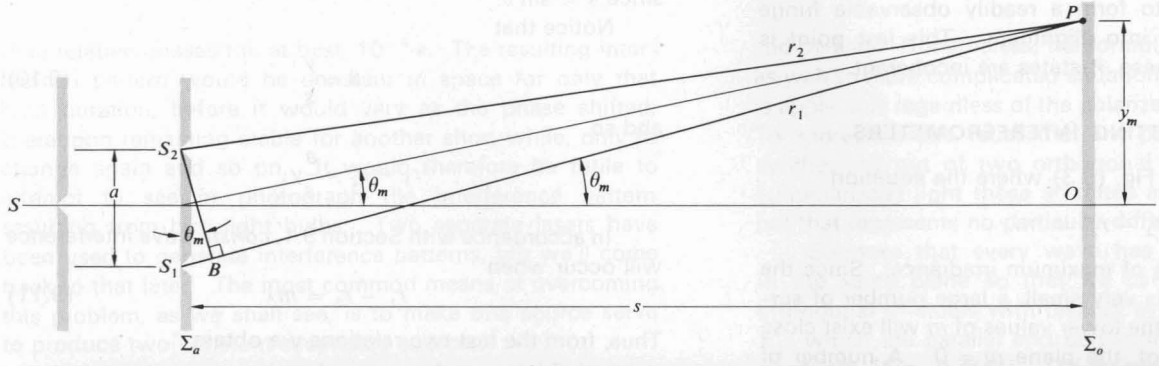
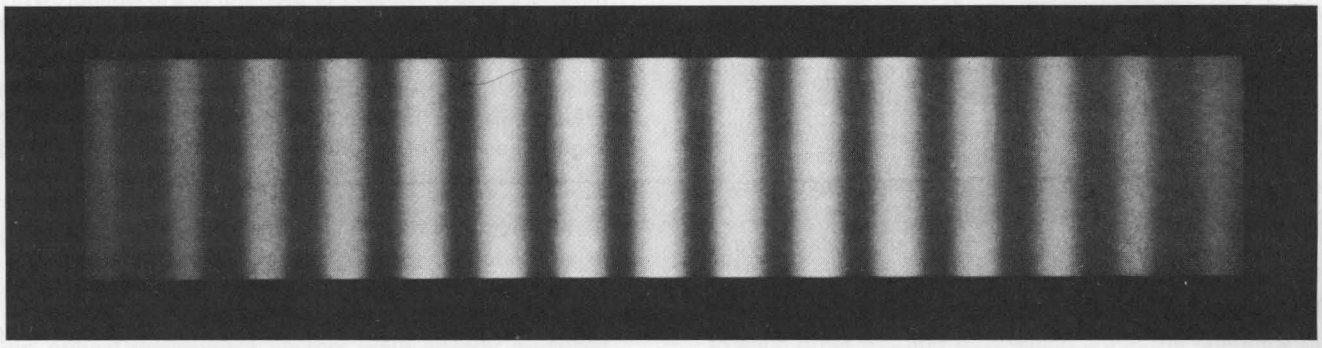


Fig. 9.5 Young's experiment. [Photo courtesy M. Cagnet, M. Francon, and J. C. Thrierr: *Atlas optischer Erscheinungen,* Berlin–Heidelberg–New York: Springer, 1962.]





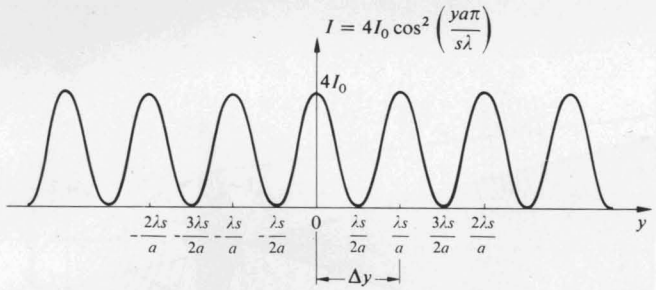


Fig. 9.6 Irradiance versus distance.

should fit within the distance $r_1 - r_2$. Therefore, from the triangle S_1S_2B,

$$\theta_m = m\lambda/a.$$

The spacing of the fringes on the screen can be gotten readily from Eq. (9.12). The difference in the positions of two consecutive maxima is

$$y_{m+1} - y_m = \frac{s}{a}(m+1)\lambda - \frac{s}{a}m\lambda$$

or

$$\Delta y = \frac{s}{a}\lambda. \tag{9.14}$$

Since this pattern is equivalent to that obtained for two overlapping spherical waves (at least in the $r_1 \approx r_2$ region), we can apply Eq. (9.6). Using the phase difference

$$\delta = k(r_1 - r_2).$$

Equation (9.6) can be rewritten as

$$I = 4I_0 \cos^2 \frac{k(r_1 - r_2)}{2},$$

provided, of course, that the two beams are coherent and have equal irradiances I_0. With

$$r_1 - r_2 = ya/s$$

the resultant irradiance becomes

$$I = 4I_0 \cos^2 \frac{ya\pi}{s\lambda}.$$

As shown in Fig. 9.6 consecutive maxima are separated by the Δy given in Eq. (9.14).*

* Modifications of this pattern arising as a result of increasing the width of either the primary S, or secondary-source slits will be considered in later chapters (10 and 12). In the former case, fringe contrast will be used as a measure of the degree of coherence (Section 12.1). In the latter, diffraction effects become significant.

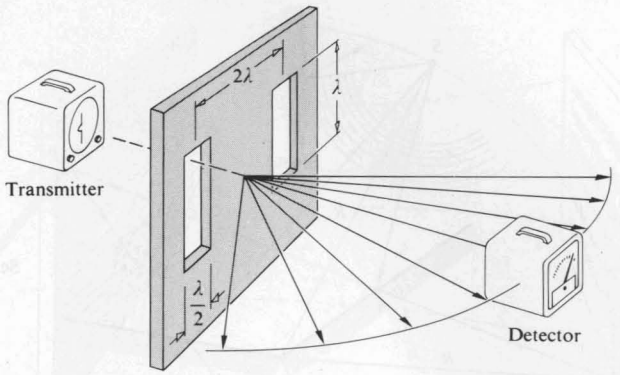


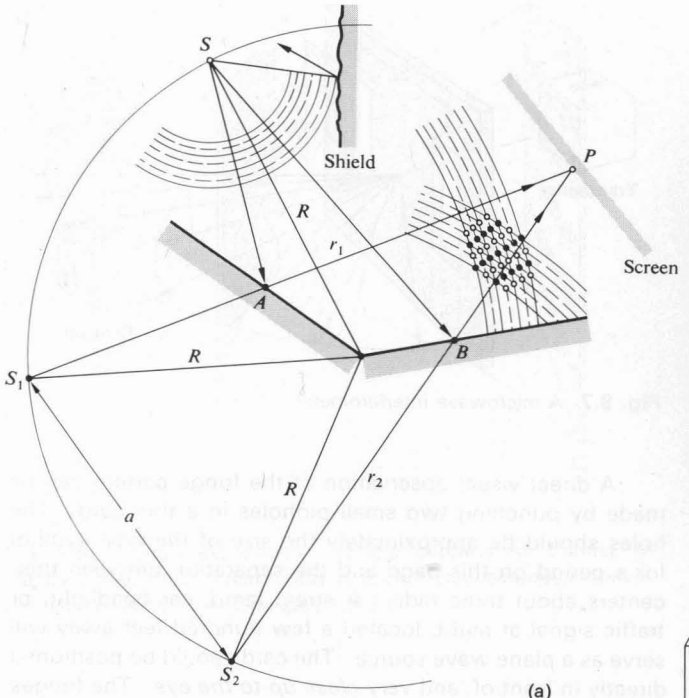
Fig. 9.7 A microwave interferometer.

A direct visual observation of the fringe pattern can be made by punching two small pinholes in a thin card. The holes should be approximately the size of the type symbol for a period on this page and the separation between their centers about three radii. A street lamp, car headlight, or traffic signal at night, located a few hundred feet away will serve as a plane wave source. The card should be positioned directly in front of, and *very close up to the eye*. The fringes will appear perpendicular to the line of centers. The pattern is much more readily seen using slits as discussed in Section 10.2.2, but you should give the pinholes a try.

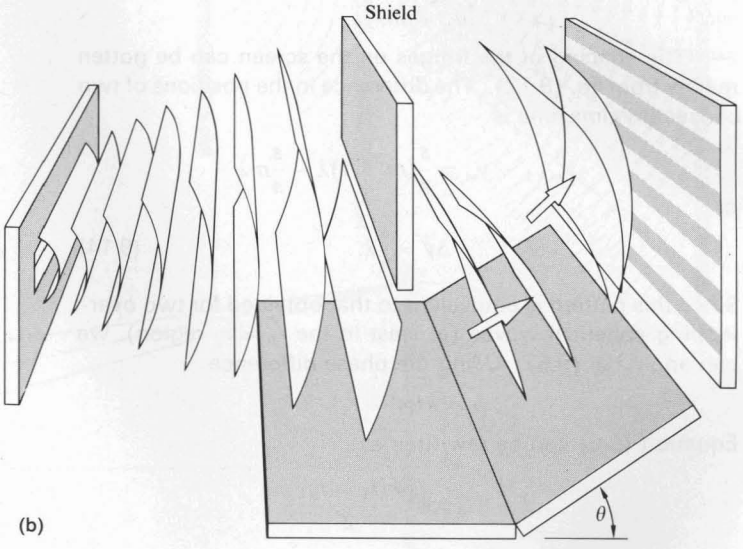
Microwaves, because of their long wavelength, also offer an easy way to observe double-slit interference. Two slits (e.g. $\lambda/2$ wide by λ long, separated by 2λ) cut in a piece of sheet metal or foil will serve quite well as secondary sources (Fig. 9.7).

The interferometric configuration discussed above, with either point or slit sources, is known as *Young's experiment*. The same physical and mathematical considerations apply directly to a number of other wavefront-splitting interferometers. Most common among these are Fresnel's double mirror, Fresnel's double prism and Lloyd's mirror.

Fresnel's double mirror consists of two plane front silvered mirrors inclined to each other at a very small angle, as shown in Fig. 9.8. One portion of the cylindrical wavefront coming from slit S is reflected from the first mirror, while another portion of the wavefront is reflected from the second mirror. An interference field exists in space in the region where the two reflected waves are superimposed on each other. The images (S_1 and S_2) of the slit S in the two mirrors can be considered as separate coherent sources, placed at a distance a apart. It follows from the laws of reflection, as



(a)



(b)

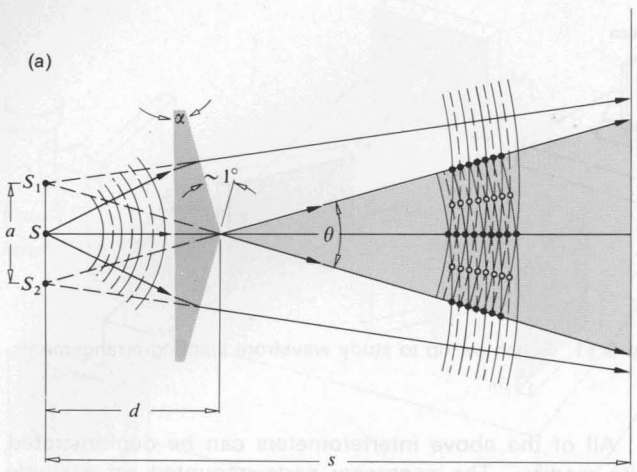
Fig. 9.8 Fresnel's double mirror.

illustrated in Fig. 9.8(a), that $\overline{SA} = \overline{S_1A}$, $\overline{SB} = \overline{S_2B}$, so that $\overline{SA} + \overline{AP} = r_1$ and $\overline{SB} + \overline{BP} = r_2$. The optical path-length difference between the two rays is then simply $r_1 - r_2$. The various maxima occur at $r_1 - r_2 = m\lambda$ as was the case for Young's interferometer. Again, the separation of the fringes is given by

$$\Delta y = \frac{s}{a}\lambda$$

where s is the distance between the plane of the two virtual sources (S_1, S_2) and the screen. The arrangement in Fig. 9.8 has again been deliberately exaggerated to make the geometry somewhat clearer. Notice that the angle θ between the mirrors must be quite small if the electric field vectors for each of the two beams are to be parallel, or nearly so. Let $\mathbf{E}_1$ and $\mathbf{E}_2$ represent the light waves emitted from the coherent virtual sources S_1 and S_2. At any instant in time at the point P in space, each of these vectors can be resolved into components, parallel and perpendicular to the plane of the figure. With $\mathbf{k}_1$ and $\mathbf{k}_2$ parallel to AP and BP respectively, it should be apparent that the components of $\mathbf{E}_1$ and $\mathbf{E}_2$ in the plane of the figure will approach being parallel only for small θ.

The Fresnel double prism or biprism consists of two thin prisms joined at their bases as shown in Fig. 9.9. A single cylindrical wavefront impinges on both prisms. The top



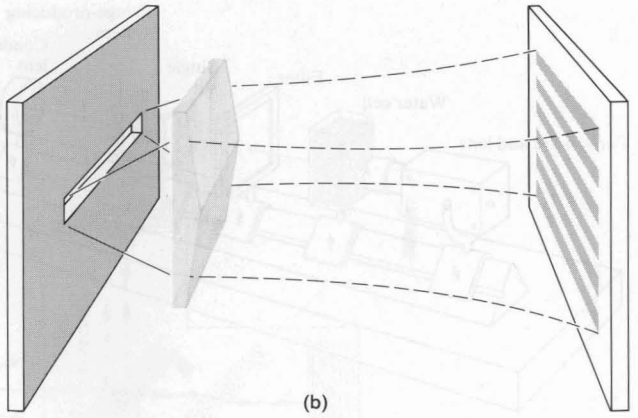


Fig. 9.9 Fresnel's biprism.

portion of the wavefront is refracted downward, while the lower segment is refracted upward. In the region of superposition, interference occurs. Here, again, two virtual sources S_1 and S_2 exist, separated by a distance a which can be expressed in terms of the prism angle α (Problem 9.5) where $s \gg a$. The expression for the separation of the fringes is the same as before.

The last wavefront-splitting interferometer which we will consider is Lloyd's mirror. It is shown in Fig. 9.10 and consists of a flat piece of either dielectric or metal which serves as a mirror, from which is reflected a portion of the cylindrical wavefront coming from slit S. Another portion of the wavefront proceeds directly from the slit to the screen. For the separation a, between the two coherent sources, we take the distance between the actual slit and its image S_1 in the mirror. The spacing of the fringes is once again given by $(s/a)\lambda$. The distinguishing feature of this device is that at glancing incidence ($\theta_i \approx \pi/2$) the reflected beam undergoes a

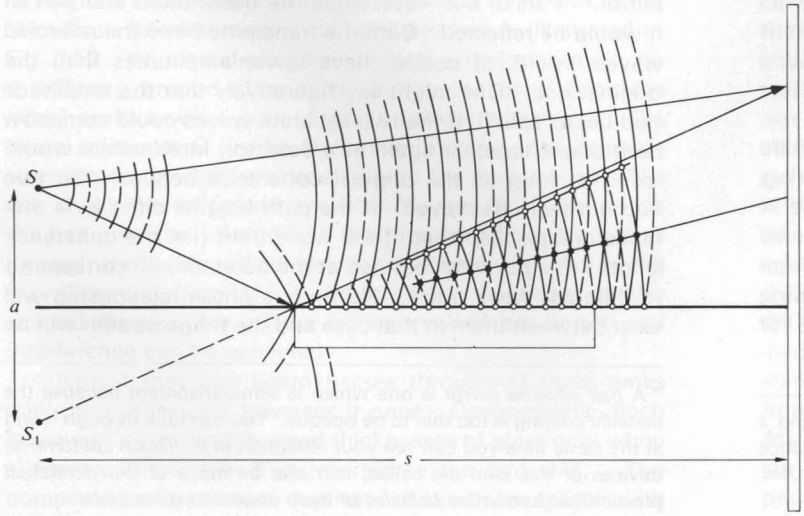


Fig. 9.10 Lloyd's mirror.

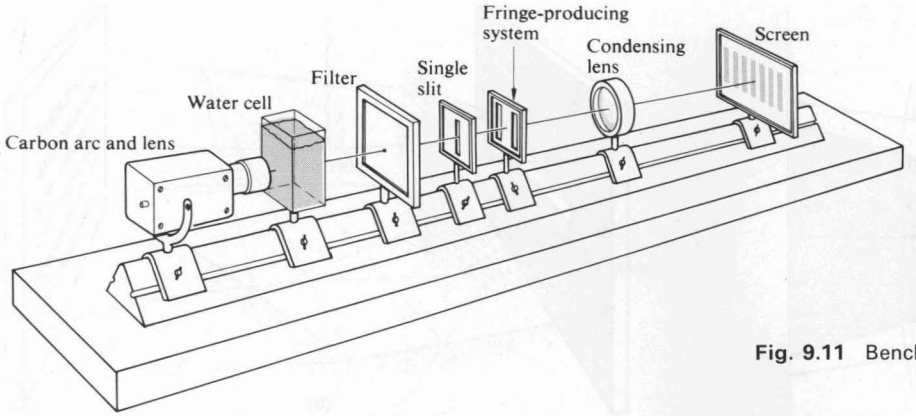


Fig. 9.11 Bench set up to study wavefront splitting arrangements.

180° phase shift (recall that the amplitude reflection coefficients are then both equal to -1). With an additional phase shift of $\pm\pi$

$$\delta = k(r_1 - r_2) \pm \pi$$

and the irradiance becomes

$$I = 4I_0 \sin^2\left(\frac{\pi a y}{s\lambda}\right).$$

The fringe pattern for Lloyd's mirror is complementary to that of Young's interferometer; the maxima of one pattern exist at values of y which correspond to minima in the other pattern. The top edge of the mirror is equivalent to $y = 0$ and will be the center of a dark fringe rather than a bright one as in Young's device. The lower half of the pattern will be obstructed by the presence of the mirror itself. Consider then, what would happen if a thin sheet of transparent material were placed in the path of the rays traveling directly to the screen. The transparent sheet would have the effect of increasing the number of wavelengths in each direct ray. The entire pattern would accordingly move upward to where the reflected rays would travel a bit farther before interfering. Because of the obvious inherent simplicity of this device, it has found use over a very wide region of the electromagnetic spectrum. The actual reflecting surfaces have varied from crystals for x-rays, to ordinary glass for light, wire screening for microwaves, to a lake or even the earth's ionosphere for radio waves.*

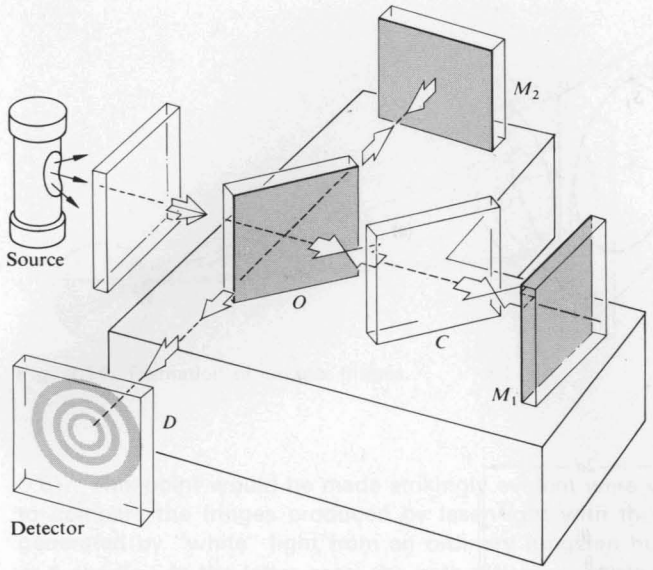
* For a discussion of the effects of having a finite slit width and a finite frequency bandwidth see R. N. Wolfe and F. C. Eisen, "Irradiance Distribution in a Lloyd Mirror Interference Pattern," *J. Opt. Soc. Am.* **38**, 706 (1948).

All of the above interferometers can be demonstrated quite readily. The necessary parts, mounted on a single optical bench, are shown diagrammatically in Fig. 9.11. The source of light should be a strong one; if a laser is not available, a discharge lamp or a carbon arc followed by a water cell, to cool things down a bit, will do nicely. The light is then not monochromatic but the fringes, which will be colored, can still be observed. A satisfactory approximation to monochromatic light can be obtained with a filter placed in front of the arc. A low-power He–Ne laser is perhaps the easiest source to work with and you won't need a water cell or filter.

9.4 AMPLITUDE-SPLITTING INTERFEROMETERS

Suppose that a light wave were incident on a half-silvered mirror.* Part of the wave would be transmitted and part of it would be reflected. Both the transmitted and the reflected waves would, of course, have lower amplitudes than the original one. One might say figuratively that the amplitude had been "split". If the two separate waves could somehow be brought together again at a detector, interference would result as long as the original coherence between the two had not been destroyed. If the path lengths differ by a distance greater than that of the wave train (i.e. the coherence length), the portions reunited at the detector will correspond to different wave trains. No unique phase relationship will exist between them in that case and the fringe pattern will be

* A *half silvered mirror* is one which is semitransparent because the metallic coating is too thin to be opaque. You can look through it and at the same time you can see your reflection in it. *Beam splitters*, as devices of this kind are called, can also be made of thin stretched plastic films known as *pellicles* or even uncoated glass plate.



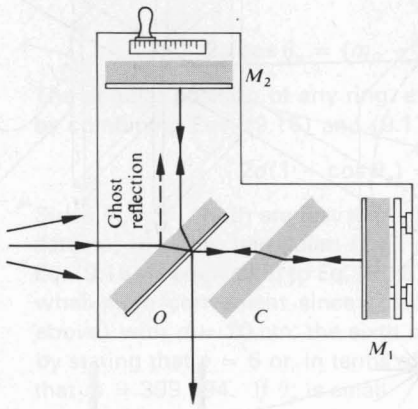


Fig. 9.12 The Michelson interferometer.

unstable to the point of being unobservable. We will get back to these ideas when we consider coherence theory in more detail. For the moment we restrict ourselves to those cases where the path difference is less than the coherence length.

There are a great number of amplitude-splitting interferometers of which we shall discuss only a very few. By far the best known, and historically the most important, is the Michelson interferometer. Its configuration is illustrated in Fig. 9.12. An extended source (which may, for example, be a diffusing ground-glass plate illuminated by a discharge lamp) emits a wave, part of which travels to the right. The beam splitter at O divides the wave into two, one segment traveling to the right and one up into the background. The two waves are reflected by mirrors M_1 and M_2 and return to the beam splitter. Part of the wave coming from M_2 passes through the beam splitter going downward and part of the wave coming from M_1 is deflected by the beam splitter toward the detector. Thus the two waves are united and interference can be expected.

Notice that one beam passes through O three times while the other only traverses it once. Consequently, each beam will pass through equal thicknesses of glass only when a *compensator plate* C is inserted in the arm OM_1. The compensator is an exact duplicate of the beam splitter with the exception of any possible silvering or thin film coating on the beam splitter. It is positioned at an angle of 45° so that O and C are parallel to each other. With the compensator in place any optical path difference arises from the actual path difference. In addition, because of the dispersion of the beam splitter, the optical path is a function of λ. Accordingly, for quantitative work, the interferometer without the compensator plate can only be used with a quasimonochromatic source. The inclusion of a compensator negates the effect of dispersion so that even a source with a very broad bandwidth will then generate discernible fringes.

To understand how fringes are formed, refer to the construction shown in Fig. 9.13 where the physical components are represented more as mathematical surfaces. An observer at the position of the detector will simultaneously see both mirrors M_1 and M_2 along with the source Σ in the beam splitter. Accordingly, we can redraw the interferometer as if all the elements were in a straight line. Here M_1' corresponds to the image of mirror M_1 in the beam splitter and Σ has been swung over in line with O and M_2. The positions of these elements in the diagram depend on their relative distances from O (e.g. M_1' can be in front of, behind, or coincident with M_2 and can even pass through it). The surfaces Σ_1 and Σ_2 are the images of the source Σ in mirrors M_1 and M_2, respectively. Now consider a single point S on the source

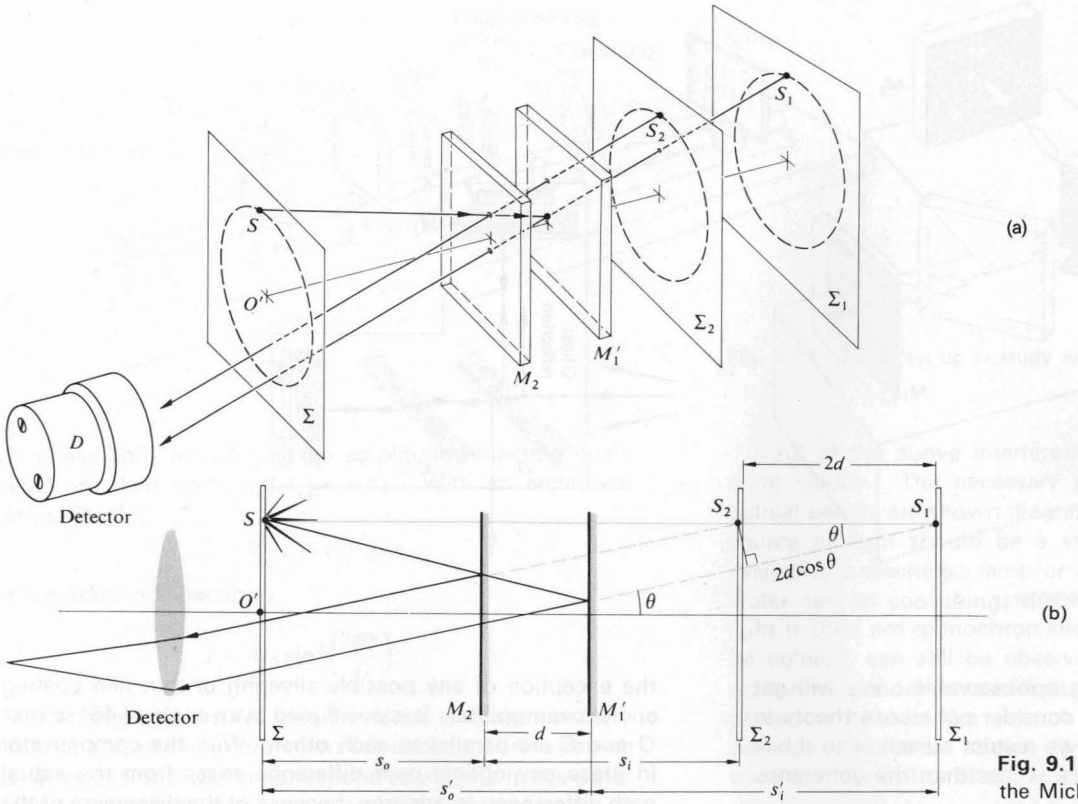


Fig. 9.13 A conceptual rearrangement of the Michelson interferometer.

emitting light in all directions; let's follow the course of one emerging ray. In actuality a wave from S will be split at O and its segments will thereafter be reflected by M_1 and M_2. In our schematic diagram we represent this by reflecting the ray off both M_2 and M_1'. To an observer at D the two reflected rays will appear to have come from the image points S_1 and S_2 [note that all rays shown in Fig. 9.13(a) and (b) share a common plane of incidence]. For all practical purposes S_1 and S_2 are coherent point sources and we can anticipate a flux-density distribution obeying Eq. (9.5). As can be seen from the figure, the optical path difference for these rays is very nearly $2d\cos\theta$ which represents a phase difference of $k_0 2d\cos\theta$. There is an additional phase term arising from the fact that the wave traversing the arm OM_2 is internally reflected in the beam splitter, whereas the OM_1-wave is externally reflected at O. If the beam splitter is simply an uncoated glass plate, the relative phase shift resulting from the two reflections will (Section 4.5) be π radians. *Destructive*, rather than constructive, interference will then exist when

$$2d\cos\theta_m = m\lambda_0 \tag{9.15}$$

where m is an integer. If this condition is fulfilled for the point S, then it will be equally well fulfilled for any point on Σ which lies on the circle of radius $O'S$ where O' is located on the axis of the detector. As illustrated in Fig. 9.14, an observer will see a circular fringe system concentric with the central axis of her eyes' lens. Because of the small aperture of the eye the observer will not be able to see the entire pattern without the use of a large lens near the beam splitter to collect most of the emergent light.

The dependence of θ_m on λ_0 in Eq. (9.15) requires that if we use a source containing a number of frequency components (e.g. a mercury discharge lamp) each such component will generate a fringe system of its own. Note too, that since $2d\cos\theta_m$ must be less than the coherence length of the source, it follows that laser light will be particularly easy to use in demonstrating the interferometer (see Section

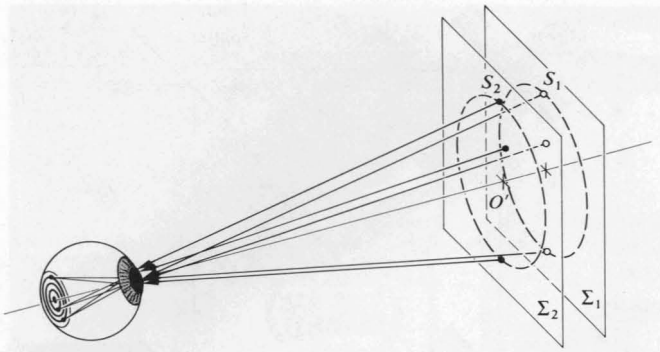


Fig. 9.14 Formation of circular fringes.

9.6). This point would be made strikingly evident were we to compare the fringes produced by laser light with those generated by "white" light from an ordinary tungsten bulb or a candle. In the latter case, the path difference must be very nearly zero if we are to see any fringes at all while in the former instance a difference of 10 cm has little noticeable effect.

An interference pattern in quasimonochromatic light typically consists of a large number of alternatively bright and dark rings. A particular ring corresponds to a fixed *order m*. As M_2 is moved toward M_1', d decreases and according to Eq. (9.15) $\cos\theta_m$ increases while θ_m therefore decreases. The rings shrink toward the center with the highest-order one disappearing whenever d decreases by $\lambda_0/2$. Each remaining ring broadens as more and more fringes vanish at the center, until only a few fill the whole screen. By the time $d = 0$ has been reached, the central fringe will have spread out filling the entire field of view. With a phase shift of π resulting from reflection off the beam splitter, the whole screen will then be an interference minimum (lack of perfection in the optical elements can render this unobservable). Moving M_2 still farther causes the fringes to reappear at the center and move outward.

Notice that a central dark fringe for which $\theta_m = 0$ in Eq. (9.15) can be represented by

$$2d = m_0\lambda_0. \tag{9.16}$$

(Keep in mind that this is a special case. The central region might correspond to neither a maximum nor a minimum.) Even if d is 10 cm, which is fairly modest in laser light, and $\lambda_0 = 500$ nm, m_0 will be quite large, namely 400,000. At a fixed value of d, successive dark rings will satisfy the expressions

$$\begin{aligned} 2d\cos\theta_1 &= (m_0 - 1)\lambda_0 \\ 2d\cos\theta_2 &= (m_0 - 2)\lambda_0 \\ &\vdots \\ 2d\cos\theta_p &= (m_0 - p)\lambda_0. \end{aligned} \tag{9.17}$$

The angular position of any ring, e.g. the pth ring, is gotten by combining Eqs. (9.16) and (9.17) to yield

$$2d(1 - \cos\theta_p) = p\lambda_0. \tag{9.18}$$

Since $\theta_m \equiv \theta_p$, both are just the half-angle subtended at the detector by the particular ring, and since $m = m_0 - p$, Eq. (9.18) is equivalent to Eq. (9.15). The new form is somewhat more convenient since (using the same example as above) with $d = 10$ cm, the sixth dark ring can be specified by stating that $p = 6$ or, in terms of the *order* of the pth ring, that $m = 399{,}994$. If θ_p is small

$$\cos\theta_p = 1 - \frac{\theta_p^2}{2}$$

and Eq. (9.18) yields

$$\theta_p = \left(\frac{p\lambda_0}{d}\right)^{\frac{1}{2}} \tag{9.19}$$

for the angular radius of the pth fringe.

The construction of Fig. 9.13 represents one possible configuration, the one where we consider only pairs of parallel emerging rays. Since these rays do not actually meet, they cannot form an image without a condensing lens of some sort. Indeed that lens is most often provided by the observer's eye focused at infinity. The resulting *fringes of equal inclination* ($\theta_m = $ const.) located at infinity, are also sometimes referred to as *Haidinger fringes* after the Austrian physicist Wilhelm Karl Haidinger (1795–1871). A comparison of Figs. 9.13(b) and 9.3(a), both showing two coherent point sources, rather suggests that in addition to these (virtual) fringes at infinity there might also be (real) fringes formed by converging rays. These fringes do in fact exist. Hence, if you illuminate the interferometer with a *broad source* and shield out all extraneous light, you can easily see the projected pattern on a screen in a darkened room (see Section 9.6). The fringes will appear in the space in front of the interferometer (i.e. where the detector is shown) and their size will increase with increasing distance from the beam splitter. We will consider the (real) fringes arising from point source illumination a little later on.

When the mirrors of the interferometer are inclined with respect to each other making a small angle, i.e. when M_1 and M_2 are not quite perpendicular, *Fizeau fringes* are observed. The resultant wedge-shaped air film between M_2 and M'_1 creates a pattern (which will be discussed in Section 9.5.2) of straight parallel fringes. The interfering rays appear to diverge from a point behind the mirrors. The eye would have to focus on this point in order to make these *localized fringes* observable. It can be shown analytically* that by appropriately adjusting the orientation of the mirrors M_1 and M_2 fringes can be produced which are straight, circular, elliptical, parabolic or hyperbolic—this holds as well for the real and virtual fringes.

It is apparent that the Michelson interferometer can be used to make extremely accurate length measurements. As the moveable mirror is displaced by $\lambda_0/2$, each fringe will move to the position previously occupied by an adjacent fringe. Using a microscope arrangement, one need only count the number of fringes N, or portions thereof, which have moved past a reference point in order to determine the distance traveled by the mirror Δd, i.e.

$$\Delta d = N\frac{\lambda_0}{2}$$

and, of course, nowadays this can be done electronically fairly easily. Michelson used the method to measure the number of wavelengths of the red cadmium line corresponding to the standard meter in Sèvres near Paris.†

The Michelson interferometer can be used along with a few polaroid filters to verify the Fresnel–Arago laws. A polarizer inserted in each arm will allow the optical path-length difference to remain fairly constant, while the vector field directions of the two beams are easily changed.

A *microwave Michelson interferometer* can be constructed with sheet-metal mirrors and a chicken-wire beam splitter. With the detector located at the central fringe, it can easily measure shifts from maxima to minima as one of the mirrors is moved, thereby determining λ. A few sheets of plywood, plastic or glass inserted in one arm will change the central fringe. Counting the number of fringe shifts yields a value for the index of refraction and from that we can compute the dielectric constant of the material.

The *Mach–Zehnder* interferometer is yet another amplitude-splitting device. As shown in Fig. 9.15, it consists

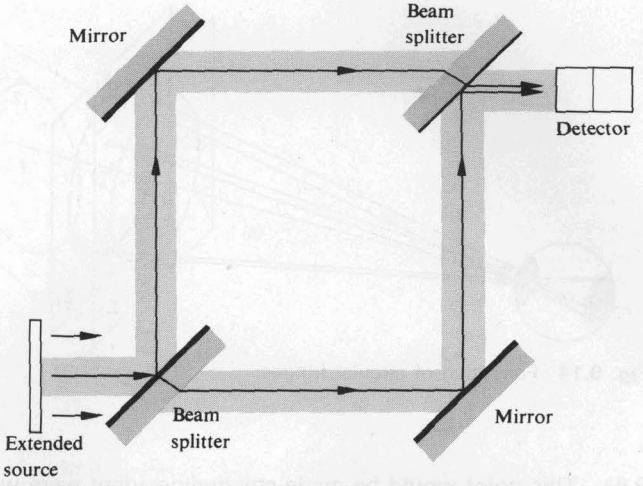


Fig. 9.15 The Mach–Zehnder interferometer.

of two beam splitters and two totally reflecting mirrors. The two waves within the apparatus travel along separate paths. A difference between the optical paths can be introduced by a slight tilt of one of the beam splitters. Since the two paths are separated, the interferometer is relatively difficult to align. For the very same reason, however, the interferometer finds myriad applications. It has even been used, in a rather altered yet conceptually similar form, to obtain electron interference fringes.*

Interposing an object in one beam will alter the optical path-length difference, thereby changing the fringe pattern. A common application of the device is to observe the density variations in gas-flow patterns within research chambers, e.g. wind tunnels, shock tubes, etc. One beam passes through the optically flat windows of the test chamber, while the other beam traverses appropriate compensator plates. The beam within the chamber will propagate through regions having a spatially varying index of refraction. The resulting distortions in the wavefront generate the fringe contours. A particularly nice application is shown in Fig. 9.16 which is a photograph of the magnetic compression device known as Scylla IV. It's used to study controlled thermonuclear reactions at the Los Alamos Scientific Laboratory. In this application the Mach–Zehnder interferometer has the form of a parallelogram as illustrated in Fig. 9.17. The two ruby laser *interferograms*, as these photos are called, show (Fig. 9.18) the background pattern without a plasma in the tube

* See e.g. Valasek, *Optics*, p. 135.

† A discussion of the procedure he used to avoid counting the 3,106,327 fringes directly can be found in Strong, *Concepts of Classical Optics*, p. 238 or Williams, *Applications of Interferometry*, p. 51.

* L. Marton, J. Arol Simpson, and J. A. Suddeth, *Rev. Sci. Instr.* **25**, 1099, (1954) and *Phys. Rev.* **90**, 490, (1953).

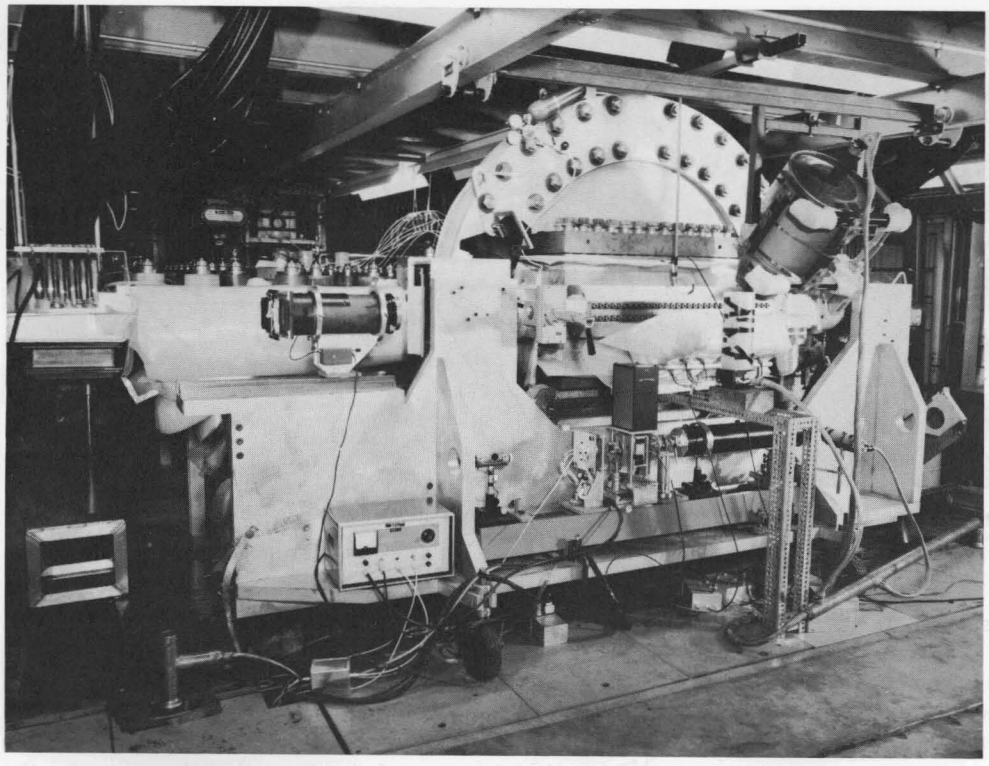


Fig. 9.16 Scylla IV.

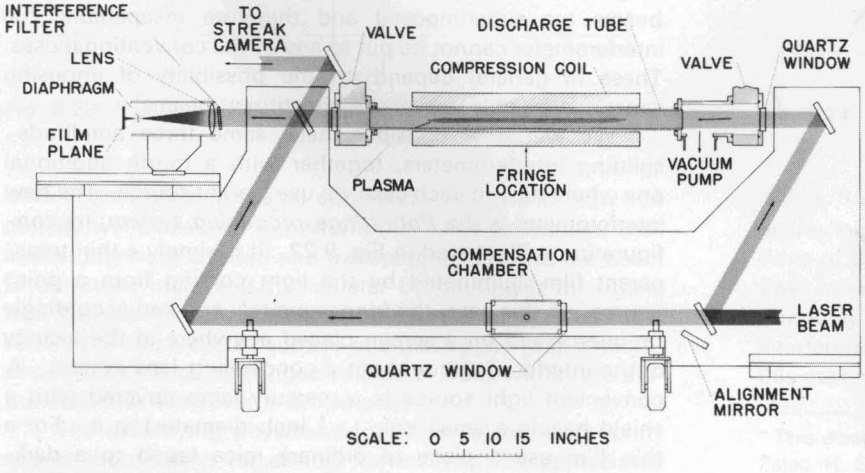


Fig. 9.17 Schematic of Scylla IV.

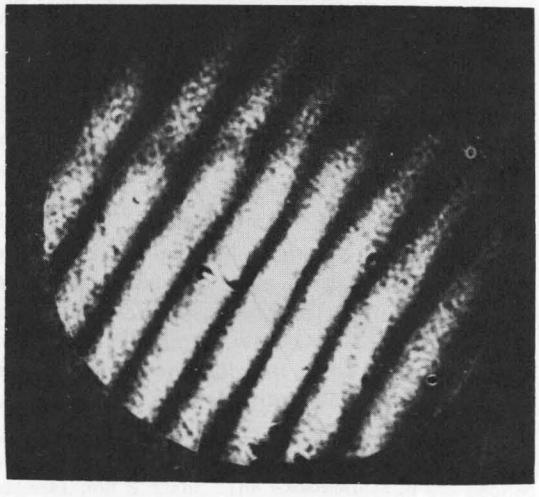


Fig. 9.18 Interferogram sans plasma.

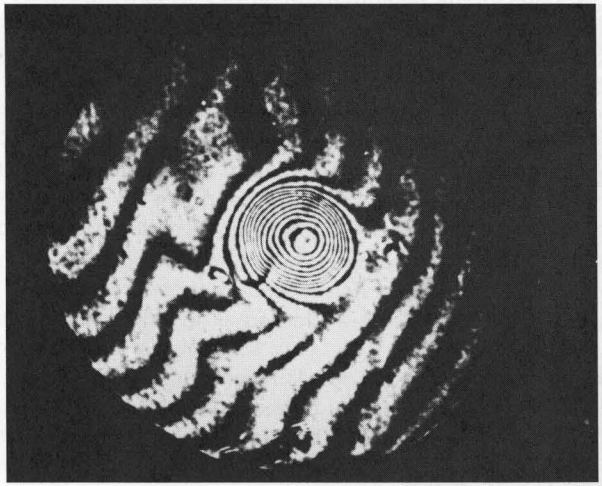


Fig. 9.19 Interferogram with plasma. [Photos courtesy Los Alamos Scientific Laboratory.]

and the density contours within the plasma during a reaction (Fig. 9.19).

Another amplitude-splitting device which contrasts with the previous instrument in many respects is one which we shall refer to as the *Sagnac interferometer*. It is very easy to

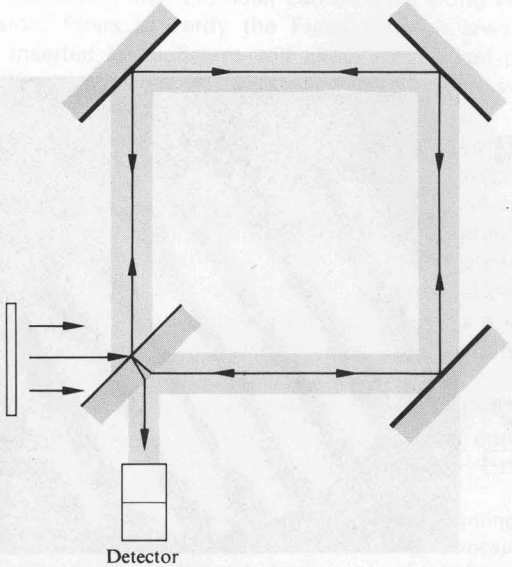


Fig. 9.20 A Sagnac interferometer.

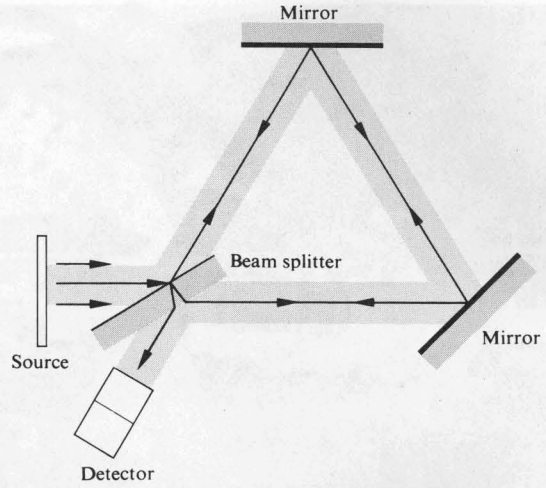


Fig. 9.21 Another variation of the Sagnac interferometer.

align, quite stable and yet rather difficult to make practical use of. One very interesting application is discussed in the last section of this chapter where we consider its use as a gyroscope. A form of the Sagnac interferometer is shown in Fig. 9.20 and another in Fig. 9.21; still others are possible. Notice that the main feature of the device is that there are two identical but oppositely directed paths taken by the beams and that both form closed loops before they are united to produce interference. A slight deliberate shift in the orientation of one of the mirrors will produce a path-length difference and a resulting fringe pattern. Since the beams are superimposed and therefore inseparable, the interferometer cannot be put to any of the conventional uses. These in general depend on the possibility of imposing variations on only one of the constituent beams.

We will now consider these same three amplitude-splitting interferometers, together with a fourth additional one where now in each case we use a point source. The new interferometer is the *Pohl fringe-producing system*; its configuration is illustrated in Fig. 9.22. It is simply a thin transparent film illuminated by the light coming from a point source. In this case, the fringes are real and can accordingly be intercepted on a screen placed anywhere in the vicinity of the interferometer without a condensing-lens system. A convenient light source is a mercury lamp covered with a shield having a small hole ($\approx \frac{1}{4}$ inch diameter) in it. For a thin film use a piece of ordinary mica taped to a dark-colored book cover which serves as an opaque backing. If you

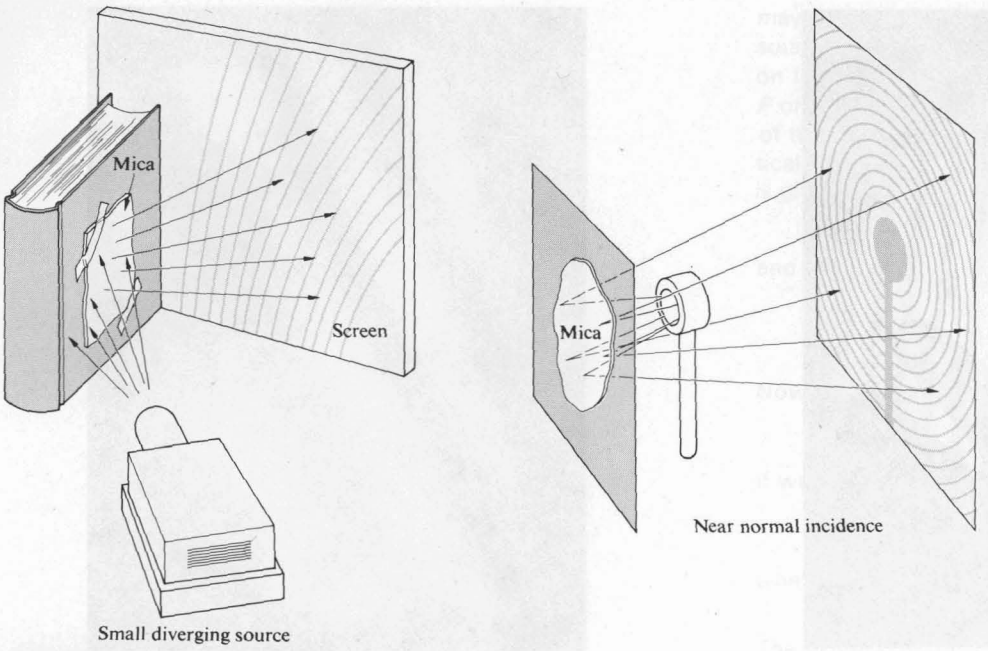


Fig. 9.22 The Pohl interferometer.

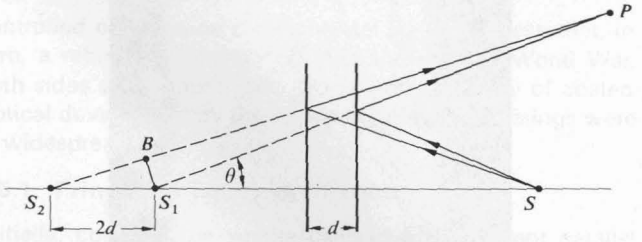


Fig. 9.23 Point-source illumination of parallel surfaces.

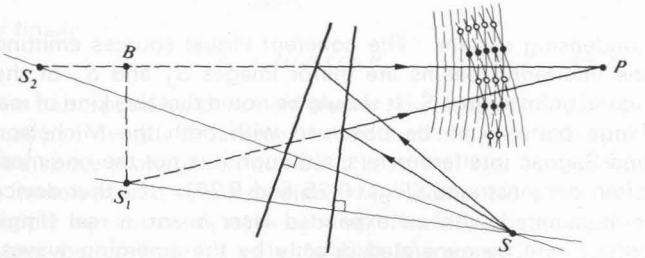


Fig. 9.24 Point-source illumination of inclined surfaces.

have a laser, its remarkable coherence length and high flux density will allow you to perform this same experiment with almost anything smooth and transparent. Expand the beam to about an inch or 2 in diameter by passing it through a lens (a focal length of 50 to 100 mm will do). Then just reflect the beam off the surface of a glass plate (e.g. a microscope slide) and the fringes will be evident within the illuminated disk wherever it strikes a screen.

The underlying physical principle involved with point-source illumination for all four devices can be appreciated with the help of a construction, variations of which are shown in Fig. 9.23 and 9.24.* The two vertical lines in Fig. 9.23, or the inclined ones in Fig. 9.24, represent either the positions of the mirrors or the two sides of the thin sheet in the Pohl interferometer. Let's assume that point P in the surrounding medium is a point at which there is constructive interference. A screen placed at that point would intercept this maximum as well as a whole fringe pattern, without any

* This discussion and part of Section 9.6 are treated at length in A. Zajac, H. Sadowski, and S. Licht, "The Real Fringes in the Sagnac and the Michelson Interferometers," *Am. J. Phys.* **29**, 669 (1961).

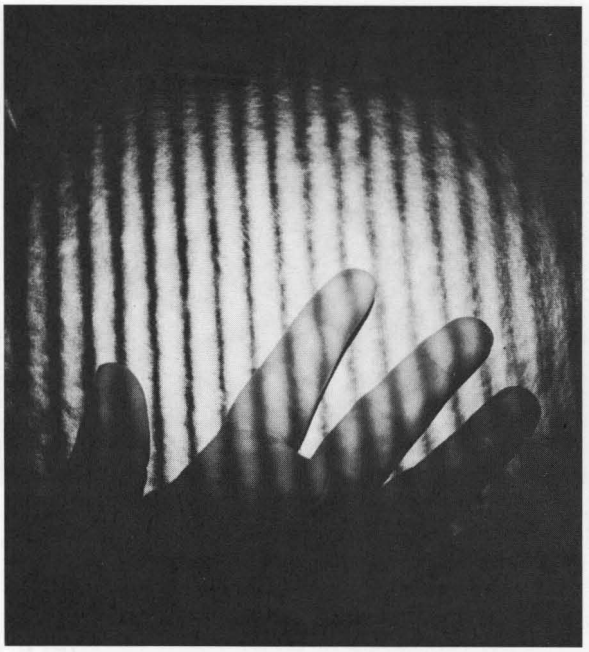


Fig. 9.25 Real Michelson fringes using He–Ne laser light. [Photo by E.H.]

condensing system. The coherent virtual sources emitting the interfering beams are mirror images S_1 and S_2 of the actual point source S. It should be noted that this kind of real fringe pattern can be observed with both the Michelson and Sagnac interferometers, although it is not the one most often demonstrated (Figs. 9.25 and 9.26). If either device is illuminated with an expanded laser beam, a real fringe pattern will be generated directly by the emerging waves. This is an extremely simple and beautiful demonstration.

9.5 DIELECTRIC FILMS—DOUBLE-BEAM INTERFERENCE

Interference effects are observable in transparent materials, the thicknesses of which vary over a very broad range. A range extending from films less than the length of a light wave (e.g. for green light λ_0 equals about $\frac{1}{150}$ the thickness of this printed page) to plates several centimeters thick. A layer of some material is referred to as a *thin film* for a given wavelength of electromagnetic radiation, when its thickness is of the order of that wavelength. Prior to the early 1940s the interference phenomena associated with thin dielectric films, although well known, had fairly limited practical appli-

Fig. 9.26 Real Sagnac fringes using a white-light point source.

cability. The rather spectacular color displays arising from oil slicks and soap films, however pleasing esthetically and theoretically, were practically mainly beautiful curiosities.

The advent of suitable vacuum deposition techniques in the 1930s brought with it the ability to produce precisely

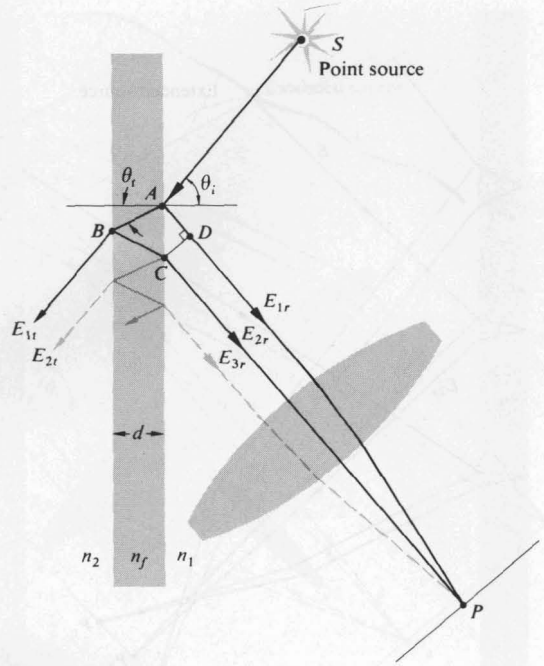


Fig. 9.27 Fringes of equal inclination.

controlled coatings on a commercial scale and with that, in turn, a rebirth of interest. During the Second World War, both sides were finding the enemy with a variety of coated optical devices and by the 1960s multilayered coatings were in widespread use.

9.5.1 Fringes of Equal Inclination

Initially, consider the simple case of a transparent parallel plate of dielectric material having a thickness d. One possible condition resulting from the reflection of two separate incident rays was depicted earlier in Fig. 9.23 in conjunction with our discussion of the Pohl interferometer. Another possibility is illustrated in Fig. 9.27. Suppose that the film is nonabsorbing and that the amplitude-reflection coefficients at the interfaces are so low that only the first two reflected beams E_{1r} and E_{2r} (both having undergone only one reflection) need be considered. In practice the amplitudes of the higher-order reflected beams (E_{3r} etc.) generally decrease very rapidly, as can be shown for the air–water and air–glass interfaces (Problem 9.10). For the moment, consider S to be a monochromatic point source. The film serves as an amplitude-splitting device, so that E_{1r} and E_{2r} may be considered as arising from two coherent virtual sources lying behind the film. The reflected rays are parallel on leaving the film and can be brought together at a point P on the focal plane of a telescope objective or on the retina of the eye when focused at infinity. From Fig. 9.27 the optical path-length difference for the first two reflected beams is given by

$$\Lambda = n_f[(\overline{AB}) + (\overline{BC})] - n_1(\overline{AD})$$

and since $(\overline{AB}) = (\overline{BC}) = d/\cos\theta_t$,

$$\Lambda = \frac{2n_f d}{\cos\theta_t} - n_1(\overline{AD}).$$

Now, to find an expression for $(\overline{AD})$, write

$$(\overline{AD}) = (\overline{AC})\sin\theta_i;$$

if we make use of Snell's law this becomes

$$(\overline{AD}) = (\overline{AC})\frac{n_f}{n_1}\sin\theta_t,$$

where

$$(\overline{AC}) = 2d\tan\theta_t. \tag{9.20}$$

The expression for Λ now becomes

$$\Lambda = \frac{2n_f d}{\cos\theta_t}(1 - \sin^2\theta_t)$$

or finally

$$\Lambda = 2n_f d\cos\theta_t. \tag{9.21}$$

The corresponding phase difference associated with the optical path-length difference is then just the product of the free-space propagation number and Λ, i.e., $k_0\Lambda$. If the film is immersed in a single medium, the index of refraction can simply be written as $n_1 = n_2 = n$. Realize, of course, that n may be less than n_f, as in the case of a soap film in air; or greater than n_f, as with an air film between two sheets of glass. In either case there will be an additional phase shift arising from the reflections themselves. Recall that, regardless of the polarization of the incoming light, the two beams, one internally and one externally reflected, will experience a *relative phase shift* of π radians. Accordingly

$$\delta = k_0\Lambda \pm \pi$$

and more explicitly

$$\delta = \frac{4\pi n_f}{\lambda_0} d\cos\theta_t \pm \pi \tag{9.22}$$

or

$$\delta = \frac{4\pi d}{\lambda_0}(n_f^2 - n^2\sin^2\theta_i)^{\frac{1}{2}} \pm \pi. \tag{9.23}$$

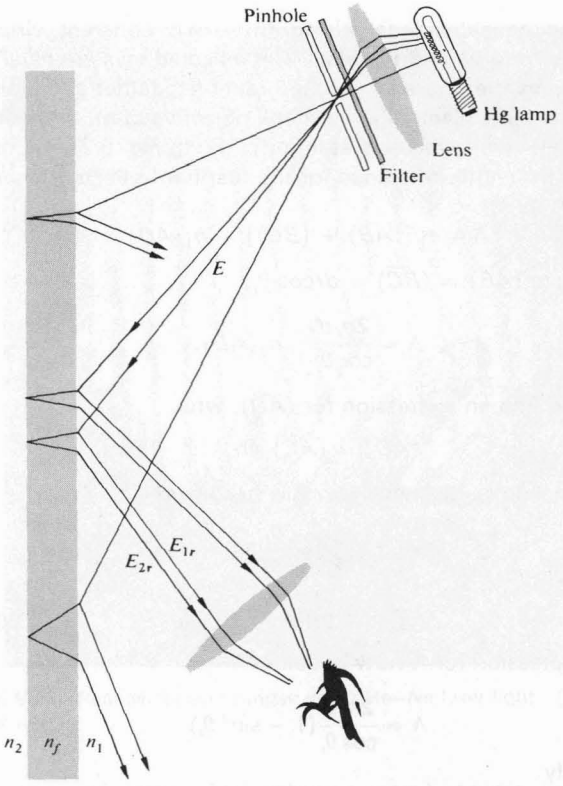


Fig. 9.28 Fringes seen on a small portion of the film.

The sign of the phase shift is immaterial so that we will choose the negative sign in order to make the equations a bit simpler in form. In reflected light an interference maximum, a bright spot, appears at P when $\delta = 2m\pi$, i.e. an even multiple of π. In that case Eq. (9.22) can be rearranged to yield

$$\text{(maxima)} \quad d\cos\theta_t = (2m+1)\frac{\lambda_f}{4}, \quad m = 0, 1, 2, \ldots, \tag{9.24}$$

where use was made of the fact that $\lambda_f = \lambda_0/n_f$. This also corresponds to minima in the transmitted light. Interference minima in reflected light (maxima in transmitted) result when $\delta = (2m \pm 1)\pi$, i.e. odd multiples of π. For such cases Eq. (9.22) yields

$$\text{(minima)} \qquad d\cos\theta_t = 2m\frac{\lambda_f}{4}. \tag{9.25}$$

The appearance of odd and even multiples of $\lambda_f/4$ in Eqs. (9.24) and (9.25) is rather significant, as will be seen later.

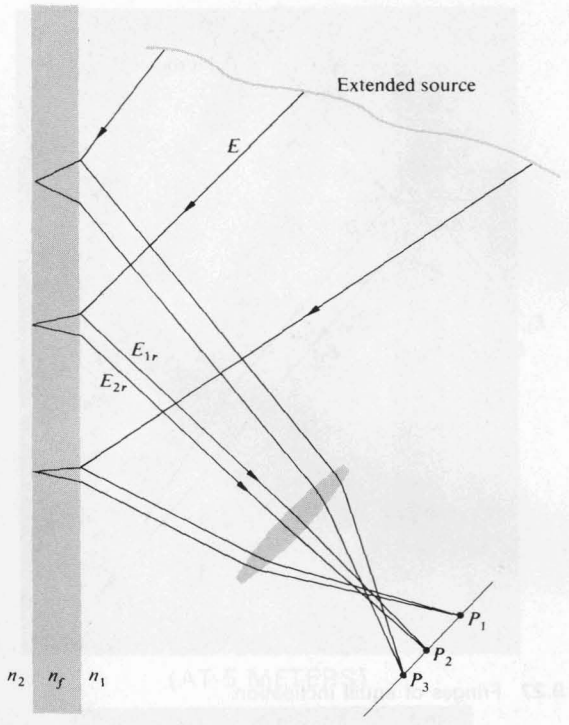


Fig. 9.29 Fringes seen on a large region of the film.

We could, of course, have a situation where $n_1 > n_f > n_2$ or where $n_1 < n_f < n_2$ as with a fluoride film deposited on an optical element of glass immersed in air. The π phase shift would then not be present and the above equations would simply be modified appropriately.

If the lens used to focus the rays has a small aperture, interference fringes will appear on a small portion of the film. Only the rays leaving the point source which are reflected directly into the lens will be seen (Fig. 9.28). For an extended source, light will reach the lens from various directions and the fringe pattern will spread out over a large area of the film (Fig. 9.29).

The angle θ_i or equivalently θ_t, determined by the position of P, will in turn control δ. The fringes appearing at points P_1 and P_2 in Fig. 9.30 are, accordingly, known as *fringes of equal inclination* (Problem 9.26 discusses some easy ways to see these fringes). Keep in mind that each source point on the extended source is incoherent with respect to the others.

Notice that as the film becomes thicker, the separation $(\overline{AC})$ between E_{1r} and E_{2r} also increases since

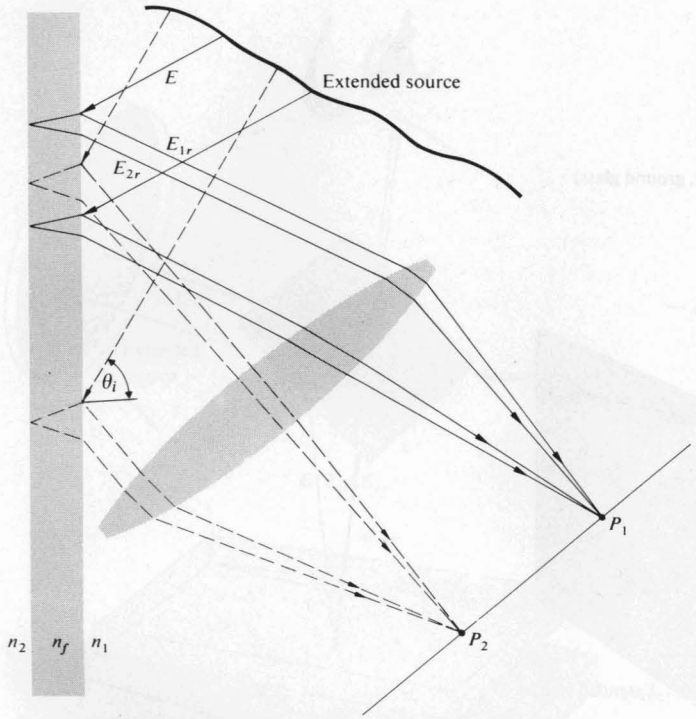


Fig. 9.30 All rays inclined at the same angle arrive at the same point.

$$(\overline{AC}) = 2d \tan \theta_t. \qquad [9.20]$$

When only one of the two rays is able to enter the pupil of the eye, the interference pattern will disappear. The larger lens of a telescope can then be used to gather in both rays, once again making the pattern visible. The separation can also be reduced by reducing θ_t and therefore θ_i, i.e. by viewing the film at nearly normal incidence. The equal-inclination fringes which are seen in this manner for thick plates are known as *Haidinger fringes*. With an extended source they consist of a series of concentric circular bands centered on the perpendicular drawn from the eye to the film (Fig. 9.31). As the observer moves, the interference pattern follows along.

9.5.2 Fringes of Equal Thickness

A whole class of interference fringes exists, for which the optical thickness, $n_f d$, is the dominant parameter rather than θ_i. These are referred to as *fringes of equal thickness*. Under white-light illumination the iridescence of soap bubbles, oil slicks (a few wavelengths thick), and even oxidized metal surfaces, are all the result of variations in film thickness. Interference bands of this kind are analogous to the constant-height contour lines of a topographical map. Each fringe is the locus of all points in the film for which the optical thickness is a constant. In general, n_f does not vary, so that the fringes actually correspond to regions of constant film thickness. As such, they can be quite useful in determining the surface features of optical elements; lenses, prisms, etc. For example, a surface to be examined may be put into contact with an *optical flat*.* The air in the space between the two generates a thin-film interference pattern. If the test surface is flat, a series of straight, equally spaced bands indicates a wedge-shaped air film, usually resulting from dust between the flats. Two pieces of plate glass separated at one end by a strip of paper will form a satisfactory wedge with which to observe these bands.

When viewed at nearly normal incidence in the manner illustrated in Fig. 9.32, the contours arising from a nonuniform film are called *Fizeau fringes*. For a thin wedge of small angle α the optical path length difference between two reflected rays may be approximated by Eq. (9.21), where d is the thickness at a particular point, i.e.

$$d = x\alpha.$$

For small values of θ_i the condition for an interference maximum becomes

$$(m + \tfrac{1}{2})\lambda_0 = 2n_f d_m$$

or

$$(m + \tfrac{1}{2})\lambda_0 = 2\alpha x_m n_f.$$

Since $n_f = \lambda_0/\lambda_f$, x_m may be written as

$$x_m = \left(\frac{m + 1/2}{2\alpha}\right)\lambda_f.$$

Maxima occur at distances from the apex given by $\lambda_f/4\alpha$, $3\lambda_f/4\alpha$, etc. and consecutive fringes are separated by a distance Δx, given by

$$\Delta x = \lambda_f/2\alpha.$$

Notice that the difference in film thickness between adjacent maxima is simply $\lambda_f/2$. Since the beam reflected from the

* A surface is said to be optically flat when it deviates by not more than about $\lambda/4$ from a perfect plane. In the past, the best flats were made of clear fused quartz. Now glass–ceramic materials (e.g. CER-VIT) having extremely small thermal coefficients of expansion (about one sixth that of quartz) are available. Individual flats of $\lambda/200$ or a bit better can be made.

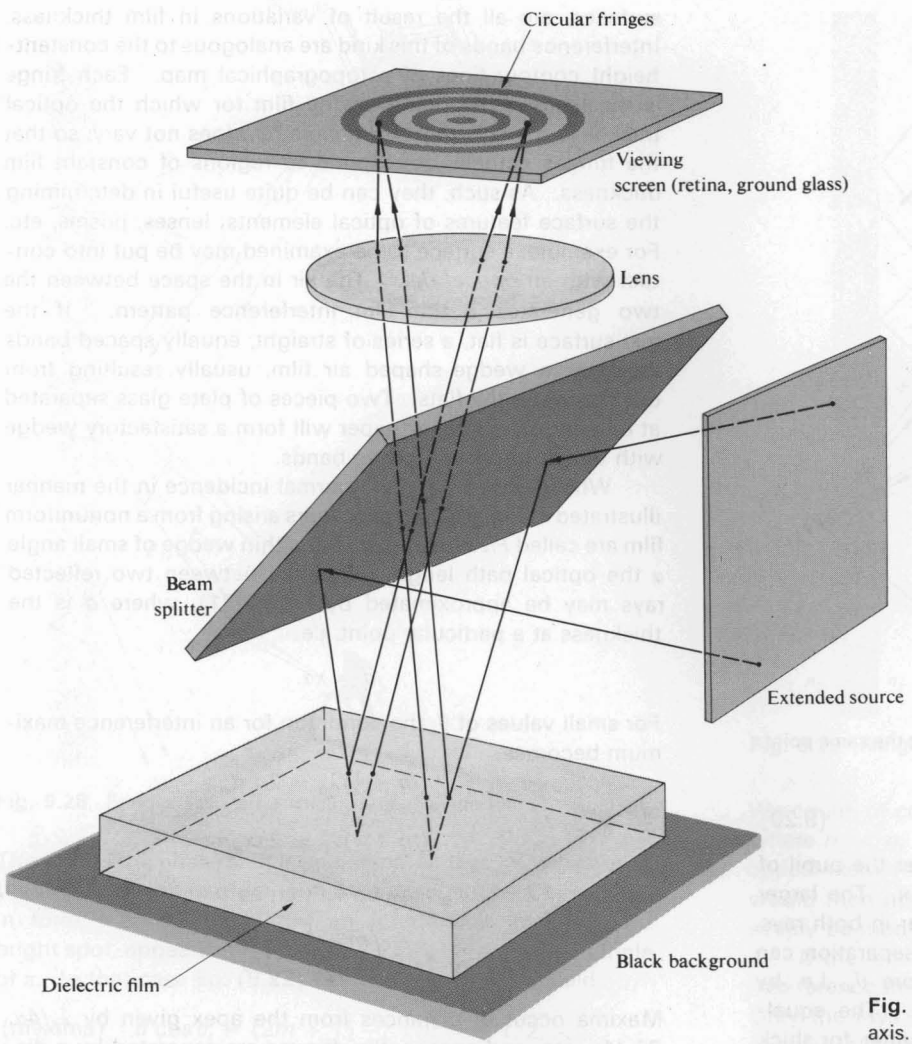


Fig. 9.31 Circular Haidinger fringes centered on the lens axis.

lower surface traverses the film twice ($\theta_i \approx \theta_t \approx 0$), adjacent maxima differ in optical path length by λ_f. Note too, that the film thickness at the various maxima is given by

$$d_m = (m + \tfrac{1}{2})\frac{\lambda_f}{2},$$

which is an odd multiple of a quarter wavelength. Traversing the film twice yields a phase shift of π which, when added to the shift of π resulting from reflection, puts the two rays back in phase.

Figure 9.33 is a photograph of a soap film held vertically so that it settles into a wedge shape under the influence of gravity. When illuminated with white light the bands are of various colors. The black region at the top is a portion where the film is less than $\lambda_f/4$ thick. Twice this, plus an additional shift of $\lambda_f/2$ due to the reflection is less than a whole wavelength. The reflected rays are therefore out of phase. As the thickness decreases still further, the total phase difference approaches π. The irradiance at the observer goes to a minimum (Eq. 9.5) and the film appears black in reflected light.*

* The relative phase shift of π between internal and external reflection is required if the reflected flux density is to go to zero smoothly, as the film gets thinner and finally disappears.

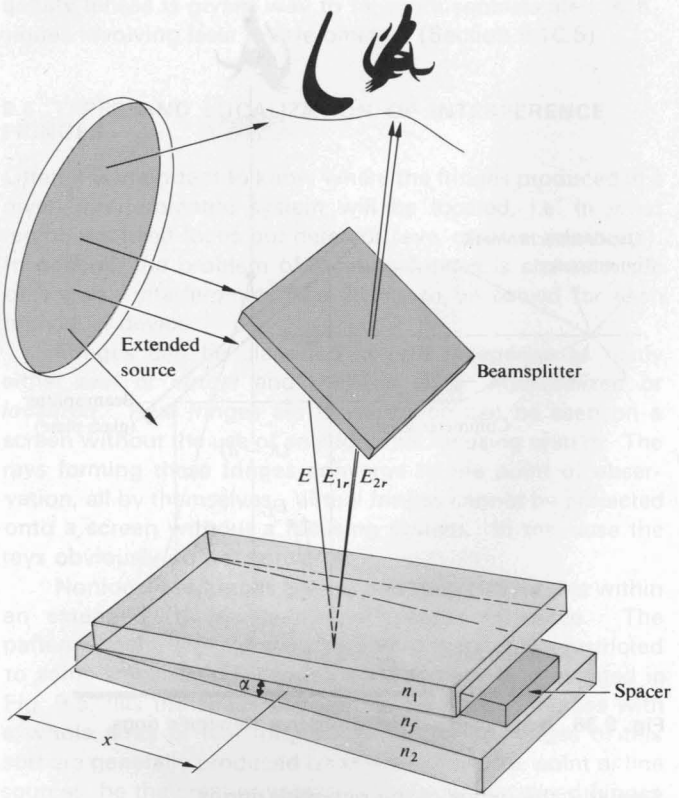


Fig. 9.32 Fringes from a wedge-shaped film.

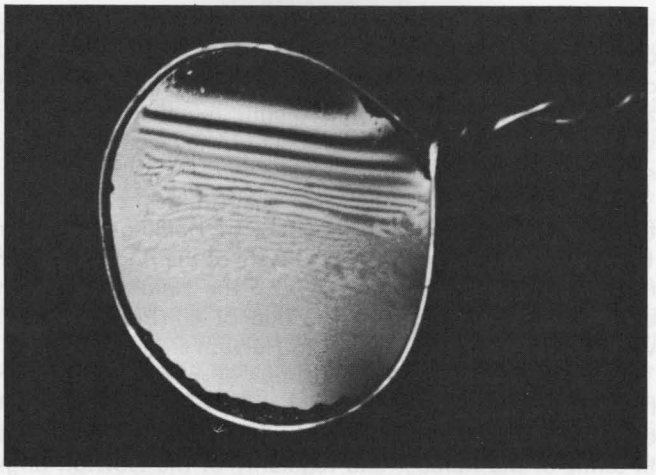


Fig. 9.33 A wedge-shaped film made of liquid dishwashing soap. [Photo by E.H.]

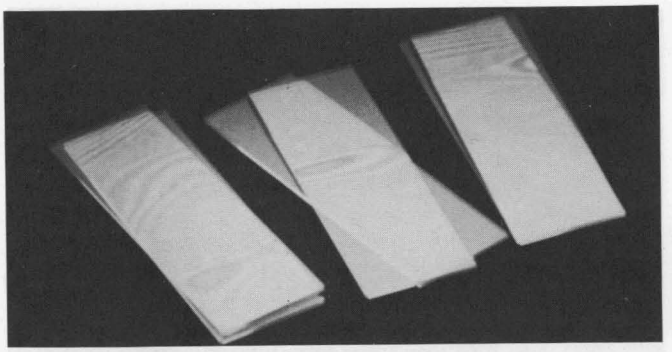


Fig. 9.34 Fringes in an air film between two microscope slides. [Photo by E.H.]

Press two well-cleaned microscope slides together. The enclosed air film will usually not be uniform. In ordinary room light a series of irregular, colored bands (fringes of equal thickness) will be clearly visible across the surface (Fig. 9.34). The thin glass slides distort under pressure and the fringes move and change accordingly. Indeed, if the two pieces of glass are forced together at a point as might be done by pressing on them with a sharp pencil a series of concentric, nearly circular, fringes are formed about that point (Fig. 9.35). Known as Newton's rings,* this pattern is more precisely examined using the arrangement of Fig. 9.36. Here a lens is placed on an optical flat and illuminated at normal incidence with quasimonochromatic light. The amount of uniformity in the concentric circular pattern is a measure of the degree of perfection in the shape of the lens. With R as the radius of curvature of the convex lens, the relation between the distance x and the film thickness d is given by

$$x^2 = R^2 - (R - d)^2,$$

or more simply by

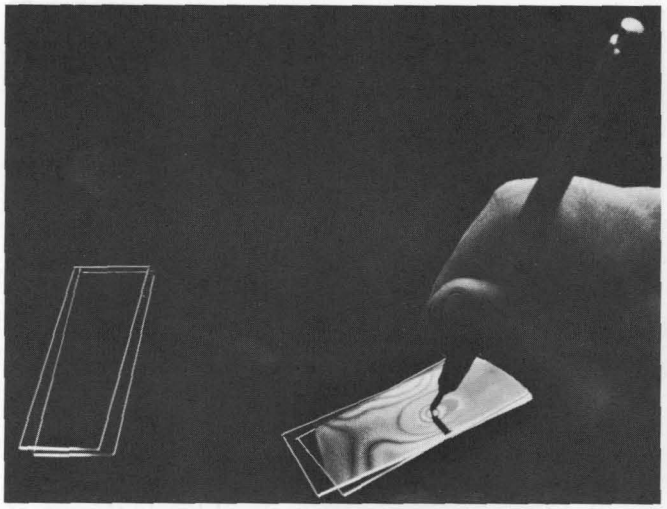
$$x^2 = 2Rd - d^2$$

Since $R \gg d$ this becomes

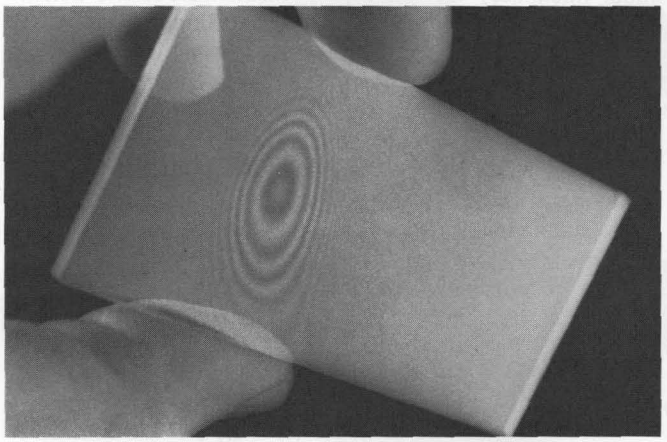
$$x^2 = 2Rd.$$

* Robert Hooke (1635–1703) and Isaac Newton both independently studied a whole range of thin-film phenomena from soap bubbles to the air film between lenses. Quoting from Newton's *Opticks*:

> I took two Object-glasses, the one a Planoconvex for a fourteen Foot Telescope, and the other a large double Convex for one of about fifty Foot; and upon this, laying the other with its plane side downwards, I pressed them slowly together to make the Colours successively emerge in the middle of the Circles.



(a)



(b)

Fig. 9.35 Newton's rings with two microscope slides. [Photos by E.H.]

We again approximate by assuming that we need only examine the first two reflected beams E_{1r} and E_{2r}. The mth-order interference maximum will occur in the thin film when its thickness is in accord with the relationship

$$2n_f d_m = (m + 1/2)\lambda_0.$$

The radius of the mth bright ring is therefore found by combining the last two expressions to yield

$$x_m = [(m + \tfrac{1}{2})\lambda_f R]^{1/2}. \qquad (9.26)$$

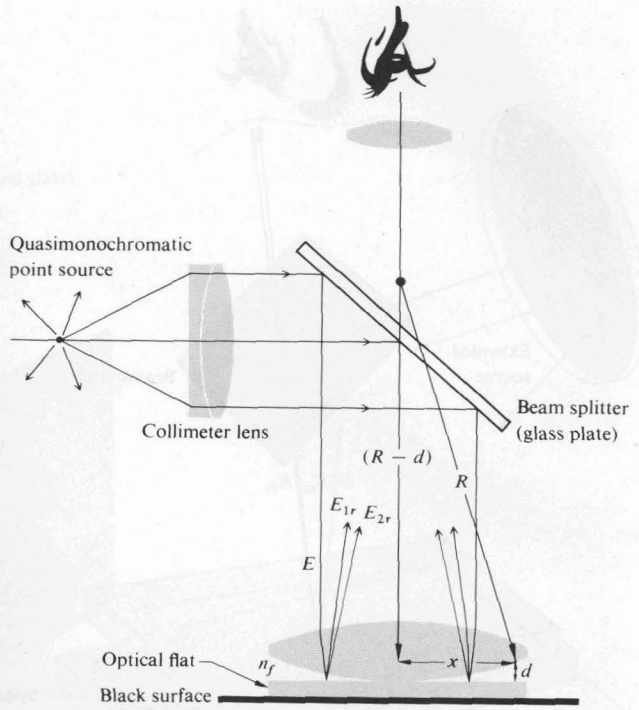


Fig. 9.36 A standard set-up to observe Newton's rings.

Similarly, the radius of the mth dark ring is

$$x_m = (m\lambda_f R)^{1/2}. \qquad (9.27)$$

If the two pieces of glass are in good contact (no dust), the central fringe at that point ($x_0 = 0$) will clearly be the zeroth-order minimum, an understandable result since d goes to zero at that point. In transmitted light, the observed pattern will be the complement of the reflected one discussed above, so that the center will now appear bright.

Newton's rings, which are Fizeau fringes, can be distinguished from the circular pattern of Haidinger's fringes by the manner in which the diameters of the rings vary with the order m. The central region in the Haidinger pattern corresponds to the maximum value of m (Problem 9.11) while just the opposite applies to Newton's rings.

An optical shop, in the business of making lenses, will have a set of precision spherical test plates or gauges. A designer may then specify the surface accuracy of a new lens in terms of the number and regularity of the Newton rings which will be seen with a particular test gauge. We should mention that the use of test plates in the manufacture of high-

quality lenses is giving way to far more sophisticated techniques involving laser interferometers (Section 9.10.5).

9.6 TYPES AND LOCALIZATION OF INTERFERENCE FRINGES

Often it is important to know where the fringes produced in a given interferometric system will be located, i.e. in what region we need focus our detector (eye, camera, telescope). In general, the problem of locating fringes is characteristic of a given interferometer, i.e. it has to be solved for each individual device.

Fringes can be classified in two categories as firstly either *real* or *virtual* and then as either *nonlocalized* or *localized*. Real fringes are those which can be seen on a screen without the use of an additional focusing system. The rays forming these fringes converge to the point of observation, all by themselves. Virtual fringes cannot be projected onto a screen without a focusing system. In this case the rays obviously do not converge.

Nonlocalized fringes are real and exist everywhere within an extended (three-dimensional) region of space. The pattern is quite literally nonlocalized in that it is not restricted to some small region. Young's experiment, as illustrated in Fig. 9.5, fills the space beyond the secondary sources with a whole array of real fringes. Nonlocalized fringes of this sort are generally produced by small sources, i.e. point or line sources, be they real or virtual. In contrast, localized fringes are clearly observable only over a particular surface. The pattern is quite literally localized, whether near a thin film or at infinity. This type of fringe will always result from the use of extended sources but can be generated with a point source as well.

The Pohl interferometer (Fig. 9.21) is particularly useful in illustrating these principles since with a point source it will produce both real nonlocalized and virtual localized fringes. The real nonlocalized fringes (Fig. 9.37 upper half) can be intercepted on a screen almost anywhere in front of the mica film.

For the nonconverging rays, realize that since the aperture of the eye is quite small it will intercept only those rays which are directed almost exactly at it. For this small pencil of rays, the eye, at a particular position, sees either a bright or dark spot but not much more. To perceive an extended fringe pattern formed by parallel rays of the type shown on the bottom in Fig. 9.37, a large lens will have to be used to gather in light entering at other orientations. In practice, however, the source is usually somewhat extended and fringes can generally be seen by looking into the film with the eye focused at infinity. These virtual fringes are localized at infinity and are equivalent to the *equal-inclination fringes* of Section 9.5. Similarly, if the mirrors M_1 and M_2 in the Michelson interferometer are parallel, the usual circular, virtual, equal-inclination fringes localized at infinity will be seen. We can imagine a thin air film between the surfaces of the mirrors M_2 and M_1' acting to generate these fringes. As with the configuration of Fig. 9.37 for the Pohl device, real nonlocalized fringes will also be present.

The geometry of the fringe pattern seen in reflected light from a transparent wedge of small angle α is shown in Fig. 9.38. The fringe location P will be determined by the direction of incidence of the incoming light. Newton's rings have this same kind of localization as do the Michelson, Sagnac and other interferometers for which the equivalent interference system consists of two reflecting planes inclined slightly to each other. Figures 9.25 and 9.26 are photographs of the real, equal-thickness fringes taken from the Sagnac and Michelson devices under point-source illumination. The wedge set-up of the Mach–Zehnder interferometer is distinctive in that the resulting virtual fringes can be localized on any plane within the region generally occupied by the test chamber by rotating the mirrors (Fig. 9.39).

9.7 MULTIPLE-BEAM INTERFERENCE

Thus far we have examined a number of situations in which two coherent beams have been combined under a diversity of conditions to produce interference patterns. There are, however, other circumstances under which a much larger number of mutually coherent waves are made to interfere. In fact, if the amplitude-reflection coefficients, the r's, for the parallel plate illustrated in Fig. 9.27 are not small, as was previously the case, the higher-order reflected waves $\mathbf{E}_{3r}$, $\mathbf{E}_{4r}$, . . . become quite significant. A glass plate, slightly silvered on both sides so that the r's approach unity, will generate a large number of multiply internally reflected rays. For the moment, we will only consider situations where the film, substrate, and surrounding medium are transparent dielectrics. This avoids the more complicated phase changes resulting from metal-coated surfaces.

To begin the analysis as simply as possible, let the film be nonabsorbing and let $n_1 = n_2$. The notation will be in accord with that of Section 4.5, i.e. the amplitude-transmission coefficients are represented by t, the fraction of the amplitude of a wave transmitted on entering into the film, and t', the fraction transmitted when a wave leaves the film.

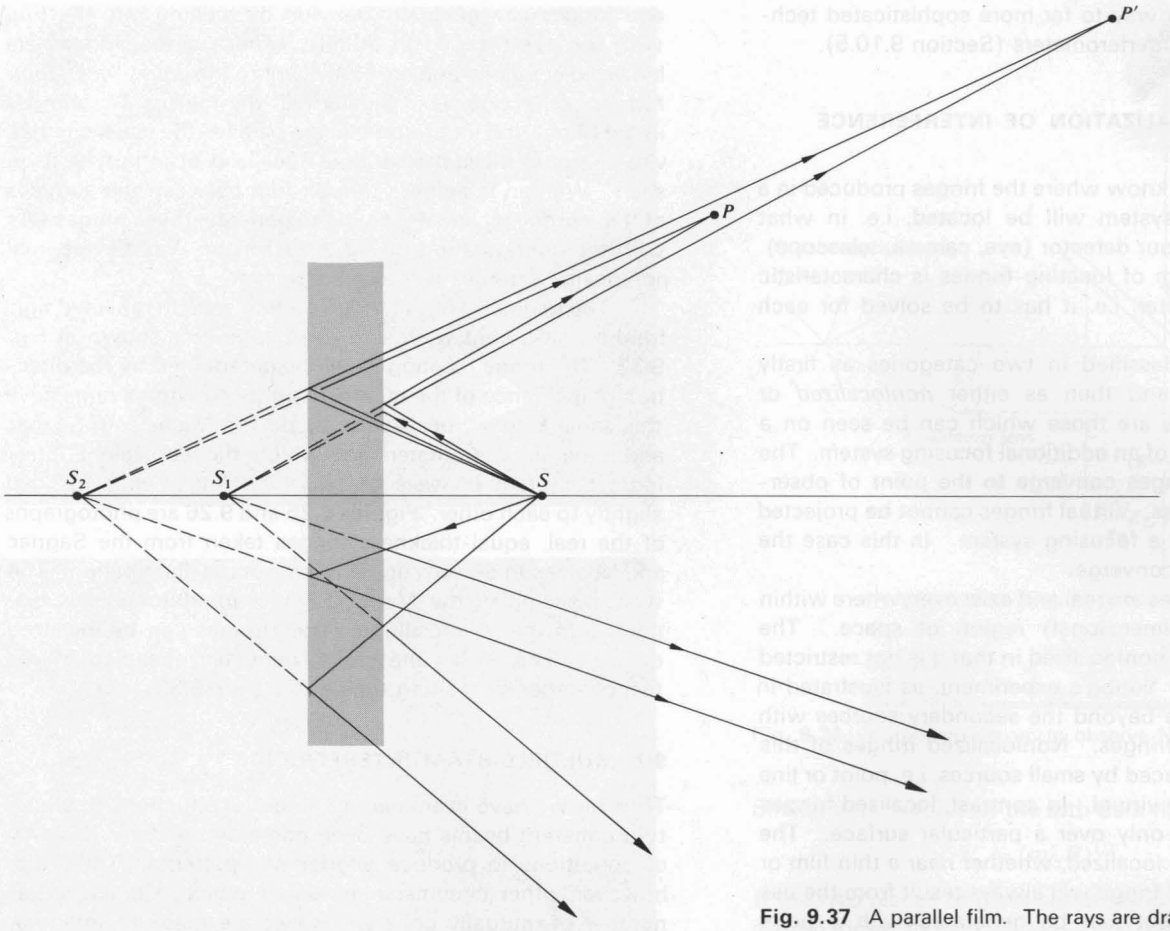


Fig. 9.37 A parallel film. The rays are drawn neglecting refraction.

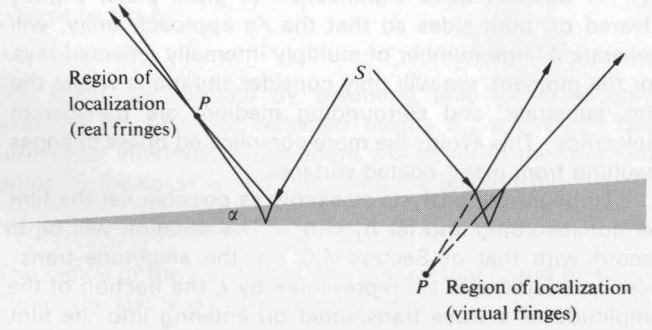


Fig. 9.38 Fringes formed by a wedge-shaped film.

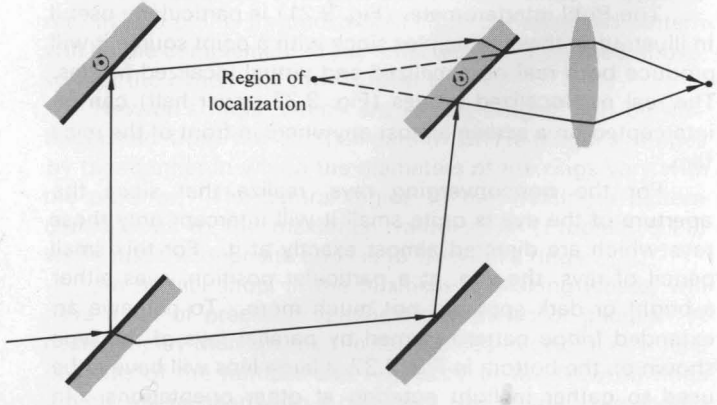


Fig. 9.39 Fringes in the Mach–Zehnder interferometer.

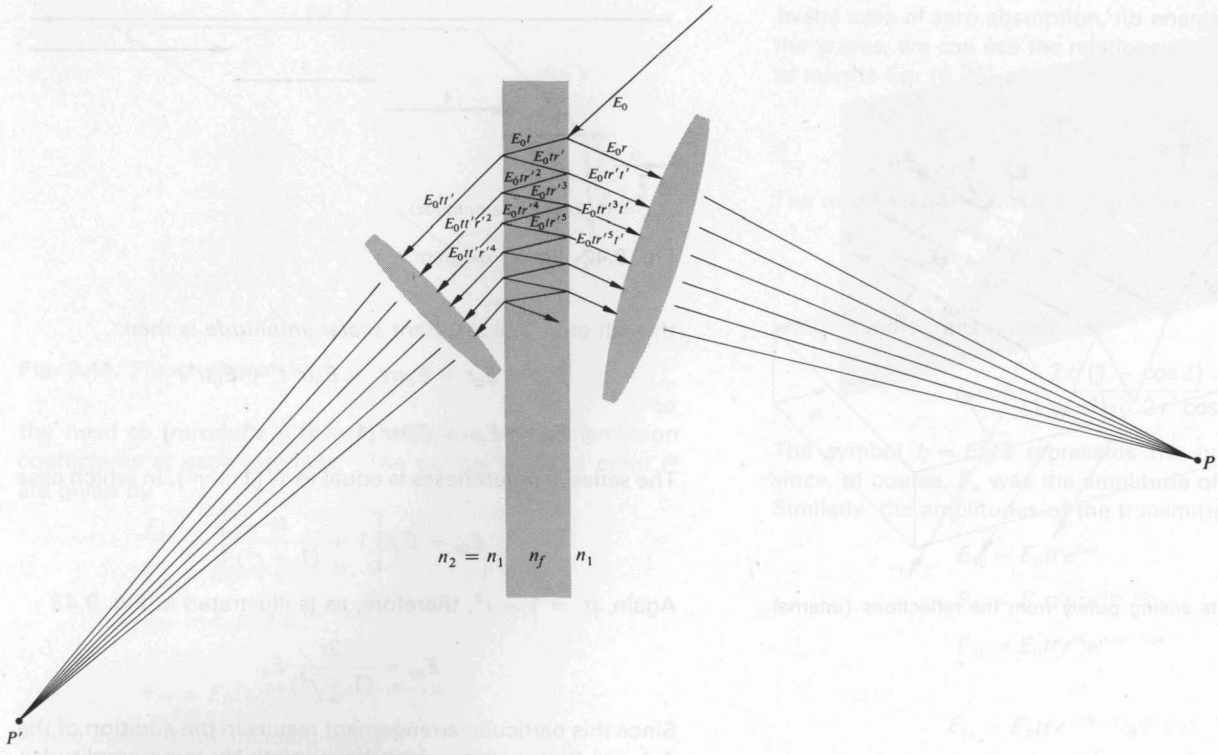


Fig. 9.40 Multiple-beam interference from a parallel film.

Keep in mind that the rays are actually lines drawn perpendicular to the wavefronts and therefore are also perpendicular to the optical fields $\mathbf{E}_{1r}$, $\mathbf{E}_{2r}$, etc. Since the rays will remain nearly parallel, the scalar theory will suffice as long as we are careful to account for any possible phase shifts. As shown in Fig. 9.40, the scalar amplitudes of the reflected waves $\mathbf{E}_{1r}$, $\mathbf{E}_{2r}$, $\mathbf{E}_{3r}$, . . . , are respectively E_0r, $E_0tr't'$, $E_0tr'^3t'$, . . . , where E_0 is the amplitude of the initial incoming wave and $r = -r'$ via Eq. (4.89). The minus sign indicates a phase shift which we will consider later. Similarly, the transmitted waves $\mathbf{E}_{1t}$, $\mathbf{E}_{2t}$, $\mathbf{E}_{3t}$, . . . will have amplitudes E_0tt', $E_0tr'^2t'$, $E_0tr'^4t'$, ... Consider the set of parallel reflected rays. Each ray bears a fixed phase relationship to all of the other reflected rays. The phase differences arise from a combination of optical path-length differences and phase shifts occurring at the various reflections. Nonetheless, the waves are mutually coherent and if they are collected and brought to focus at a point P by a lens, they will all interfere. The resultant irradiance expression has a particularly simple form for two special cases.

The difference in optical path length between adjacent rays is given by

$$\Lambda = 2n_f d \cos \theta_t. \qquad [9.21]$$

All of the waves except for the first, $\mathbf{E}_{1r}$, undergo an odd number of reflections *within* the film. It follows from Fig. 4.23 that at each internal reflection the component of the field parallel to the plane of incidence changes phase by either 0 or π depending on the internal incident angle, $\theta_i < \theta_c$. The component of the field perpendicular to the plane of incidence suffers no change in phase on internal reflection when $\theta_i < \theta_c$. Clearly then, no relative change in phase amongst these waves results from an odd number of such reflections (Fig. 9.41). As the *first special case*, if $\Lambda = m\lambda$, the 2nd, 3rd, 4th, etc. waves will all be in phase at P. The wave $\mathbf{E}_{1r}$, however, because of its reflection at the top surface of the film, will be out of phase by 180° with respect to all the other waves. The phase shift is embodied in the fact that $r = -r'$ and r' occurs only in odd powers. The sum of the scalar amplitudes, i.e. the total *reflected amplitude*

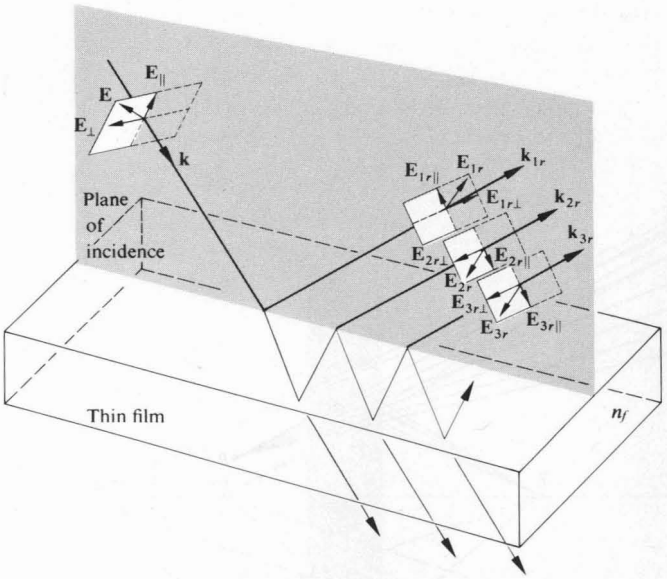


Fig. 9.41 Phase shifts arising purely from the reflections (internal $\theta_i < \theta_p'$).

at point P, is then

$$E_{0r} = E_0 r - (E_0 trt' + E_0 tr^3 t' + E_0 tr^5 t' + \cdots)$$

or

$$E_{0r} = E_0 r - E_0 trt'(1 + r^2 + r^4 + \cdots),$$

where since $\Lambda = m\lambda$, we've just replaced r' by $-r$. The geometric series in parentheses converges to the finite sum $1/(1 - r^2)$ as long as $r^2 < 1$ so that

$$E_{0r} = E_0 r - \frac{E_0 trt'}{(1 - r^2)}.$$

It was shown in Section 4.5 when considering Stokes' treatment of the principle of reversibility (Eq. 4.86) that $tt' = 1 - r^2$, and so it follows that

$$E_{0r} = 0.$$

Thus when $\Lambda = m\lambda$ the 2nd, 3rd, 4th, etc. waves exactly cancel the first reflected wave as is shown in Fig. 9.42. In this case no light is reflected; all of the incoming energy is transmitted. The *second special case* arises when $\Lambda = (m + \frac{1}{2})\lambda$. Now the first and second rays are in phase while all other adjacent waves are $\lambda/2$ out of phase, i.e. the 2nd is out of phase with the 3rd, the 3rd is out of phase with

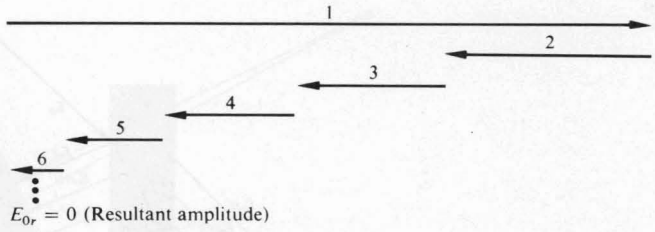


Fig. 9.42 Phasor diagram.

the 4th etc. The resultant *scalar amplitude* is then

$$E_{0r} = E_0 r + E_0 trt' - E_0 tr^3 t' + E_0 tr^5 t' - \cdots$$

or

$$E_{0r} = E_0 r + E_0 rtt'(1 - r^2 + r^4 - \cdots).$$

The series in parentheses is equal to $1/(1 + r^2)$, in which case

$$E_{0r} = E_0 r\left[1 + \frac{tt'}{(1 + r^2)}\right].$$

Again, $tt' = 1 - r^2$, therefore, as is illustrated in Fig. 9.43

$$E_{0r} = \frac{2r}{(1 + r^2)} E_0.$$

Since this particular arrangement results in the addition of the 1st and 2nd waves, which have relatively large amplitudes, it should yield a large reflected flux density. Since the irradiance is proportional to $E_{0r}^2/2$, via Eq. (3.52)

$$I_r = \frac{4r^2}{(1 + r^2)^2}\left(\frac{E_0^2}{2}\right) \tag{9.28}$$

That this is in fact the maximum, $(I_r)_{\max}$ will be shown later.

We will now consider the problem of multiple-beam interference in a more general fashion, making use of the complex representation. Again let $n_1 = n_2$, thereby avoiding

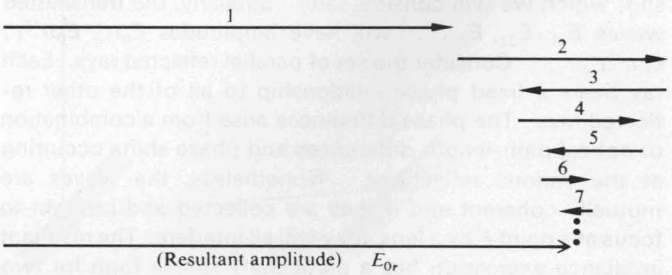


Fig. 9.43 Phasor diagram.

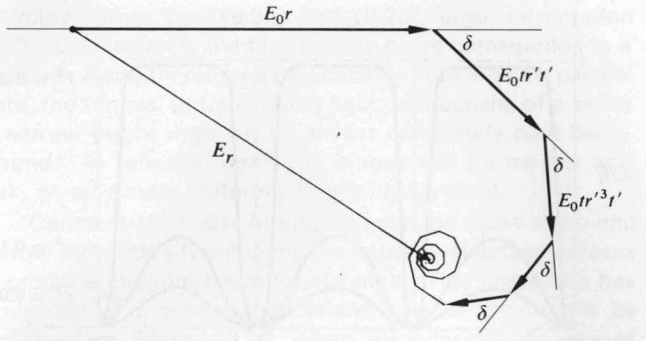


Fig. 9.44 Phasor diagram.

the need to introduce different reflection and transmission coefficients at each interface. The optical fields at point P are given by

$$\begin{aligned} E_{1r} &= E_0 r e^{i\omega t} \\ E_{2r} &= E_0 t r' t' e^{i(\omega t - \delta)} \\ E_{3r} &= E_0 t r'^3 t' e^{i(\omega t - 2\delta)} \\ &\vdots \\ E_{Nr} &= E_0 t r'^{(2N-3)} t' e^{i[\omega t - (N-1)\delta]}, \end{aligned}$$

where $E_0 e^{i\omega t}$ is the incident wave.

The terms $\delta, 2\delta, \ldots, (N-1)\delta$ are the contributions to the phase arising from an optical path-length difference between adjacent rays ($\delta = k_0\Lambda$). There is an additional phase contribution arising from the optical distance traversed in reaching point P, but this is common to each ray and has been omitted. The relative phase shift undergone by the 1st ray as a result of the reflection is embodied in the quantity r'. The resultant *reflected scalar wave* is then

$$E_r = E_{1r} + E_{2r} + E_{3r} + \cdots + E_{Nr},$$

or upon substitution (Fig. 9.44)

$$E_r = E_0 r e^{i\omega t} + E_0 t r' t' e^{i(\omega t - \delta)} + \cdots + E_0 t r'^{(2N-3)} t' e^{i[\omega t - (N-1)\delta]}.$$

This can be rewritten as

$$\begin{aligned} E_r = E_0 e^{i\omega t}\{r + r' t t' e^{-i\delta}[1 + (r'^2 e^{-i\delta}) \\ + (r'^2 e^{-i\delta})^2 + \cdots + (r'^2 e^{-i\delta})^{N-2}]\}. \end{aligned}$$

If $r'^2 e^{-i\delta} < 1$, and if the number of terms in the series approaches infinity, the series converges. The resultant wave becomes

$$E_r = E_0 e^{i\omega t}\left[r + \frac{r' t t' e^{-i\delta}}{1 - r'^2 e^{-i\delta}}\right]. \tag{9.29}$$

In the case of zero absorption, no energy being taken out of the waves, we can use the relations $r = -r'$ and $tt' = 1 - r^2$ to rewrite Eq. (9.29) as

$$E_r = E_0 e^{i\omega t}\left[\frac{r(1 - e^{-i\delta})}{1 - r^2 e^{-i\delta}}\right].$$

The reflected flux density at P is then $I_r = E_r E_r^*/2$, that is

$$I_r = \frac{E_0^2 r^2 (1 - e^{-i\delta})(1 - e^{+i\delta})}{2(1 - r^2 e^{-i\delta})(1 - r^2 e^{+i\delta})},$$

which can be transformed into

$$I_r = I_i \frac{2r^2(1 - \cos\delta)}{(1 + r^4) - 2r^2\cos\delta}. \tag{9.30}$$

The symbol $I_i = E_0^2/2$ represents the incident flux density since, of course, E_0 was the amplitude of the incident wave. Similarly, the amplitudes of the transmitted waves given by

$$\begin{aligned} E_{1t} &= E_0 t t' e^{i\omega t} \\ E_{2t} &= E_0 t t' r'^2 e^{i(\omega t - \delta)} \\ E_{3t} &= E_0 t t' r'^4 e^{i(\omega t - 2\delta)} \\ &\vdots \\ E_{Nt} &= E_0 t t' r'^{2(N-1)} e^{i[\omega t - (N-1)\delta]} \end{aligned}$$

add to yield

$$E_t = E_0 e^{i\omega t}\left[\frac{tt'}{1 - r^2 e^{-i\delta}}\right]. \tag{9.31}$$

Multiplying this by its complex conjugate we obtain (Problem 9.18) the irradiance of the transmitted beam

$$I_t = \frac{I_i (tt')^2}{(1 + r^4) - 2r^2\cos\delta}. \tag{9.32}$$

If we use trigonometric identity $\cos\delta = 1 - 2\sin^2(\delta/2)$, Eqs. (9.30) and (9.32) become

$$I_r = I_i \frac{[2r/(1 - r^2)]^2 \sin^2(\delta/2)}{1 + [2r/(1 - r^2)]^2 \sin^2(\delta/2)}. \tag{9.33}$$

and

$$I_t = I_i \frac{1}{1 + [2r/(1 - r^2)]^2 \sin^2(\delta/2)}, \tag{9.34}$$

where energy is not absorbed, i.e., $tt' + r^2 = 1$. If indeed none of the incident energy is absorbed, the flux density of the incoming wave should exactly equal the sum of the flux density reflected off the film and total transmitted flux

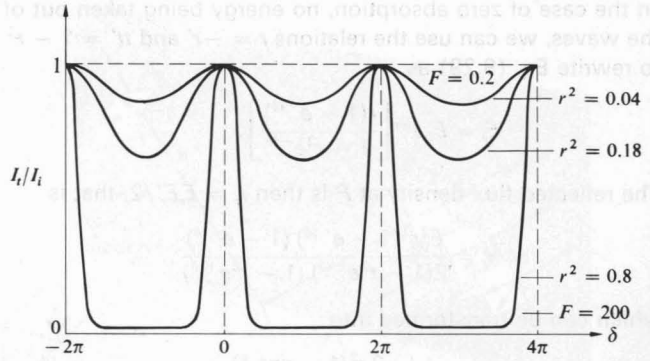


Fig. 9.45 Airy function.

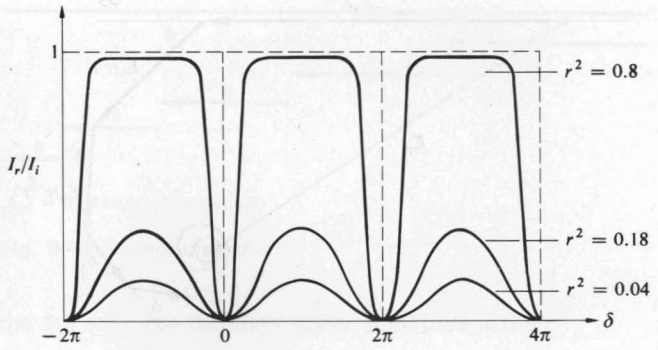


Fig. 9.46 One minus the Airy function.

density emerging from the film. It follows from Eqs. (9.33) and (9.34), that this is indeed the case, that is

$$I_i = I_r + I_t. \tag{9.35}$$

This will not be true, however, if the dielectric film is coated with a thin layer of semitransparent metal. Surface currents induced in the metal will dissipate a portion of the incident electromagnetic energy (see Section 4.3.5).

Consider the transmitted waves as described by Eq. (9.32). A maximum will exist when the denominator is as small as possible, i.e. when $\cos\delta = 1$, in which case $\delta = 2\pi m$ and

$$(I_t)_{\max} = I_i.$$

Under these conditions Eq. (9.30) indicates that

$$(I_r)_{\min} = 0$$

as was to be expected from Eq. (9.35). Again, from Eq. (9.32) it is clear that a minimum transmitted flux density will exist when the denominator is a maximum, i.e. when $\cos\delta = -1$. In that case $\delta = (2m + 1)\pi$ and

$$(I_t)_{\min} = I_i \frac{(1 - r^2)^2}{(1 + r^2)^2}. \tag{9.36}$$

The corresponding maximum in the reflected flux density is

$$(I_r)_{\max} = I_i \frac{4r^2}{(1 + r^2)^2}. \tag{9.37}$$

Notice that the constant-inclination fringe pattern has its maxima when $\delta = (2m + 1)\pi$ or

$$\frac{4\pi n_f}{\lambda_0} d\cos\theta_t = (2m + 1)\pi,$$

which is the same result arrived at previously, Eq. (9.24), by only using the first two reflected waves. Note too, that Eq. (9.37) verifies that Eq. (9.28) was indeed a maximum.

The form of Eqs. (9.33) and (9.34) suggests that we introduce a new quantity known as the *coefficient of finesse* F, such that

$$F \equiv \left(\frac{2r}{1 - r^2}\right)^2 \tag{9.38}$$

whereupon these equations can be written as

$$\frac{I_r}{I_i} = \frac{F\sin^2(\delta/2)}{1 + F\sin^2(\delta/2)} \tag{9.39}$$

and

$$\frac{I_t}{I_i} = \frac{1}{1 + F\sin^2(\delta/2)}. \tag{9.40}$$

The term $[1 + F\sin^2(\delta/2)]^{-1} \equiv \mathscr{A}(\theta)$ is known as the *Airy function*. It represents the transmitted flux density distribution, and is plotted in Fig. 9.45. The complementary function $[1 - \mathscr{A}(\theta)]$, i.e. Eq. (9.39), is plotted as well, in Fig. 9.46. When $\delta/2 = m\pi$ the Airy function is equal to unity for all values of F and therefore r. When r approaches one, the transmitted flux density is very small, except within the sharp spikes centered about the points $\delta/2 = m\pi$. Multiple-beam interference has resulted in a redistribution of the energy density in comparison to the sinusoidal two-beam pattern (of which the curves corresponding to a small reflectance are reminiscent). This effect will be further demonstrated when we consider the diffraction grating. At that time we will clearly see this same peaking effect, resulting from increasing the number of coherent sources contributing to the interference pattern. Remember that the Airy function is, in fact, a function of θ_t or θ_i by way of its dependence on δ,

as follows from Eqs. (9.22) and (9.23), ergo the notation $\mathscr{A}(\theta)$. Each spike in the flux-density curve corresponds to a particular δ and therefore a particular θ_i. For a plane-parallel plate, the fringes, in transmitted light, will consist of a series of narrow bright rings on an almost completely dark background. In reflected light, the fringes will be narrow and dark, on an almost uniformly bright background.

Constant-thickness fringes can also be made sharp and narrow by lightly silver coating the relevant reflecting surfaces to produce multiple-beam interference. This procedure has a number of practical applications, one of which will be discussed in Section 9.10, when we consider the use of multiple-beam Fizeau fringes to examine surface topography.

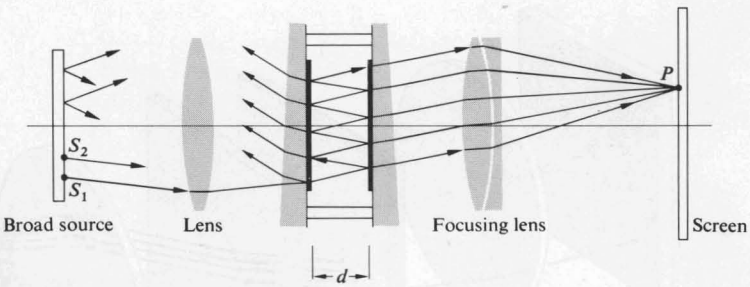


Fig. 9.47 Fabry–Perot etalon.

9.8 THE FABRY–PEROT INTERFEROMETER

9.8.1 Modus Operandi

The multiple-beam interferometer, first constructed by Charles Fabry and Alfred Perot in the late eighteen hundreds, is of considerable importance in modern optics. Its particular value arises from the fact that besides being a spectroscopic device of extremely high resolving power, it also serves as the basic laser resonant cavity. In principle, the device consists of two plane, parallel, highly reflecting surfaces separated by some distance d. This is the simplest configuration and as we shall see, other forms are also widely in use. In practice, two semisilvered or aluminized glass optical flats form the reflecting boundary surfaces. The enclosed air gap generally varies from several millimeters to several centimeters when the apparatus is used interferometrically, and often to considerably greater lengths when serving as a laser resonant cavity. If the gap can be mechanically varied by moving one of the mirrors, it's referred to as an interferometer. When the mirrors are held fixed and adjusted for parallelism by screwing down on some sort of spacer (invar or quartz are commonly used), it's said to be an *etalon* (although it is, of course, still an interferometer in the broad sense). Indeed, if the two surfaces of a single quartz plate are appropriately polished and silvered, it too will serve as an etalon; the gap need not be air. The unsilvered sides of the plates are often made to have a slight wedge shape (a few minutes of arc) to reduce the interference pattern arising from reflections off these sides. The etalon in Fig. 9.47 is shown illuminated by a broad source which might be a mercury arc or a Helium–Neon laser beam spread out in diameter to several centimeters. This can be done rather nicely by sending the beam into the back end of a telescope focused at infinity. The light can then be made diffuse by passing it through a sheet of ground glass. Only one ray emitted from some point S_1 on the source is traced through the etalon. Entering by way of the partially silvered plate, it is multiply reflected within the gap. The transmitted rays are collected by a lens and brought to a focus on a screen, where they interfere to form either a bright or dark spot. Consider this particular plane of incidence which contains all of the reflected rays. Any other ray emitted from a different point S_2, parallel to the original ray and in that plane of incidence, will form a spot at the same point P on the screen. As we shall see, the discussion of the previous section is again applicable, so that Eq. (9.32) determines the transmitted flux density I_t. The multiple waves generated in the cavity, arriving at P from either S_1 or S_2, are coherent amongst themselves. But the rays arising from S_1 are completely incoherent with respect to those from S_2, so that there is no sustained mutual interference. The contribution to the irradiance I_t at P is just the sum of the two irradiance contributions.

All of the rays incident on the gap at a given angle will result in a single circular fringe of uniform irradiance (Fig. 9.48). With a broad diffuse source, the interference bands will be narrow concentric rings, corresponding to the multiple-beam transmission pattern.

The fringe system can be observed visually, by looking directly into the etalon, while focusing at infinity. The job of the focusing lens, which is no longer needed, is done by the eye. At large values of d, the rings will be close together and a telescope might be needed to magnify the pattern. A relatively inexpensive monocular will serve the same purpose and will allow for photographing the fringes localized at infinity. As might be expected from the considerations of Section 9.6, it is possible to produce real nonlocalized fringes using a bright point source.

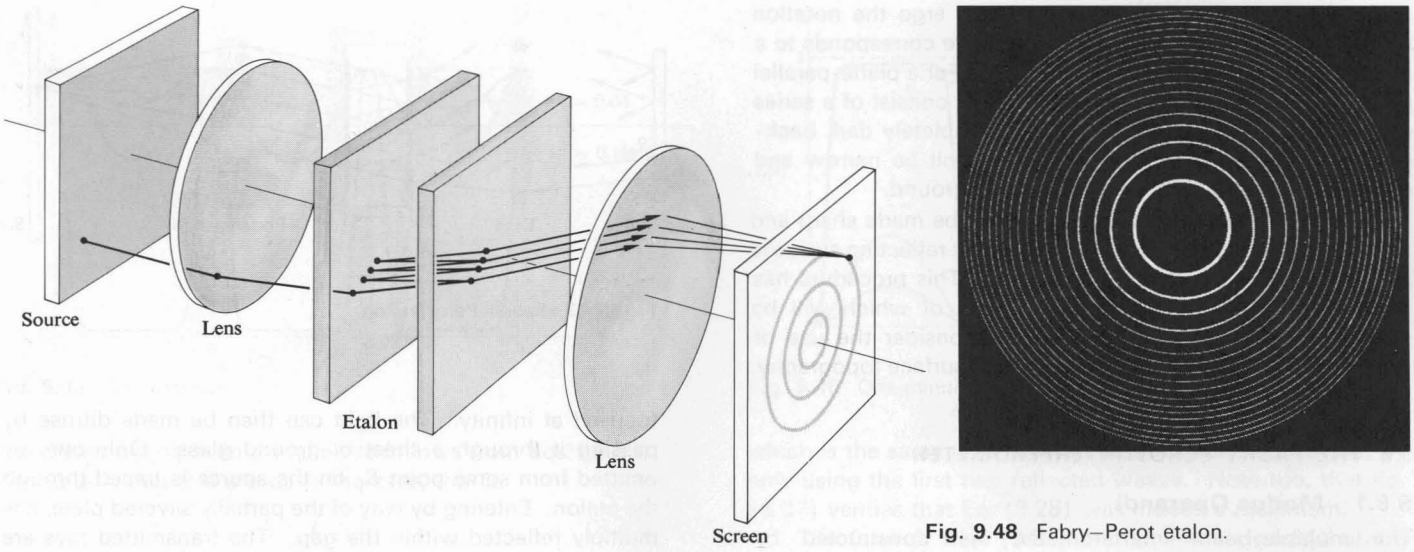


Fig. 9.48 Fabry–Perot etalon.

The partially transparent metal films which are often used to increase the reflectance ($R = r^2$), will absorb a fraction A of the flux density; this fraction is referred to as the absorptance.

The expression

$$tt' + r^2 = 1$$

or

$$T + R = 1, \qquad [4.60]$$

where T is the transmittance, must now be rewritten as

$$T + R + A = 1. \qquad (9.41)$$

One further complication is introduced by the metallic films and that is, an additional phase shift $\phi(\theta_i)$, which can differ from either zero or π. The phase difference between two successively transmitted waves is then

$$\delta = \frac{4\pi n_f}{\lambda_0} d \cos\theta_t + 2\phi. \qquad (9.42)$$

For the conditions under consideration, θ_i is small and ϕ may be considered to be constant. In general, d is so large, and λ_0 so small, that ϕ can be neglected. We can now express Eq. (9.32) as

$$\frac{I_t}{I_i} = \frac{T^2}{1 + R^2 - 2R\cos\delta},$$

or equivalently

$$\frac{I_t}{I_i} = \left(\frac{T}{1-R}\right)^2 \frac{1}{1 + [4R/(1-R)]^2 \sin^2(\delta/2)}. \qquad (9.43)$$

Making use of Eq. (9.41) and the definition of the Airy function, we obtain

$$\frac{I_t}{I_i} = \left[1 - \frac{A}{(1-R)}\right]^2 \mathscr{A}(\theta) \qquad (9.44)$$

as compared with the equation for zero absorption

$$\frac{I_t}{I_i} = \mathscr{A}(\theta). \qquad [9.40]$$

Inasmuch as the absorbed portion A is never zero, the transmitted flux-density maxima $(I_t)_{\max}$ will always be somewhat less than I_i. [Recall that for $(I_t)_{\max}$ $\mathscr{A}(\theta) = 1$.]

Accordingly, the *peak transmission* is defined as $(I_t/I_i)_{\max}$:

$$\frac{(I_t)_{\max}}{I_i} = \left[1 - \frac{A}{(1-R)}\right]^2. \qquad (9.45)$$

A silver film 50 nm thick would be approaching its maximum value of R, e.g. about 0.94, while T and A might be respectively 0.01 and 0.05. In this case, the peak transmission will be down to $\frac{1}{36}$. The relative irradiance of the fringe

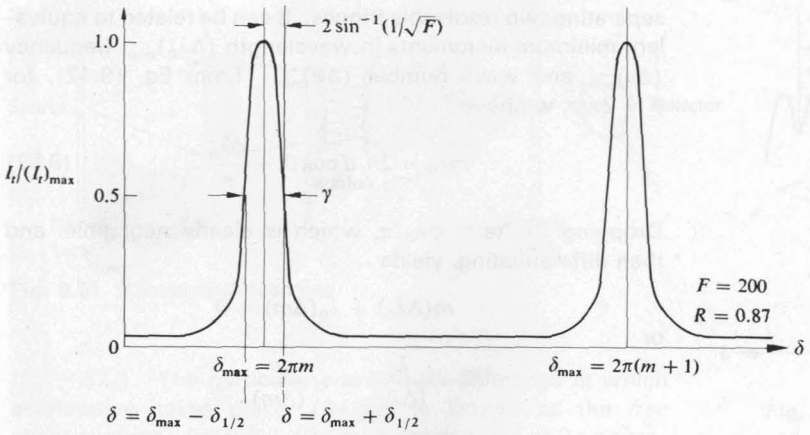


Fig. 9.49 Fabry–Perot fringes.

pattern will still be determined by the Airy function, since

$$\frac{I_t}{(I_t)_{max}} = \mathscr{A}(\theta). \tag{9.46}$$

A measure of the sharpness of the fringes, i.e. how rapidly the irradiance drops off on either side of the maximum, is given by the half-width γ. Shown in Fig. 9.49, γ is the width of the peak, in radians, when $I_t = (I_t)_{max}/2$.

Peaks in the transmission occur at specific values of the phase difference $\delta_{max} = 2\pi m$. Accordingly, the irradiance will drop to half its maximum value, i.e. $\mathscr{A}(\theta) = \frac{1}{2}$ whenever $\delta = \delta_{max} \pm \delta_{1/2}$. Inasmuch as

$$\mathscr{A}(\theta) = [1 + F\sin^2(\delta/2)]^{-1},$$

then when

$$[1 + F\sin^2(\delta_{1/2}/2)]^{-1} = \tfrac{1}{2}$$

it follows that

$$\delta_{1/2} = 2\sin^{-1}(1/\sqrt{F}).$$

Since F is generally rather large, $\sin^{-1}(1/\sqrt{F}) \approx 1/\sqrt{F}$ and therefore the half-width, $\gamma = 2\delta_{1/2}$, becomes

$$\gamma = 4/\sqrt{F}. \tag{9.47}$$

Recall that $F = 4R/(1-R)^2$, so that the larger R is, the sharper will be the transmission peaks.

Another quantity of particular interest is the ratio of the separation of adjacent maxima to the half-width. Known as the *finesse*, $\mathscr{F} \equiv 2\pi/\gamma$ or, from Eq. (9.47),

$$\mathscr{F} = \frac{\pi\sqrt{F}}{2}. \tag{9.48}$$

Over the visible spectrum, the finesse of most ordinary Fabry–Perot instruments is about 30. The physical limitation on $\mathscr{F}$ is set by deviations in the mirrors from plane parallelism. Keep in mind that as the finesse increases, the half-width decreases, but so too does the peak transmission. Incidently, finesse of about 1000 is attainable with curved-mirror systems using dielectric thin-film coatings.*

9.8.2 Fabry–Perot Spectroscopy

The Fabry–Perot interferometer is frequently used to examine the detailed structure of spectral lines. We will not attempt a complete treatment of interference spectroscopy, but rather will define the relevant terminology, briefly outlining appropriate derivations.†

As has been seen, a hypothetical purely monochromatic light wave generates a particular circular fringe system. But δ is a function of λ_0, so that if the source were made up of two such monochromatic components, two superimposed ring systems would result. When the individual fringes partially overlap, a certain amount of ambiguity exists in deciding when the two systems are individually discernible, i.e. when they are said to be *resolved*. Lord Rayleigh's‡

* The paper "Multiple Beam Interferometry" by H. D. Polster, *Appl. Opt.* **8**, 522 (1969) should be of interest.

† A more complete treatment can be found in Born and Wolf, *Principles of Optics*, or W. E. Williams, *Applications of Interferometry*, to name only two.

‡ The criterion will be reconsidered with respect to diffraction in the next chapter (see Fig. 10.40).

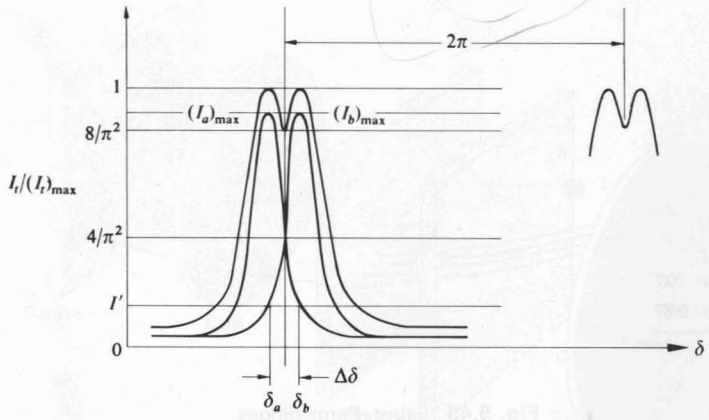


Fig. 9.50 Overlapping fringes.

criterion for resolving two equal irradiance overlapping slit images is well accepted, even if somewhat arbitrarily in the present application. Using it, however, will allow a comparison with prism or grating instruments. The essential feature of this criterion, is that the fringes are *just resolvable* when the combined irradiance of both fringes at the center, or saddle point, of the resultant broad fringe is $8/\pi^2$ times the maximum irradiance. This just means that one would see a broad bright fringe with a grey central region. To be a bit more analytic about it, examine Fig. 9.50, keeping in mind the previous derivation of the half-width. Consider the case when both constituent fringes have equal irradiances, $(I_a)_{max} = (I_b)_{max}$. The peaks in the resultant, occurring at $\delta = \delta_a$ and $\delta = \delta_b$, will have equal irradiances,

$$(I_t)_{max} = (I_a)_{max} + I'. \tag{9.49}$$

At the saddle point, the irradiance, $(8/\pi^2)\,(I_t)_{max}$ is the sum of the two constituent irradiances, so that, recalling Eq. (9.46),

$$(8/\pi^2)\,\frac{(I_t)_{max}}{(I_a)_{max}} = [\mathscr{A}(\theta)]_{\delta=\delta_a+\Delta\delta/2} + [\mathscr{A}(\theta)]_{\delta=\delta_b+\Delta\delta/2}. \tag{9.50}$$

Using $(I_t)_{max}$ given by Eq. (9.49), along with the fact that

$$\frac{I'}{(I_a)_{max}} = [\mathscr{A}(\theta)]_{\delta=\delta_a+\Delta\delta}$$

Eq. (9.50) can be solved for $\Delta\delta$. For large values of F,

$$(\Delta\delta) \approx \frac{4.2}{\sqrt{F}}. \tag{9.51}$$

This then represents the smallest phase increment, $(\Delta\delta)_{min}$, separating two resolvable fringes. It can be related to equivalent minimum increments in wavelength $(\Delta\lambda_0)_{min}$, frequency $(\Delta\nu)_{min}$, and wave number $(\Delta\varkappa)_{min}$. From Eq. (9.42), for $\delta = 2\pi m$, we have

$$m\lambda_0 = 2n_f d\cos\theta_t + \frac{\phi\lambda_0}{\pi}. \tag{9.52}$$

Dropping the term $\phi\lambda_0/\pi$, which is clearly negligible, and then differentiating, yields

$$m(\Delta\lambda_0) + \lambda_0(\Delta m) = 0$$

or

$$\frac{\lambda_0}{(\Delta\lambda_0)} = -\frac{m}{(\Delta m)}.$$

The minus will be omitted, since it only means that the order increases when λ_0 decreases. When δ changes by 2π, m changes by 1 and so

$$\frac{2\pi}{(\Delta\delta)} = \frac{1}{(\Delta m)}.$$

Now then

$$\frac{\lambda_0}{(\Delta\lambda_0)} = \frac{2\pi m}{(\Delta\delta)}. \tag{9.53}$$

The ratio of λ_0 to the least resolvable wavelength difference $(\Delta\lambda_0)_{min}$, is known as the *chromatic resolving power* $\mathscr{R}$, of any spectroscope. And so at nearly normal incidence

$$\mathscr{R} \equiv \frac{\lambda_0}{(\Delta\lambda_0)_{min}} \approx \mathscr{F}\frac{2n_f d}{\lambda_0} \tag{9.54}$$

or

$$\mathscr{R} \approx \mathscr{F}m.$$

For a wavelength of 500 nm, $n_f d = 10$ mm, and $R = 90\%$, the resolving power is well over a million, a range only recently achieved by the finest diffraction gratings. It follows as well, in this example, that $(\Delta\lambda_0)_{min}$ is less than a millionth of λ_0. In terms of frequency, the *minimum resolvable bandwidth* is

$$(\Delta\nu)_{min} = \frac{c}{\mathscr{F}2n_f d} \tag{9.55}$$

inasmuch as $|\Delta\nu| = |c\Delta\lambda_0/\lambda_0^2|$.

As the two components present in the source become increasingly different in wavelength, the peaks shown overlapping in Fig. 9.50 separate. As the wavelength difference increases, the mth-order fringe for one wavelength λ_0 will approach the $(m + 1)$th-order for the other wavelength

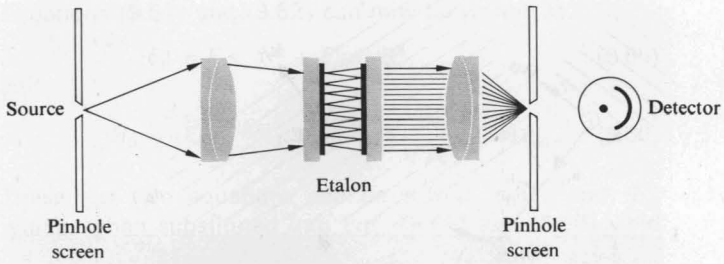


Fig. 9.51 Central spot scanning.

$(\lambda_0 - \Delta\lambda_0)$. The particular wavelength difference at which overlapping takes place, $(\Delta\lambda_0)_{\text{fsr}}$, is known as the *free spectral range*. From Eq. (9.53), a change in δ of 2π corresponds to $(\Delta\lambda_0)_{\text{fsr}} = \lambda_0/m$, or at near normal incidence,

$$(\Delta\lambda_0)_{\text{fsr}} \approx \lambda_0^2/2n_f d, \tag{9.56}$$

and similarly

$$(\Delta\nu)_{\text{fsr}} \approx c/2n_f d. \tag{9.57}$$

Continuing the above example, i.e. $\lambda_0 = 500$ nm and $n_f d = 10$ mm, $(\Delta\lambda_0)_{\text{fsr}} = 0.0125$ nm. Clearly, if we attempt to increase the resolving power by merely increasing d, the *free spectral range* will decrease, bringing with it the resulting confusion from the overlapping of orders. What is needed is that $(\Delta\lambda_0)_{\text{min}}$ *be as small as possible and* $(\Delta\lambda_0)_{\text{fsr}}$ *as large as possible*. But lo and behold

$$\frac{(\Delta\lambda)_{\text{fsr}}}{(\Delta\lambda)_{\text{min}}} = \mathscr{F}. \tag{9.58}$$

This result should not be too surprising in view of the original definition of $\mathscr{F}$.

Both the applications and configurations of the Fabry–Perot interferometer are numerous indeed. Etalons have been arranged in series with other etalons, as well as with grating and prism spectroscopes and multilayer dielectric films have been used to replace the metallic mirror coatings.

Scanning techniques are now widely in use. These take advantage of the superior linearity of photoelectric detectors over photographic plates, to obtain more reliable flux-density measurements. The basic set-up for *central-spot scanning* is illustrated in Fig. 9.51. Scanning is accomplished by varying δ, either by changing n_f or d rather than $\cos\theta_t$. In some arrangements, n_f is smoothly varied by altering the air pressure within the etalon. Alternatively, mechanically vibrating one mirror with a displacement of $\lambda/2$ will be enough to scan the free spectral range, corresponding as it does to $\Delta\delta = 2\pi$.

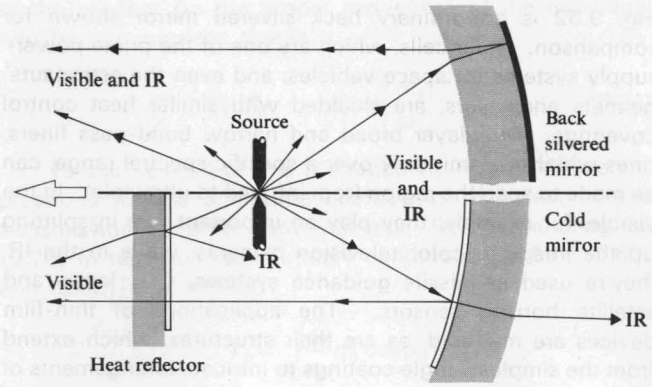


Fig. 9.52 A composite drawing showing an ordinary system on top and a coated one on the bottom.

A popular technique for accomplishing this utilizes a piezoelectric mirror mount. This kind of material will change its length, and therefore d, as a voltage is applied to it. The voltage profile determines the mirror motion.

Rather than photographically recording irradiance over a large region in space, at a single point in time, this method records irradiance over a large region in time, at a single point in space.

The actual configuration of the etalon itself has also undergone some significant variations. Pierre Connes in 1956 first described the *spherical-mirror Fabry–Perot interferometer*. Since then curved-mirror systems have become prominent as laser cavities and in addition are finding increasing use as spectrum analyzers.

9.9 APPLICATIONS OF SINGLE AND MULTILAYER FILMS

The optical uses to which coatings of thin dielectric films have been put in recent times are many indeed. Coatings to eliminate unwanted reflections off a diversity of surfaces, from showcase glass to quality camera lenses, are now commonplace. Multilayer, nonabsorbing beam splitters and *dichroic* mirrors (color selective beam splitters which transmit and reflect particular wavelengths) can be purchased commercially. Figure 9.52 is a segmented diagram illustrating the use of a *cold mirror* in combination with a *heat reflector* to channel infrared radiation to the rear of a motion-picture projector. The intense unwanted infrared (IR) radiation emitted by the source is removed from the beam to avoid heating problems at the photographic film. The top half of

Fig. 9.52 is an ordinary back silvered mirror shown for comparison. Solar cells, which are one of the prime power-supply systems for space vehicles, and even the astronauts' helmets and visors, are shielded with similar heat control coverings. Multilayer broad and narrow band-pass filters, ones which transmit only over a specific spectral range, can be made to span the region from infrared to ultraviolet. In the visible, for example, they play an important role in splitting up the image in color television cameras, while in the IR, they're used in missile guidance systems, CO_2 lasers and satellite horizon sensors. The applications of thin-film devices are manifold, as are their structures, which extend from the simplest single coatings to intricate arrangements of 100 or more layers.

The treatment of multilayer film theory used here will deal with the *total* electric and magnetic fields and their boundary conditions in the various regions. This is a far more practical approach for many-layered systems than is the multiple-wave technique used earlier.*

9.9.1 Mathematical Treatment

Consider the linearly polarized wave shown in Fig. 9.53, impinging on a thin dielectric film between two semi-infinite transparent media. In practice, this might correspond to a dielectric layer a fraction of a wavelength thick, deposited on the surface of a lens, a mirror, or a prism. One point must be made clear at the outset: each wave $E_{r\mathrm{I}}$, $E'_{r\mathrm{II}}$, $E_{t\mathrm{II}}$, etc. represents the resultant of all possible waves traveling in that direction, at that point in the medium. The summation process is, therefore, built in. As discussed in Section 4.3.2 the boundary conditions require that the tangential components of both the electric (**E**) and magnetic ($\mathbf{H} = \mathbf{B}/\mu$) fields be continuous across the boundaries (i.e. equal on both sides). At boundary I

$$E_{\mathrm{I}} = E_{i\mathrm{I}} + E_{r\mathrm{I}} = E_{t\mathrm{I}} + E'_{r\mathrm{II}} \tag{9.59}$$

and

$$H_{\mathrm{I}} = \sqrt{\frac{\epsilon_0}{\mu_0}}(E_{i\mathrm{I}} - E_{r\mathrm{I}})n_0 \cos\theta_{i\mathrm{I}} = \sqrt{\frac{\epsilon_0}{\mu_0}}(E_{t\mathrm{I}} - E'_{r\mathrm{II}})n_1 \cos\theta_{i\mathrm{II}}, \tag{9.60}$$

where use is made of the fact that **E** and **H** in nonmagnetic media are related by way of the index of refraction and the unit propagation vector:

$$\mathbf{H} = \sqrt{\frac{\epsilon_0}{\mu_0}} n\hat{\mathbf{k}} \times \mathbf{E}.$$

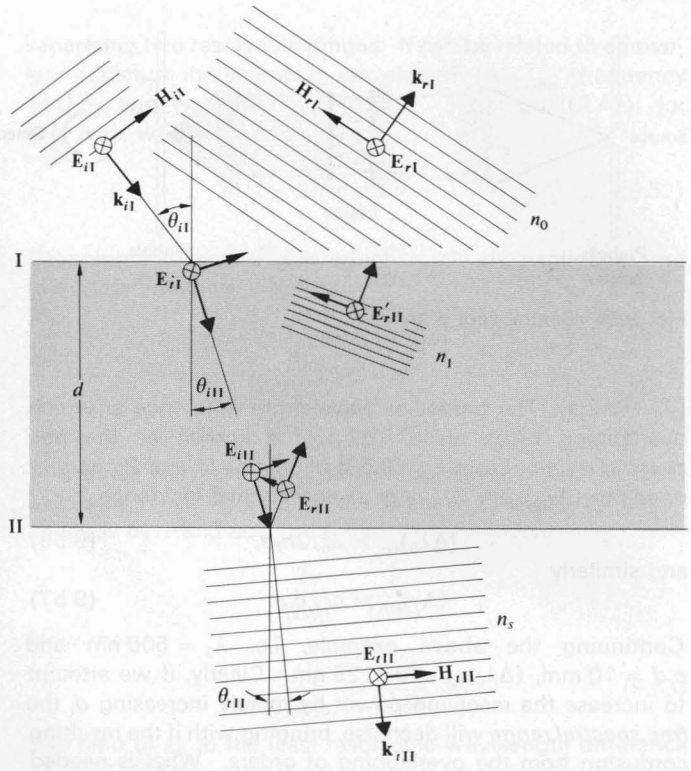


Fig. 9.53 Fields at the boundaries.

At boundary II

$$E_{\mathrm{II}} = E_{i\mathrm{II}} + E_{r\mathrm{II}} = E_{t\mathrm{II}} \tag{9.61}$$

and

$$H_{\mathrm{II}} = \sqrt{\frac{\epsilon_0}{\mu_0}}(E_{i\mathrm{II}} - E_{r\mathrm{II}})n_1 \cos\theta_{i\mathrm{II}} = \sqrt{\frac{\epsilon_0}{\mu_0}} E_{t\mathrm{II}} n_s \cos\theta_{t\mathrm{II}}, \tag{9.62}$$

the substrate having an index n_s. In accord with Eq. (9.21), a wave which traverses the film once undergoes a shift in phase of $k_0(2n_1 d \cos\theta_{i\mathrm{II}})/2$ which will be denoted by $k_0 h$ so that

$$E_{i\mathrm{II}} = E_{t\mathrm{I}} e^{-ik_0 h} \tag{9.63}$$

and

$$E_{r\mathrm{II}} = E'_{r\mathrm{II}} e^{+ik_0 h} \tag{9.64}$$

* For a very readable nonmathematical discussion, see P. Bowmeister and G. Pincus, "Optical Interference Coatings," *Sci. Amer.* **223**, 59 (December 1970).

Equations (9.61) and (9.62) can now be written as

$$E_{\text{II}} = E_{t\text{I}}e^{-ik_0h} + E'_{r\text{II}}e^{+ik_0h} \tag{9.65}$$

and

$$H_{\text{II}} = (E_{t\text{I}}e^{-ik_0h} - E'_{r\text{II}}e^{+ik_0h})\sqrt{\frac{\epsilon_0}{\mu_0}}n_1\cos\theta_{i\text{II}}. \tag{9.66}$$

These last two equations can be solved for $E_{t\text{I}}$ and $E'_{r\text{II}}$ which, when substituted into Eqs. (9.59) and (9.60) yield

$$E_{\text{I}} = E_{\text{II}}\cos k_0h + H_{\text{II}}(i\sin k_0h)/\Upsilon_1 \tag{9.67}$$

and

$$H_{\text{I}} = E_{\text{II}}\Upsilon_1 i\sin k_0h + H_{\text{II}}\cos k_0h, \tag{9.68}$$

where

$$\Upsilon_1 \equiv \sqrt{\frac{\epsilon_0}{\mu_0}}n_1\cos\theta_{i\text{III}}.$$

The above calculations carried out for the case when **E** is in the plane of incidence result in similar equations, provided that now

$$\Upsilon_1 \equiv \sqrt{\frac{\epsilon_0}{\mu_0}}n_1/\cos\theta_{i\text{III}}.$$

In matrix notation, the above linear relations take the form

$$\begin{bmatrix} E_{\text{I}} \\ H_{\text{I}} \end{bmatrix} = \begin{bmatrix} \cos k_0h & (i\sin k_0h)/\Upsilon_1 \\ \Upsilon_1 i\sin k_0h & \cos k_0h \end{bmatrix}\begin{bmatrix} E_{\text{II}} \\ H_{\text{II}} \end{bmatrix} \tag{9.69}$$

or

$$\begin{bmatrix} E_{\text{I}} \\ H_{\text{I}} \end{bmatrix} = \mathcal{M}_{\text{I}}\begin{bmatrix} E_{\text{II}} \\ H_{\text{II}} \end{bmatrix}. \tag{9.70}$$

The *characteristic matrix* $\mathcal{M}_{\text{I}}$ relates the fields at the two adjacent boundaries. It follows therefore, that if two overlaying films were deposited on the substrate, there would be three boundaries or interfaces, and now

$$\begin{bmatrix} E_{\text{II}} \\ H_{\text{II}} \end{bmatrix} = \mathcal{M}_{\text{II}}\begin{bmatrix} E_{\text{III}} \\ H_{\text{III}} \end{bmatrix}. \tag{9.71}$$

Multiplying both sides of this expression by $\mathcal{M}_{\text{I}}$, we obtain

$$\begin{bmatrix} E_{\text{I}} \\ H_{\text{I}} \end{bmatrix} = \mathcal{M}_{\text{I}}\mathcal{M}_{\text{II}}\begin{bmatrix} E_{\text{III}} \\ H_{\text{III}} \end{bmatrix}. \tag{9.72}$$

In general, if p is the number of layers, each with a particular value of n and h, then the first and the last boundaries are related by

$$\begin{bmatrix} E_{\text{I}} \\ H_{\text{I}} \end{bmatrix} = \mathcal{M}_{\text{I}}\mathcal{M}_{\text{II}}\cdots\mathcal{M}_p\begin{bmatrix} E_{(p+1)} \\ H_{(p+1)} \end{bmatrix}. \tag{9.73}$$

The characteristic matrix of the entire system is the resultant of the product (in the proper sequence) of the individual two-by-two matrices, that is

$$\mathcal{M} = \mathcal{M}_{\text{I}}\mathcal{M}_{\text{II}}\cdots\mathcal{M}_p = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}. \tag{9.74}$$

To see how all of this fits together, we will derive expressions for the amplitude coefficients of reflection and transmission using the above scheme. By reformulating Eq. (9.70) in terms of the boundary conditions [(9.59), (9.60) and (9.62)] and setting

$$\Upsilon_0 = \sqrt{\frac{\epsilon_0}{\mu_0}}n_0\cos\theta_{i\text{I}}$$

and

$$\Upsilon_s = \sqrt{\frac{\epsilon_0}{\mu_0}}n_s\cos\theta_{t\text{III}},$$

we obtain

$$\begin{bmatrix} (E_{i\text{I}} + E_{r\text{I}}) \\ (E_{i\text{I}} - E_{r\text{I}})\Upsilon_0 \end{bmatrix} = \mathcal{M}_{\text{I}}\begin{bmatrix} E_{t\text{III}} \\ E_{t\text{III}}\Upsilon_s \end{bmatrix}.$$

On expanding the matrices, the last relation becomes

$$1 + r = m_{11}t + m_{12}\Upsilon_s t$$

and

$$(1 - r)\Upsilon_0 = m_{21}t + m_{22}\Upsilon_s t$$

in as much as

$$r = E_{r\text{I}}/E_{i\text{I}} \qquad \text{and} \qquad t = E_{t\text{III}}/E_{i\text{I}}.$$

Consequently,

$$r = \frac{\Upsilon_0 m_{11} + \Upsilon_0\Upsilon_s m_{12} - m_{21} - \Upsilon_s m_{22}}{\Upsilon_0 m_{11} + \Upsilon_0\Upsilon_s m_{12} + m_{21} + \Upsilon_s m_{22}} \tag{9.75}$$

and

$$t = \frac{2\Upsilon_0}{\Upsilon_0 m_{11} + \Upsilon_0\Upsilon_s m_{12} + m_{21} + \Upsilon_s m_{22}}. \tag{9.76}$$

To find either r or t for any configuration of films, we need only compute the characteristic matrices for each film, multiply them, and then substitute the resulting matrix elements into the above equations.

9.9.2 Antireflection Coatings

Now consider the extremely important case of normal incidence, that is

$$\theta_{i\text{I}} = \theta_{i\text{III}} = \theta_{t\text{III}} = 0,$$

which in addition to being the simplest, is quite frequently approximated in practical situations. If we put a subscript

on r to indicate the number of layers present, the reflection coefficient for a single film becomes

$$r_1 = \frac{n_1(n_0 - n_s)\cos k_0 h + i(n_0 n_s - n_1^2)\sin k_0 h}{n_1(n_0 + n_s)\cos k_0 h + i(n_0 n_s + n_1^2)\sin k_0 h}. \quad (9.77)$$

Multiplying r_1 by its complex conjugate leads to the reflectance

$$R_1 = \frac{n_1^2(n_0 - n_s)^2\cos^2 k_0 h + (n_0 n_s - n_1^2)^2 \sin^2 k_0 h}{n_1^2(n_0 + n_s)^2\cos^2 k_0 h + (n_0 n_s + n_1^2)^2 \sin^2 k_0 h}. \quad (9.78)$$

This formula becomes particularly simple when $k_0 h = \frac{1}{2}\pi$, which is equivalent to saying that the optical thickness h of the film is an odd multiple of $\frac{1}{4}\lambda_0$. In this case $d = \frac{1}{4}\lambda_f$, and

$$R_1 = \frac{(n_0 n_s - n_1^2)^2}{(n_0 n_s + n_1^2)^2} \quad (9.79)$$

which, quite remarkably, will equal zero when

$$n_1^2 = n_0 n_s. \quad (9.80)$$

Generally, d is chosen so that h equals $\frac{1}{4}\lambda_0$ in the yellow-green portion of the visible spectrum, where the eye is most sensitive. Cryolite ($n = 1.35$), a sodium aluminum fluoride compound, and magnesium fluoride ($n = 1.38$) are common low-index films. Since MgF_2 is by far the more durable, it is most frequently used. On a glass substrate, ($n_s \approx 1.5$), both these films would have indices which are still somewhat too large to satisfy Eq. (9.80). Nonetheless, a single $\frac{1}{4}\lambda_0$ layer of MgF_2, will reduce the reflectance of glass from about 4% to a bit more than 1%, over the visible spectrum. It is now common practice to apply antireflection coatings to the elements of optical instruments. On camera lenses, such coatings produce a decrease in the haziness caused by stray internally scattered light, as well as a marked increase in image brightness. At wavelengths on either side of the central yellow-green region, R increases and the lens surface will appear blue-red in reflected light.

For a double layer, quarter-wavelength antireflection coating

$$\mathscr{M} = \mathscr{M}_{\mathrm{I}}\mathscr{M}_{\mathrm{II}}$$

or more specifically

$$\mathscr{M} = \begin{bmatrix} 0 & i/\Upsilon_1 \\ i\Upsilon_1 & 0 \end{bmatrix}\begin{bmatrix} 0 & i/\Upsilon_2 \\ i\Upsilon_2 & 0 \end{bmatrix}. \quad (9.81)$$

At normal incidence this becomes

$$\mathscr{M} = \begin{bmatrix} -n_2/n_1 & 0 \\ 0 & -n_1/n_2 \end{bmatrix}. \quad (9.82)$$

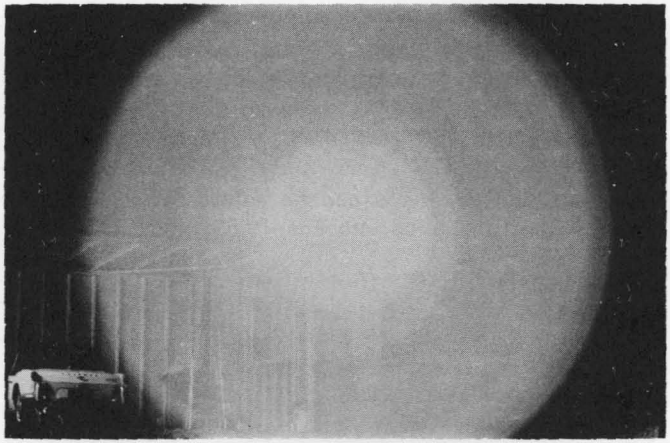


Fig. 9.54 Lens elements coated with a single layer of MgF_2.

Substituting the appropriate matrix elements into Eq. (9.15) yields r_2 which, when squared, leads to the reflectance

$$R_2 = \left[\frac{n_2^2 n_0 - n_s n_1^2}{n_2^2 n_0 + n_s n_1^2}\right]^2. \quad (9.83)$$

In order that R_2 be exactly zero at a particular wavelength, we need

$$\left(\frac{n_2}{n_1}\right)^2 = \frac{n_s}{n_0}. \quad (9.84)$$

This kind of film is referred to as a *double-quarter, single-minimum* coating. When n_1 and n_2 are as small as possible, the reflectance will have its single broadest minimum equal to zero at the chosen frequency. It should be clear from Eq. (9.84) that $n_2 > n_1$; accordingly, it is now a common practice to designate a (glass)–(high index)–(low index)–(air) system as *gHLa*. Zirconium dioxide ($n = 2.1$), titanium dioxide ($n = 2.40$), and zinc sulfide ($n = 2.32$) are commonly used for H-layers while magnesium fluoride ($n = 1.38$) and cerium fluoride ($n = 1.63$) often serve as L-layers.

Other double- and triple-layer schemes can be designed to satisfy specific requirements on spectral response, incident angle, cost, etc. Fig. 9.54 is a scene photographed through a fifteen-element zoom lens, with a 150 W lamp pointing directly into the camera. The lens elements were covered with a single layer of MgF_2. For Fig. 9.55 a triple-layer antireflection coating was used. The improved contrast and glare reduction is apparent.

Fig. 9.55 Lens elements coated with a multi-layer film structure. [Photos courtesy Optical Coating Laboratory, Inc., Santa Rosa, California.]

9.9.3 Multilayer Periodic Systems

The simplest kind of periodic system is the *quarter-wave stack* which, not unreasonably, is made up of a number of quarter-wave layers. The periodic structure of alternately high- and low-index materials, illustrated in Fig. 9.56, is designated by

$$g(HL)^3a.$$

Figure 9.57 illustrates the general form of a portion of the spectral reflectance for a few multilayer filters. The width of the high-reflectance central zone increases with increasing values of the index ratio n_H/n_L while its height increases with the number of layers. Note that the maximum reflectance of a periodic structure such as $g(HL)^m a$ can be increased further by adding another H-layer such that it now has the form $g(HL)^m Ha$. Very high reflectance mirror surfaces can be produced using this arrangement.

The small peak on the short-wavelength side of the central zone can be decreased by adding an eighth-wave low-index film to both ends of the stack, in which case the whole arrangement will be denoted by

$$g(0.5L)\,(HL)^m H(0.5L)a.$$

This has the effect of increasing the short-wavelength high-frequency transmittance and is therefore known as a *high-pass filter*. Similarly, the structure

$$g(0.5H)L(HL)^m(0.5H)a$$

merely corresponds to the case when the end H-layers are

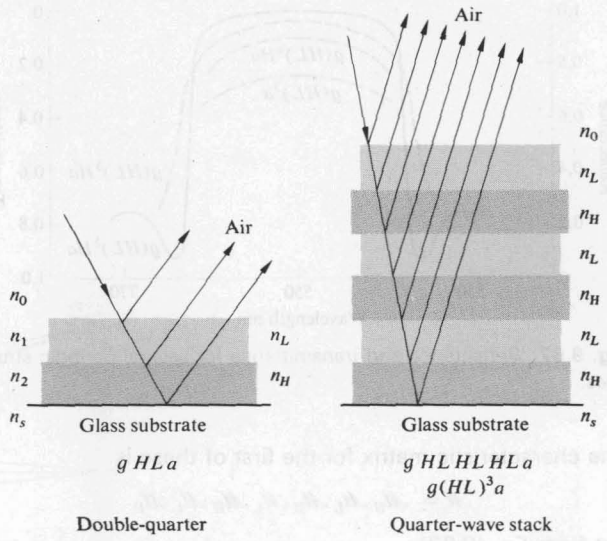


Fig. 9.56 A periodic structure.

$\lambda_0/8$ thick. It has a higher transmittance at the long-wavelength, low-frequency range and serves as a *low-pass filter*.

At nonnormal incidence, up to about 30°, there is quite frequently little degradation in the response of thin-film coatings. In general, the effect of increasing the incident angle is a shift in the whole reflectance curve down to slightly shorter wavelengths. This kind of behavior is evidenced by several naturally occurring periodic structures, e.g. peacock and hummingbird feathers, butterflies' wings, as well as the backs of several varieties of beetles.

The last multilayer system to be considered is the *interference*, or more precisely the *Fabry–Perot, filter*. If the separation between the plates of an etalon is of the order of λ, the transmission peaks will be widely separated in wavelength. It will then be possible to block all of the peaks but one, by using absorbing filters of colored glass or gelatin. The transmitted light then corresponds to a single sharp peak and the etalon serves as a narrow band-pass filter. Such devices can be fabricated by depositing a semitransparent metal film onto a glass support, followed by a MgF_2 spacer and another metal coating.

All-dielectric, essentially nonabsorbing Fabry–Perot filters have an analogous structure, two possible examples of which are

$$g\,HLH\,LL\,HLH\,a$$

and

$$g\,HLHL\,HH\,LHLH\,a.$$

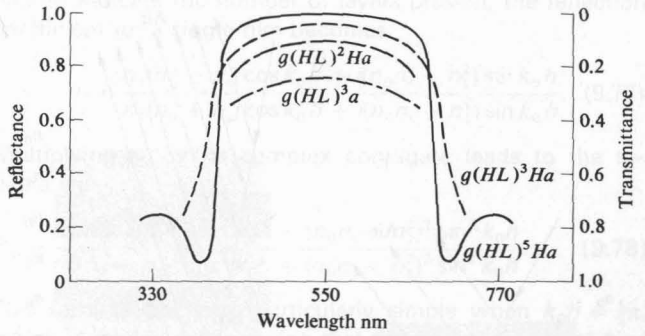


Fig. 9.57 Reflectance and transmittance for several periodic structures.

The characteristic matrix for the first of these is

$$\mathscr{M} = \mathscr{M}_H \mathscr{M}_L \mathscr{M}_H \mathscr{M}_L \mathscr{M}_H \mathscr{M}_L \mathscr{M}_H$$

but from Eq. (9.82)

$$\mathscr{M}_L \mathscr{M}_L = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix},$$

or

$$\mathscr{M}_L \mathscr{M}_L = -\mathscr{I},$$

where $\mathscr{I}$ is the unity matrix. The central double layer, corresponding to the Fabry–Perot cavity, is a half-wavelength thick ($d = \frac{1}{2}\lambda_f$). It therefore has no effect on the reflectance *at the particular wavelength under consideration*. Thus, it is said to be an absentee layer, and as a consequence

$$\mathscr{M} = -\mathscr{M}_H \mathscr{M}_L \mathscr{M}_H \mathscr{M}_H \mathscr{M}_L \mathscr{M}_H$$

The same conditions prevail again at the center, and will obviously finally result in

$$\mathscr{M} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

At the special frequency for which the filter was designed, r, according to Eq. (9.75) at normal incidence reduces to

$$r = \frac{n_0 - n_s}{n_0 + n_s},$$

the value for the uncoated substrate. In particular, for glass ($n_s = 1.5$), in air ($n_0 = 1$) the theoretical peak transmission is 96% (neglecting reflections from the back surface of the substrate, as well as losses in both the blocking filter and the films themselves).

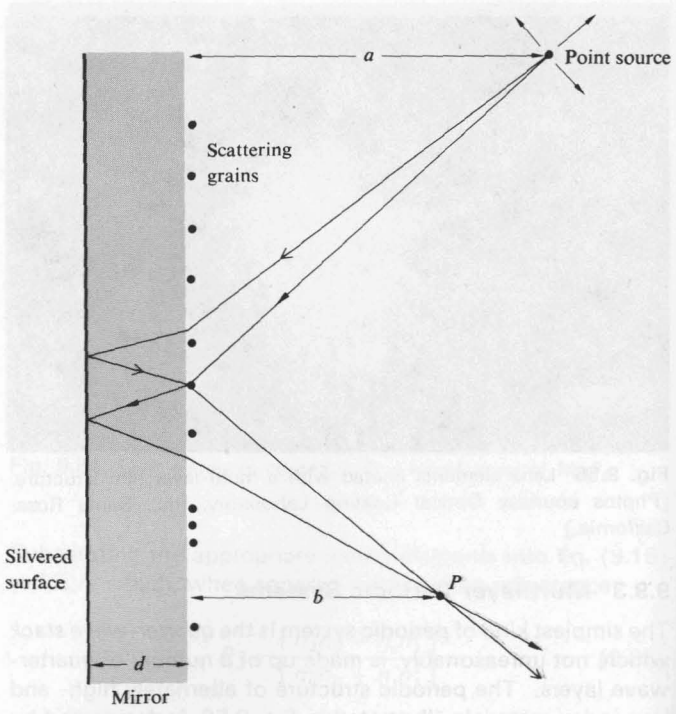


Fig. 9.58 Interference of scattered light.

9.10 APPLICATIONS OF INTERFEROMETRY

There are a great many physical applications utilizing the principles of interferometry. Some of these are now only of historical or pedagogical significance, while others are presently being used extensively. The advent of the laser and the resultant availability of highly coherent quasimonochromatic light has made it particularly easy to create new interferometer configurations.

9.10.1 Scattered-light Interference

Probably the earliest recorded study of interference fringes arising from scattered light is to be found in Sir Isaac Newton's *Optiks* (1704, Book Two, Part IV). Our present interest in this phenomenon is twofold. Firstly, it provides an extremely easy way to see some rather beautiful colored interference fringes. Secondly, it is the basis for a remarkably simple and highly useful interferometer.

To see the fringes, lightly rub a thin layer of ordinary talcum powder onto the surface of any common back-silvered mirror (dew will do as well). Neither the thickness

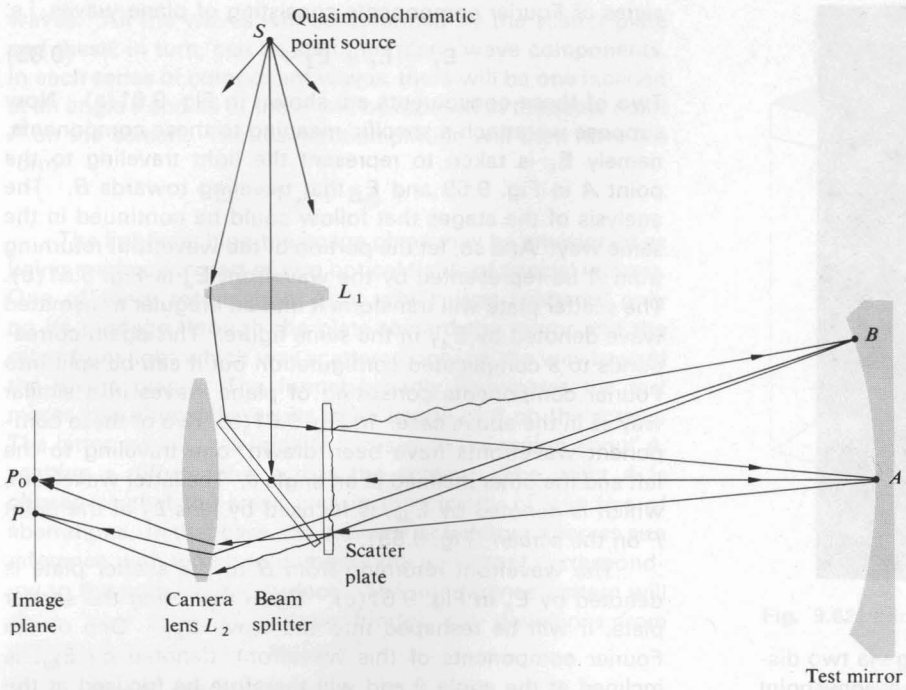


Fig. 9.59 Scatter plate set up. Adapted from R. M. Scott, *Appl. Opt.* **8**, 531 (1969).

nor the uniformity of the coating is particularly important. The use of a bright point source, however, is crucial. A satisfactory source can be made by taping a heavy piece of cardboard having a hole about $\frac{1}{4}$ inch in diameter over a good flashlight. Initially, stand back from the mirror about three or four feet; the fringes will be too fine and closely spaced to see if you stand very much nearer. Hold the flashlight alongside your cheek and illuminate the mirror so that you can see the brightest reflection of the bulb in it. The fringes will then be clearly seen as a number of alternately bright and dark bands.

Two coherent rays leaving the point source are shown in Fig. 9.58 arriving at point P after traveling different routes. One ray is reflected from the mirror and then scattered by a single transparent talcum grain toward P. The second ray is first scattered downward by the grain after which it crosses the mirror and is reflected back toward P. The resulting optical path-length difference determines the interference at P. At normal incidence, the pattern is a series of concentric rings of radius*

$$\rho \approx \left[\frac{nm\lambda a^2 b^2}{d(a^2 - b^2)}\right]^{\frac{1}{2}}.$$

Now consider a related device which is very useful in testing optical systems. Known as a *scatter plate*, it generally consists of a slightly rough-surfaced, transparent sheet. In an arrangement such as the one shown in Fig. 9.59, it serves as an amplitude-splitting element. In this application it must have a center of symmetry, i.e. each scattering site is required to have a duplicate, symmetrically located about a central point.

In the system under consideration, a point source of quasimonochromatic light S is imaged, by means of lens L_1 on the surface, at point A of the mirror being tested. A portion of the light coming from the source is scattered by the scatter plate and thereafter illuminates the entire surface of the mirror. The mirror, in turn, reflects light back to the scatter plate. This wave, as well as the light forming the image of the pinhole at point A, passes through the scatter plate again, and finally reaches the image plane (either on a screen or in a camera). Fringes are formed on this latter plane. The interference process, which is manifest in the formation of these fringes, occurs because each point in the

* For more of the details, see A. J. deWitte, "Interference in Scattered Light," *Am. J. Phys.* **35**, 301 (1967).

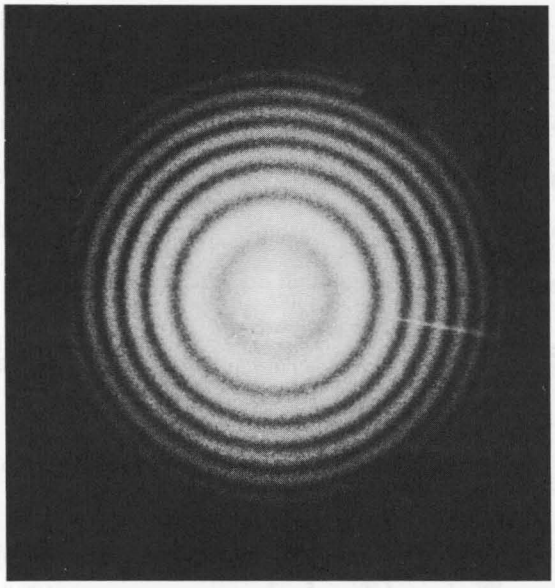


Fig. 9.60 Fringes in scattered light.

final image plane is illuminated by light arriving via two dissimilar routes; one originating at A and the other at some point B which reflects scattered light. Indeed, as strange as it may look at first sight, well-defined fringes do result as may be seen from Fig. 9.60.

Examining the passage of light through the system in a bit more detail, consider the light initially incident on the scatter plate and assume that the wave is planar as shown in Fig. 9.61. After it passes through the scatter plate, the incident plane wavefront $\mathbf{E}_i$ will be distorted into a transmitted wavefront $\mathbf{E}_T$. We envision this wave, in turn, split into a series of Fourier components consisting of plane waves, i.e.

$$\mathbf{E}_T = \mathbf{E}_1 + \mathbf{E}_2 + \cdots . \tag{9.85}$$

Two of these constituents are shown in Fig. 9.61(a). Now suppose we attach a specific meaning to these components, namely $\mathbf{E}_1$ is taken to represent the light traveling to the point A in Fig. 9.59 and $\mathbf{E}_2$ that traveling towards B. The analysis of the stages that follow could be continued in the same way. And so, let the portion of the wavefront returning from A be represented by the wavefront $\mathbf{E}_A$ in Fig. 9.61(b). The scatter plate will transform it into an irregular transmitted wave denoted by $\mathbf{E}_{AT}$ in the same figure. This again corresponds to a complicated configuration but it can be split into Fourier components consisting of plane waves in a similar way as in the above case. In Fig. 9.61(b), two of these component wavefronts have been drawn, one traveling to the left and the other inclined at an angle θ. The latter wavefront, which is denoted by $\mathbf{E}_{A\theta}$, is focused by lens L_2 at the point P on the screen (Fig. 9.59).

The wavefront returning from B to the scatter plate is denoted by $\mathbf{E}_B$ in Fig. 9.61(c). Upon traversing the scatter plate, it will be reshaped into the wave $\mathbf{E}_{BT}$. One of the Fourier components of this wavefront, denoted by $\mathbf{E}_{B\theta}$, is inclined at the angle θ and will therefore be focused at the same point P on the screen.

Some of the waves arriving at P will be coherent in the sense that interference occurs. To obtain the resultant irradiance I_P, first add the amplitudes of all the waves arriving at P, i.e. $\mathbf{E}_P$ and then square and time average $\mathbf{E}_P$.

In the discussion above, only two point sources at the mirror were considered. Actually, of course, the whole surface of the mirror is illuminated by the ongoing light, and every point of it will serve as a secondary source of returning

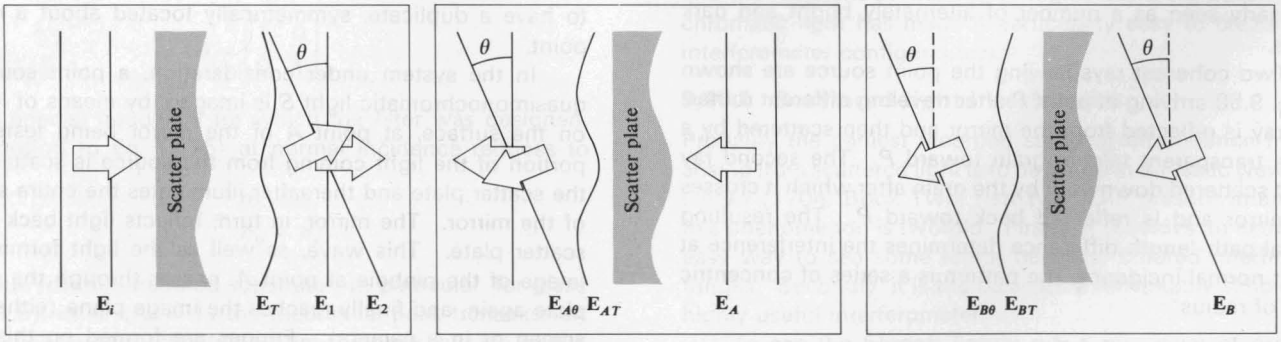


Fig. 9.61 Wavefronts passing through the scatter plate.

waves. All the waves will be deformed by the scatter plate and these, in turn, can be split into plane wave components. In each series of component waves, there will be one inclined at an angle θ and all of these will be focused at the same point P on the screen. The resultant amplitude will then have the form

$$\mathbf{E}_P = \mathbf{E}_{A\theta} + \mathbf{E}_{B\theta} + \cdots .$$

The light reaching the image plane may be envisioned as being made up in part of two optical fields of special interest. One of these results from light which was scattered only on its passage through the plate toward the mirror and the other from light which was scattered only on the way toward the image plane. The former broadly illuminates the test mirror and ultimately results in an image of it on the screen. The latter, which was initially focused to the region about A, scatters a diffuse blur across the screen. The point A is chosen so that the small area in the vicinity of it is free of aberrations. In that case, the wave reflected from it serves as a reference with which to compare the wavefront corresponding to the entire mirror surface. The interference pattern will show, as a series of contour fringes, any deviations from perfection in the mirror surface.*

9.10.2 Thin-film Measurements by Multiple-beam Interferometry

Return to Fig. 9.32 and now suppose that the wedge has a step in it. Figure 9.62 illustrates the fringe pattern that might be seen under these circumstances. If the wedge angle is the same for each surface, i.e. if the top surfaces are parallel, the fringes will be equally spaced.

When the separation of the fringes is b, and the shift is a, then the height of the step is given by

$$t = \frac{a}{b}\frac{\lambda_f}{2}.$$

If one of the boundaries of the film is an optical flat and the other boundary is a crystal surface or some other surface examined for flatness, then these Fizeau fringes are contours of the surface under examination.

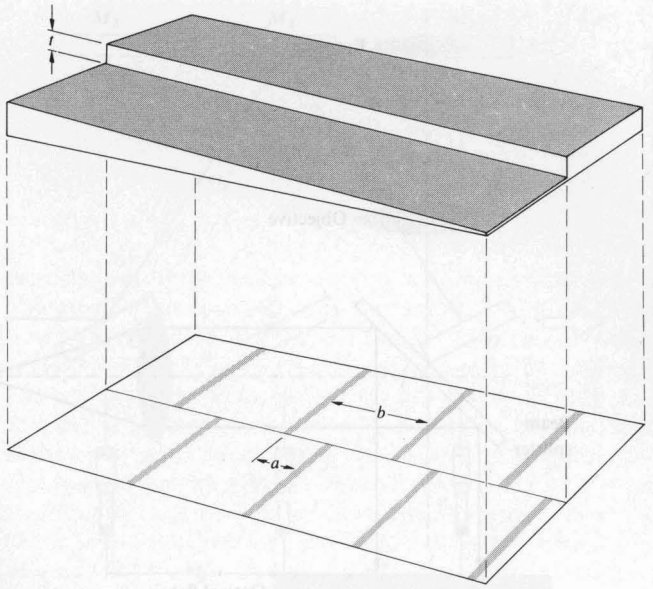


Fig. 9.62 Fringes arising from a stepped wedge-shaped film.

An actual optical system for measuring the thickness of a thin film deposited on a glass substrate, is shown in Fig. 9.63. The film whose thickness is to be determined is coated with an opaque layer of silver, about 70 nm thick, which accurately contours the undersurface. The opposing silvered surfaces generate a sharp multiple-wave Fizeau pattern. The upper plate is tilted slightly to create an air film in the form of Fig. 9.62, so that the same arrangement of fringes is now observed, Fig. 9.64. Film thicknesses of about 2.0 nm can readily be determined in this manner. Such methods yield a resolution in depth comparable to the lateral resolution of an electron microscope. Tolansky, using the multiple-beam techniques which he invented, has measured height changes of 1×10^{-8} inches, very nearly the size of a single atom.

9.10.3 The Michelson–Morley Experiment

The Michelson interferometer has, over the years since 1881, found innumerable applications, most of which are now mainly of historical interest. One of the most significant of these was its use in the Michelson–Morley experiment.

During the last century there was a common belief amongst scientists that there existed a medium, the *luminiferous* (light-carrying) *ether*, which permeated all matter, pervaded all space, was massless and neither solid, liquid, nor

* For further discussion of the scatter plate, the reader might consult the rather succinct papers by J. M. Burch *Nature* **171**, 889 (1953) and *J. Opt. Soc. Am.* **52**, 600 (1962). Reference should be made to J. Strong, *Concepts of Classical Optics*, p. 383. Also see R. M. Scott, "Scatter Plate Interferometry," *Appl. Opt.* **8**, 531 (1969) and J. B. Houston, Jr., "How to Make and Use a Scatterplate Interferometer," *Optical Spectra* (June 1970), p. 32.

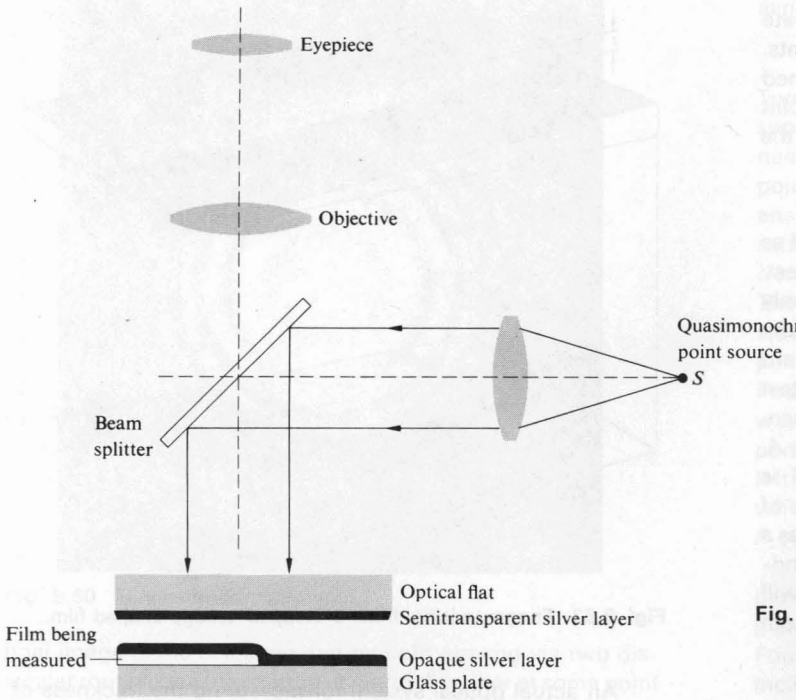


Fig. 9.63 Arrangement for measuring film thickness.

gas. As James Maxwell wrote in the *Encyclopaedia Britannica*:

> Ethers were invented for the planets to swim in, to constitute electric atmospheres and magnetic affluvia, to convey sensations from one part of our bodies to another, and so on, until all space had been filled three or four times over with ethers . . . The only ether which has survived is that which was invented by Huygens to explain the propagation of light.

It was well established that light was a wave, and so it was only natural to have a medium in which the disturbance propagated. Assuming that much, the nature of the ether had to match terrestrial and astronomical observations. At the time, there was no denying the actual existence of ether; the debate centered on its physical properties. Was the ether stationary in space, thereby providing a reference frame from which to measure the absolute motion of all other objects? Or, was it dragged along by the planets as they moved through space? If the ether were stationary an observer on the earth would be able to detect an ether wind streaming over its surface, as it moved in orbit. A. A. Michelson, later joined by E. W. Morley, set out to measure the effects of the ether

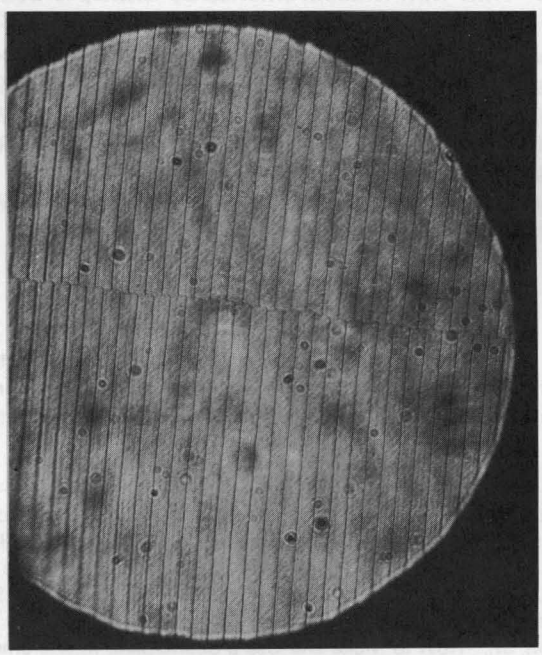


Fig. 9.64 Actual fringes from a stepped wedge.

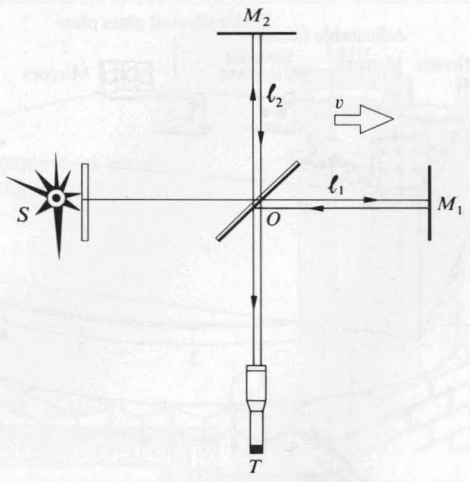


Fig. 9.65 The Michelson–Morley experiment.

wind using his interferometer, which was designed specifically for that purpose. It was oriented as shown in Fig. 9.65, with the arm OM_1 parallel to the velocity **v** of the earth through space. The basic reasoning of the Michelson–Morley approach, derived from purely classical laws of physics, was as follows: when the beam of light travels to the right, its relative speed with respect to the moving interferometer is $c - v$, it is moving against the ether wind, and the time to travel the length OM_1 is

$$t_1' = \frac{\ell_1}{c - v}.$$

For the return trip, M_1O, the beam travels with the ether wind and

$$t_1'' = \frac{\ell_1}{c + v}.$$

The total time, $t_1' + t_1''$, to traverse OM_1O is

$$t_1 = \frac{\ell_1}{c - v} + \frac{\ell_1}{c + v},$$

which can be written as

$$t_1 = \frac{2\ell_1}{c}\beta^2,$$

where

$$\beta = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

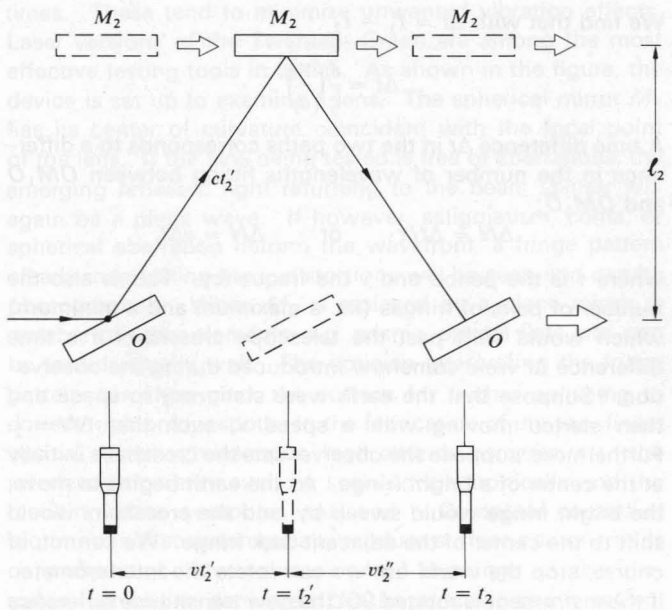


Fig. 9.66 The Michelson–Morley experiment.

The time of travel towards the second mirror can be obtained with the help of Fig. 9.66. From the right triangle, where t_2' is the transit time to cover OM_2,

$$c^2t_2'^2 = v^2t_2'^2 + \ell_2^2$$

from which it follows that

$$t_2' = \frac{\ell_2}{c}\beta.$$

But this is also the time t_2'' that it takes the beam of light to return from M_2 to O and so, since $t_2 = t_2' + t_2''$,

$$t_2 = \frac{2\ell_2}{c}\beta.$$

Notice that even when $\ell_1 = \ell_2 = \ell$, $t_1 \neq t_2$ and

$$t_1 - t_2 = \frac{2\ell}{c}(\beta^2 - \beta).$$

Using the binomial expansion with $c \gg v$,

$$\beta^2 = (1 - v^2/c^2)^{-1} = 1 + v^2/c^2$$

and

$$\beta = (1 - v^2/c^2)^{-\frac{1}{2}}$$

or

$$\beta = 1 + \tfrac{1}{2}v^2/c^2.$$

We find that with $\Delta t = t_1 - t_2$

$$\Delta t = \frac{\ell}{c}\left(\frac{v}{c}\right)^2.$$

A time difference Δt in the two paths corresponds to a difference in the number of wavelengths fitting between OM_1O and OM_2O:

$$\Delta N = \Delta t/\tau \qquad \text{or} \qquad \Delta N = \nu \Delta t,$$

where τ is the period and ν the frequency. This is also the number of pairs of fringes (i.e. a maximum and a minimum) which would shift past the telescope crosshairs if a time difference Δt were somehow introduced during the observation. Suppose that the earth were stationary in space and then started moving with a speed v, such that $\Delta N = \frac{1}{2}$. Furthermore, suppose the observer sets the crosshairs initially at the center of a bright fringe. As the earth begins to move, the bright fringe would sweep by, and the crosshairs would shift to the center of the adjacent dark fringe. We cannot, of course, stop the world but we can rotate the interferometer. If the instrument is rotated 90° the new transit time difference can be gotten by just interchanging the 1 and 2 subscripts and is therefore equal to $-\Delta t$. This means that if the observer were to rotate the interferometer 90°, a time difference of $2\,\Delta t$ would be introduced for which, in that example, $\Delta N = 1$, and the crosshairs would end up on the next bright fringe.

This is essentially what Michelson and Morley did do. Their apparatus was multimirrored to make the path length as large as possible, $\ell_1 \approx \ell_2 \approx 11.0$ m. It rested on a massive stone which floated on a trough filled with mercury (Fig. 9.67). Each man took turns slowly pushing the stone, while continuously observing the fringe pattern. Assuming a velocity equal to the earth's orbital speed of about 30 km/s and $\lambda_0 = 550$ nm, the fringe shift on rotation would be

$$\Delta N = \frac{2\ell}{\lambda}\left(\frac{v}{c}\right)^2$$

or

$$\Delta N = 0.4.$$

They made thousands of observations at different hours of the earth's daily cycle and on different days during its yearly orbit. And yet, even though they could have detected a shift of a minute fraction of a fringe, they saw none whatever. There was no ether wind; Michelson and Morley had sounded the prelude to special relativity.

Ten years later, Michelson interferometrically tested the possibility that the ether was being dragged along with the earth. His results showed that this too was not true, and the ether theory was doomed.

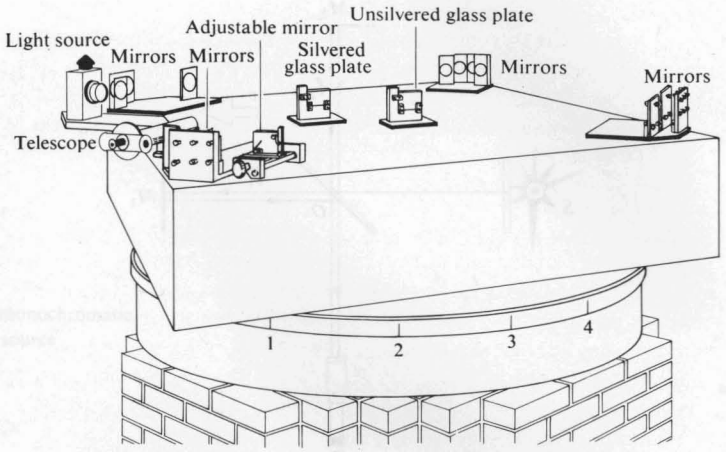


Fig. 9.67 The Michelson–Morley experiment.

A modern version of the Michelson–Morley experiment,* shown here in Fig. 9.68, compared the frequencies of two infrared lasers. (Recall that in Section 7.2.1, we considered the application of lasers to the problem of generating beats.) The combined beam reaching the photomultiplier, being the resultant of two coplanar harmonic waves, was *amplitude modulated* by a relatively slow variation. These *beats* had a frequency equal to the difference between those of the two constituent laser beams. The precise frequency of the mode in which each laser operated, was governed by the length of the particular resonant cavity, and the speed of light therein. If both lasers, functioning at about 3×10^{14} Hz, were rotated 90°, the *ether wind* would affect the speed of light in the cavities and therefore the frequency difference between them. A relative change in ν of 3 MHz would be expected from the ether wind hypothesis, due to the earth's orbital velocity. No change in the beat frequency, to within an accuracy of 3 kHz or $\frac{1}{1000}$ of that predicted, was detected.

9.10.4 The Twyman–Green Interferometer

The Twyman–Green is essentially a variation of the Michelson interferometer. It's an instrument of very great importance in the domain of modern optical testing. Amongst its

* T. S. Jaseja, A. Javan, J. Murray, and C. H. Townes, "Test of Special Relativity or of the Isotropy of Space by Use of Infrared Masers," *Phys. Rev.* **133**, A1221 (1964).

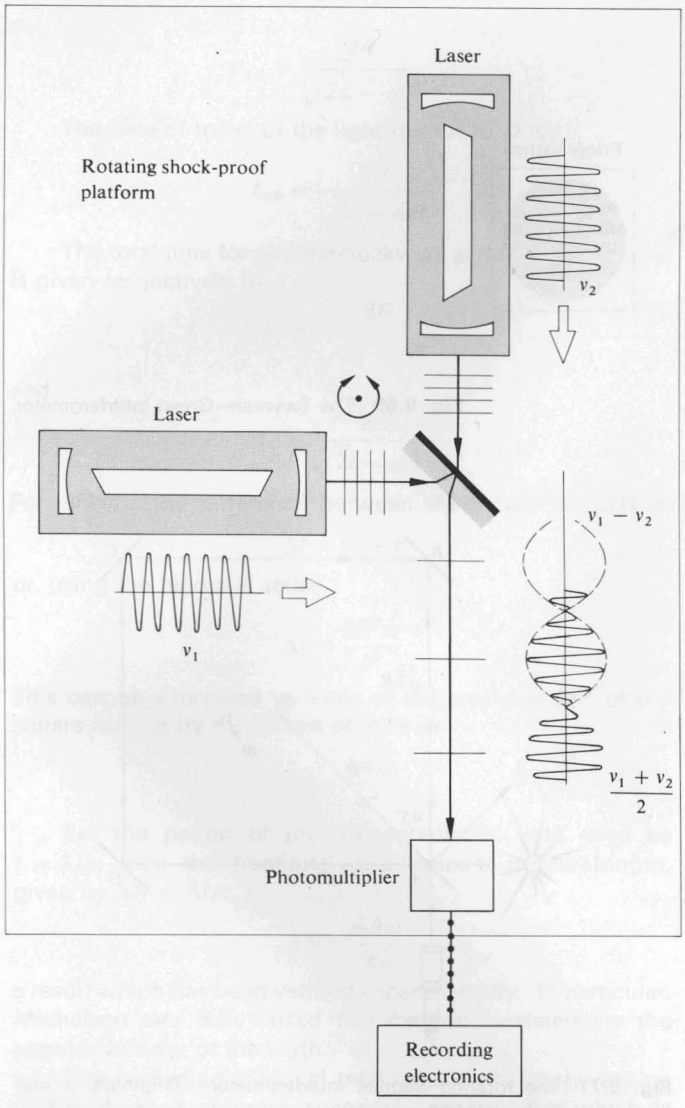


Fig. 9.68 A variation of the Michelson–Morley experiment.

distinguishing physical characteristics are (as illustrated in Fig. 9.69) a quasimonochromatic point source and lens L_1 to provide a source of incoming *plane waves*, and a lens L_2 which permits all of the light from the aperture to enter the eye so that the entire field can be seen, i.e., any portion of M_1 and M_2. A continuous laser serves as a superior source in that it provides both the convenience of long path-length differences and, in addition, short photographic exposure times. These tend to minimize unwanted vibration effects. Laser versions of the Twyman–Green are among the most effective testing tools in optics. As shown in the figure, the device is set up to examine a lens. The spherical mirror M_2 has its center of curvature coincident with the focal point of the lens. If the lens being tested is free of aberrations, the emerging reflected light returning to the beam splitter will again be a plane wave. If however, astigmatism, coma, or spherical aberration deform the wavefront, a fringe pattern clearly manifesting these distortions will be seen and can be photographed. When M_2 is replaced by a plane mirror, a number of other elements, e.g. prisms, optical flats, etc. can be tested equally well. The optician interpreting the fringe pattern can then mark the surface for further polishing to correct high or low spots. In the fabrication of the very finest optical systems, telescopes, high altitude cameras, etc., the interferograms may even be scanned electronically and the resulting data computer analyzed. Computer-controlled plotters can then automatically produce surface contour maps or perspective "three-dimensional" drawings of the distorted wavefront generated by the element being tested. These procedures can be used throughout the fabrication process to ensure the highest-quality optical instruments. Complex systems with wavefront aberrations in the fractional-wavelength range are the result of what might be called the *new technology*.*

9.11 THE ROTATING SAGNAC INTERFEROMETER

The application of the Sagnac interferometer to the task of measuring the rotational speed of a system has become of interest in recent times. In particular the *ring laser*, which is essentially a Sagnac interferometer containing a laser in one or more of its arms, was designed specifically for that purpose. The first ring laser gyroscope was introduced in 1963 and work is continuing on various devices of this sort (Fig. 9.70). The initial experiments which gave impetus to these efforts were performed by Sagnac in 1911. At that time he rotated the entire interferometer, mirrors, source, and detector, about a perpendicular axis passing through its center (Fig. 9.71). Recall, from Section 9.4, that two overlapping beams traverse the interferometer, one clockwise, the other counterclockwise. The rotation effectively shortens the path taken by one beam in comparison to that of the other. In the interferometer the result is a fringe shift proportional to the angular

* Take a look at R. Berggren, "Analysis of Interferograms," *Optical Spectra*, (Dec. 1970), p. 22.

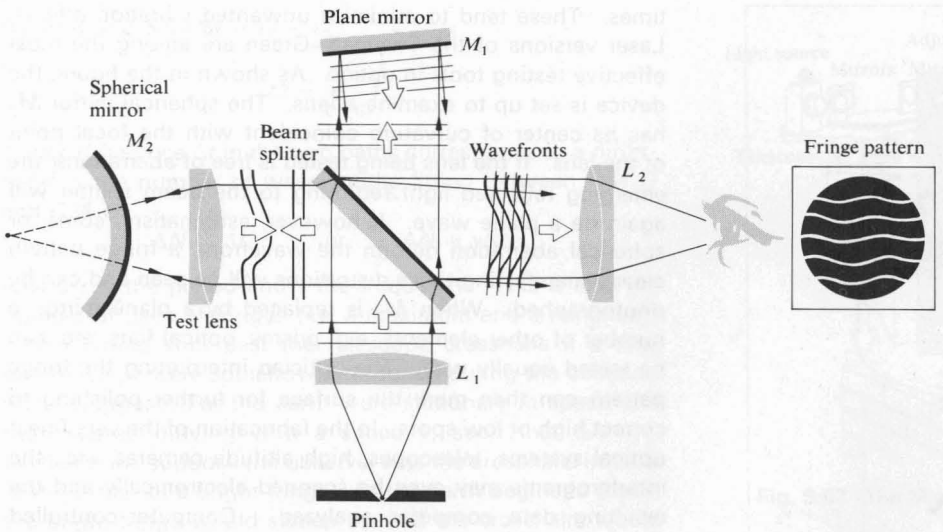


Fig. 9.69 The Twyman–Green interferometer.

Fig. 9.70 A ring laser gyro. [Photo courtesy Autonetics, a Division of North American Rockwell Corp.]

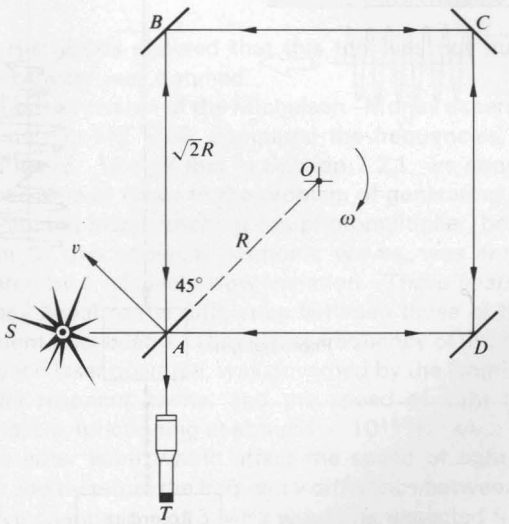


Fig. 9.71 The rotating Sagnac interferometer. Originally it was 1 m × 1 m with $\omega = 120$ rev/min.

speed of rotation ω. In the ring laser, it is a frequency difference between the two beams which is proportional to ω.

Consider the arrangement depicted in Fig. (9.71). The corner A (and every other corner) moves with a linear speed $v = R\omega$, where R is half the diagonal of the square. Using classical reasoning we find that the time of travel of light along AB is

$$t_{AB} = \frac{R\sqrt{2}}{c - v/\sqrt{2}}$$

or

$$t_{AB} = \frac{2R}{\sqrt{2}c - \omega R}$$

The time of travel of the light from A to D is

$$t_{AD} = \frac{2R}{\sqrt{2}c + \omega R}.$$

The total time for counterclockwise and clockwise travel is given respectively by

$$t_{\circlearrowleft} = \frac{8R}{\sqrt{2}c + \omega R}$$

and

$$t_{\circlearrowright} = \frac{8R}{\sqrt{2}c - \omega R}$$

For $\omega R \ll c$ the difference between these two intervals is

$$\Delta t = t_{\circlearrowleft} - t_{\circlearrowright}$$

or, using the binomial series

$$\Delta t = \frac{8R^2\omega}{c^2}.$$

This can be expressed in terms of the area $A = 2R^2$ of the square formed by the beams of light as

$$\Delta t = \frac{4A\omega}{c^2}.$$

Let the period of the monochromatic light used be $\tau = \lambda/c$; then the fractional displacement of the fringes, given by $\Delta N = \Delta t/\tau$, is

$$\Delta N = \frac{4A\omega}{c\lambda},$$

a result which has been verified experimentally. In particular, Michelson and Gale* used this method to determine the angular velocity of the earth.

The preceding classical treatment is obviously lacking inasmuch as it assumes speeds in excess of c which is contrary to the dictates of special relativity. Furthermore, it would appear that since the system is accelerating, general relativity would prevail. In fact, all of these formalisms yield the same results.

PROBLEMS

9.1 Returning to Section 9.1 let

$$\mathbf{E}_1(\mathbf{r}, t) = \boldsymbol{E}_1(\mathbf{r})e^{-i\omega t}$$

and

$$\mathbf{E}_2(\mathbf{r}, t) = \boldsymbol{E}_2(\mathbf{r})e^{-i\omega t},$$

where the wavefront shapes are not explicitly specified and $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$ are complex vectors depending on space and epoch angle. Show that the interference term is then given by

$$I_{12} = \tfrac{1}{2}(\boldsymbol{E}_1 \cdot \boldsymbol{E}_2^* + \boldsymbol{E}_1^* \cdot \boldsymbol{E}_2). \tag{9.86}$$

You will have to evaluate terms of the form

$$\langle \boldsymbol{E}_1 \cdot \boldsymbol{E}_2 e^{-2i\omega t} \rangle = \frac{\boldsymbol{E}_1 \cdot \boldsymbol{E}_2}{T} \int_t^{t+T} e^{-2i\omega t'}\, dt'$$

for $T \gg \tau$ (take another look at Problem 3.8). Show that Eq. (9.86) leads to Eq. (9.4) for plane waves.

9.2 In Section 9.1 we considered the spatial distribution of energy for two point sources. We mentioned that for the case where the separation $a \gg \lambda$, I_{12} spatially averages to zero. Why is this true? What happens when a is much less than λ?

9.3 Will we get an interference pattern in Young's experiment (Fig. 9.5) if we replace the source slit S by a single long-filament light bulb? What would occur if we replaced the slits S_1 and S_2 by these same bulbs?

9.4 To examine the conditions under which the approximations of Eq. (9.9) are valid:

a) Apply the law of cosines to triangle S_1S_2P in Fig. 9.5(c) to get

$$\frac{r_2}{r_1} = \left[1 - 2\left(\frac{a}{r_1}\right)\sin\theta + \left(\frac{a}{r_1}\right)^2\right]^{1/2}$$

b) Expand this in a Maclaurin series yielding

$$r_2 = r_1 - a\sin\theta + \frac{a^2}{2r_1}\cos^2\theta + \cdots.$$

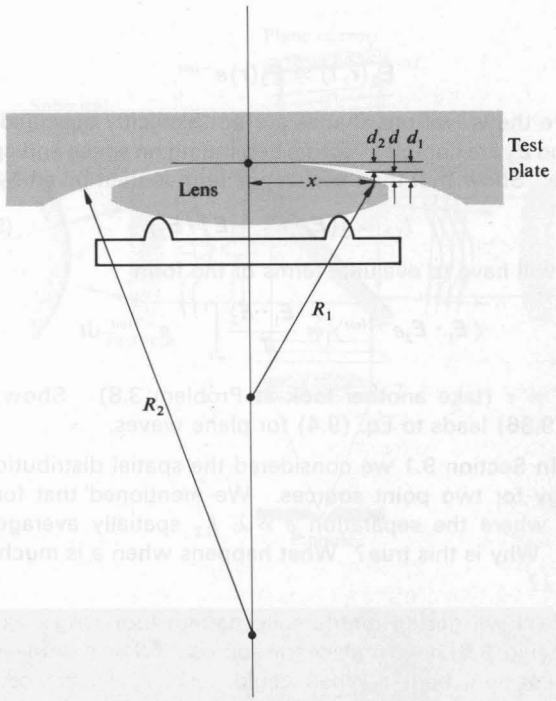
c) In light of Eq. (9.6), show that if $(r_1 - r_2)$ is to equal $a\sin\theta$ it is required that $r_1 \gg a^2/\lambda$.

9.5* Show that a for the Fresnel biprism of Fig. 9.9 is given by $a = 2d(n-1)\alpha$.

9.6* In the Fresnel double mirror $s = 2$ m, $\lambda_0 = 589$ nm, and the separation of the fringes was found to be 0.5 mm. What is the angle of inclination of the mirrors if the perpendicular distance of the actual point source to the intersection of the two mirrors is 1 m?

9.7* The Fresnel biprism is used to obtain fringes from a point source which is placed 2 m from the screen and the

* Michelson and Gale, *Astrophys. J.* **61**, 140 (1925).



prism is midway between the source and the screen. Let the light have the wavelength $\lambda_0 = 500$ nm, and the index of refraction of the glass be $n = 1.5$. What is the prism angle if the separation of the fringes is 0.5 mm?

9.8 What is the general expression for the separation of the fringes of a Fresnel biprism of index n immersed in a medium having an index of refraction n'?

9.9 Using Lloyd's mirror, x-ray fringes were observed, the spacing of which was found to be 0.0025 cm. The wavelength used was 8.33 Å. If the source–screen distance is 3 m, how high above the mirror plane was the point source of x-rays placed?

9.10* If the plate in Fig. 9.27 is glass in air, show that the amplitudes of E_{1r}, E_{2r}, and E_{3r} are respectively $0{:}2E_{0i}$, $0.192\,E_{0i}$ and $0.008\,E_{0i}$ where E_{0i} is the incident amplitude. Make use of the Fresnel coefficients at normal incidence assuming no absorption. You might repeat the calculation for a water film in air.

9.11 Consider the circular pattern of Haidinger's fringes resulting from a film of thickness 2 mm and index of refraction 1.5. For monochromatic illumination of $\lambda_0 = 600$ nm, find the order of the central fringe ($\theta_t = 0$). Will it be bright or dark?

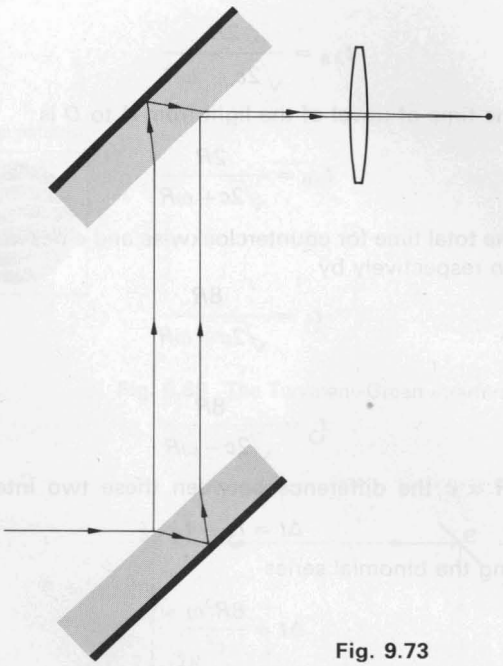


Fig. 9.73

9.12 Figure 9.72 illustrates a set-up used for testing lenses. Show that

$$d = x^2(R_2 - R_1)/2R_1R_2$$

when d_1 and d_2 are negligible in comparison with $2R_1$ and $2R_2$, respectively. (Recall the theorem from plane geometry which relates the products of the segments of intersecting chords.) Prove that the radius of the mth dark fringe is then

$$x_m = [R_1R_2m\lambda_f/(R_2 - R_1)]^{1/2}.$$

How does this relate to Eq. (9.27)?

9.13 Draw the configuration which you would use to see Newton's rings in a Twyman–Green interferometer.

9.14* Newton rings are observed with quasimonochromatic light of wavelength 500 nm. If the 20th bright ring has a radius of 1 cm, what is the radius of curvature of the lens forming one part of the interfering system?

9.15 Fringes are observed when a parallel beam of light of wavelength 500 nm is incident perpendicularly onto a wedge-shaped film of index of refraction 1.5. What is the angle of the wedge if the fringe separation is $\frac{1}{3}$ cm?

9.16* A form of the Jamin interferometer is illustrated in Fig. 9.73. How does it work? To what use might it be put?

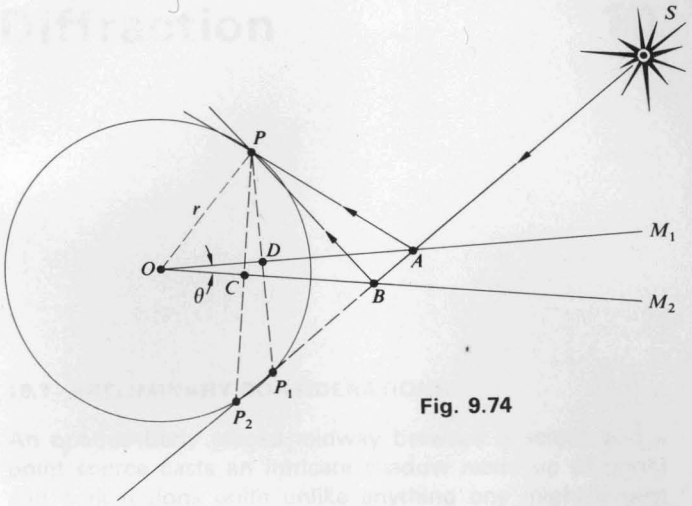


Fig. 9.74

9.17* In Searle's method of localization of fringes [*Phil. Mag.* **37**, 361 (1946)] (Fig. 9.74) the two film boundary planes $\overline{OM}_1$ and $\overline{OM}_2$ reflect a beam coming from the source. Show that the path difference for the two rays is $2r \sin \theta$ and discuss the usefulness of this construction.

9.18 Starting with Eq. (9.31) for the transmitted wave, compute the flux density, i.e. Eq. (9.32).

9.19* Satisfy yourself of the fact that a film of thickness $\lambda_f/4$ and index n_1 will always reduce the reflectance of the substrate on which it is deposited as long as $n_s > n_1 > n_0$. Consider the simplest case of normal incidence and $n_0 = 1$.

Show that this is equivalent to saying that the waves reflected back from the two interfaces cancel.

9.20 Verify that the reflectance of a substrate can be increased by coating it with a $\lambda_f/4$, high-index layer i.e. $n_1 > n_s$. Show that the reflected waves interfere constructively. The quarter-wave stack $g(HL)^m Ha$ can be thought of as a series of such structures.

9.21 Imagine that we have an antenna at the edge of a lake picking up a signal from a distant radio star (Fig. 9.75) which is just coming up above the horizon. Write expressions for δ and for the angular position of the star when the antenna detects its first maximum.

9.22 Determine the refractive index and thickness of a film to be deposited on a glass surface ($n_g = 1.54$) such that no normally incident light of wavelength 540 nm is reflected.

9.23* Figure 9.76 is a Newton's ring fringe pattern generated by a lens and an optical flat under $\lambda_0 = 600$ nm illumination.

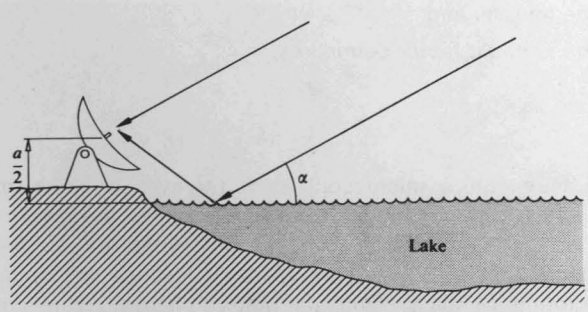


Fig. 9.75

Draw a possible contour map of the lens surface. Is it possible, with this figure alone, to determine whether the lens surface is concave or convex? Explain.

9.24 Consider the interference pattern of the Michelson interferometer as arising from two equal flux density beams. Using Equation (9.6) compute the half-width. What is the separation, in δ, between adjacent maxima? What then is the finesse?

9.25 Given that the mirrors of a Fabry–Perot interferometer have an amplitude reflection coefficient of $r = 0.8944$, find

a) the coefficient of finesse,

b) the half-width,

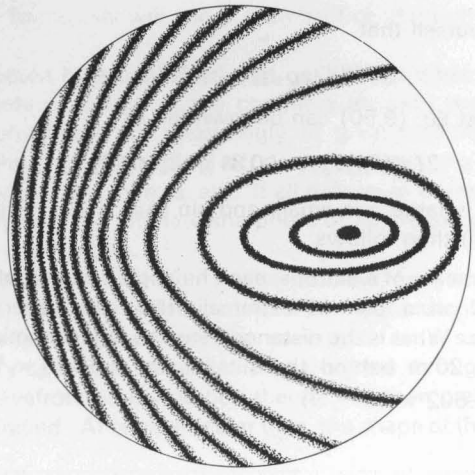


Fig. 9.76

c) the finesse, and,

d) the *contrast factor* defined by

$$C \equiv \frac{(I_t/I_i)_{\max}}{(I_t/I_i)_{\min}}.$$

9.26 Illuminate a microscope slide (or even better a thin cover glass slide). Colored fringes can easily be seen with an ordinary fluorescent lamp serving as a broad source, or a mercury street light as a point source. Describe the fringes. Now rotate the glass. Does the pattern change? Duplicate the conditions shown in Figs. 9.28 and 9.29. Try it again with a sheet of plastic food wrap stretched across the top of a cup.

9.27* In order to get some feeling for the need for antireflection coatings, consider the following somewhat oversimplified problem. If there is normal incidence, no absorption and $n = 1.5$, what fraction of the incident light will be transmitted through a twelve-element lens system?

9.28* Discuss the influence of the linewidth $\Delta\lambda$ of the source on the sharpness of fringes in an interferometer. Show that the fringes will disappear when $\Delta\lambda \approx \lambda/m$ where m, the order of interference, is a large number. Determine the order for which the fringes first vanish with the red cadmium line of Fig. 7.18. Take a look at Section 12.2 for further study.

9.29 To fill in some of the details in the derivation of the smallest phase increment separating two resolvable Fabry–Perot fringes, that is

$$(\Delta\delta) \approx 4.2/\sqrt{F}, \qquad [9.51]$$

satisfy yourself that

$$[\mathscr{A}(\theta)]_{\delta=\delta_a \pm \Delta\delta/2} = [\mathscr{A}(\theta)]_{\delta=\Delta\delta/2}.$$

Show that Eq. (9.50) can be rewritten as

$$2[\mathscr{A}(\theta)]_{\delta=\Delta\delta/2} = 0.81\{1 + [\mathscr{A}(\theta)]_{\delta=\Delta\delta}\}.$$

When F is large γ is small, and $\sin(\Delta\delta) \approx \Delta\delta$. Prove that Eq. (9.51) then follows.

9.30 A stream of electrons, each having an energy of 0.5 eV, impinges on a pair of extremely thin slits separated by 10^{-2} mm. What is the distance between adjacent minima on a screen 20 m behind the slits? ($m_e = 9.108 \times 10^{-31}$ kg, 1 eV $= 1.602 \times 10^{-19}$ J).